

Danish Technical University
41789 Introduction to 3D printing
summary

Comprehensive Study Notes from spring 2026



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1 Fused Deposition Model (FDM)



Figure 1: FDM Printers - Ultimaker, bq and Wanhao respectively

FDM can be used for, Concept models, Functional prototypes, Manufacturing aids and Low volume end-use parts.

Process sequence

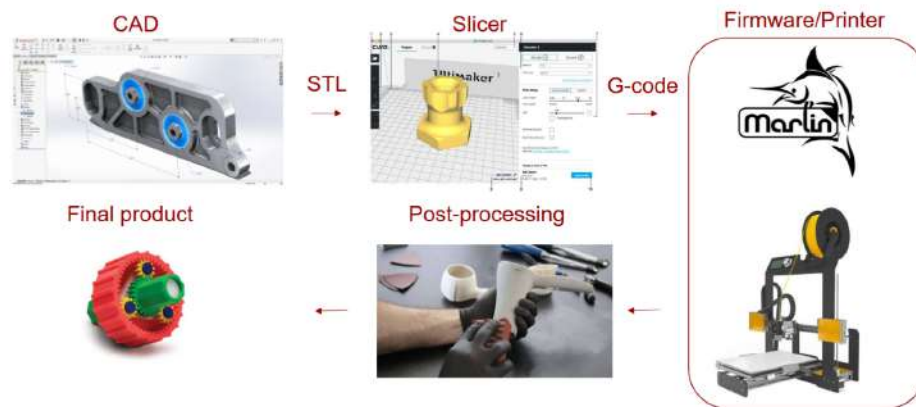


Figure 2: Process sequence of FDM

As can be seen from the diagram above the FDM process starts as a CAD design where it is exported as a STL file into a Slicer where the printing specifications are introduced, after this it is exported as G-code into the Firmware/Printer before finally going through the post processing into the final product.

CAD

- CAD software: Autodesk Fusion, SolidWorks
- Most CAD software supports STL
- Build orientation
- Freedom in the geometry
- Pre-assembled parts
- Topology optimization

Slicer

- Slicers: Cura, Slic3r, Simplify3D.

- Important parameters,
 - Build orientation
 - Layer thickness
 - Printing speed
 - Infill of part
 - Printing temperature
 - Build plate temperature

Firmware

- Internal software, which is installed on the mainboard of the 3D printer
- Interprets the G-code and is responsible for its execution
- Controls the state of the printer
- Marlin open-source firmware

Printer

Motors



Figure 3: Motors

Extrusion Systems

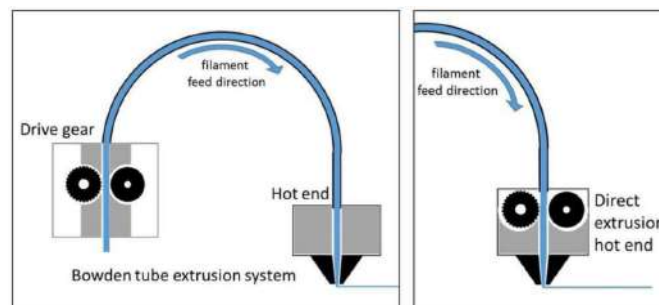


Figure 4: Extrusion Systems

Materials - Thermoplastics

- Polylactic Acid (PLA)
 - Pros: Printability, Biodegradable, Odorless, UV resistance.
 - Cons: Humidity resistance, Glueability.

- Akrylonitril-butadien-styren (ABS)
 - Pros: Heat-resistant, Abrasion resistance, Post-processed with acetone vapor.
 - Cons: UV sensitive, Oder, Fume emissions.

Materials - Other materials

- Polyetylentereftala (PET)
 - Pros: Contact with foods, humidity resistance, recyclable.
 - Cons: Heavier than PLA and ABS.
- Nylon
 - Pros: Chemical resistance, high strength
 - Cons: Absorbs moisture, Fume emissions.
- Thermoplastic polyurethane (TPU)
 - Pros: Flexible, Abrasion resistance.
 - Cons: Difficult to post process.

Post-Processing

Support Removal



Cold Welding



Gap Filling



Sanding



Painting

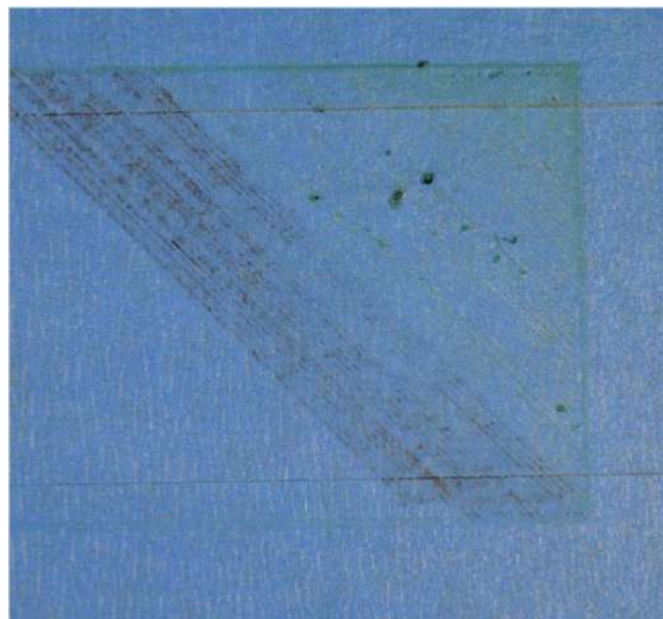


Vapor Smoothing



1.1 Print Quality Troubleshooting

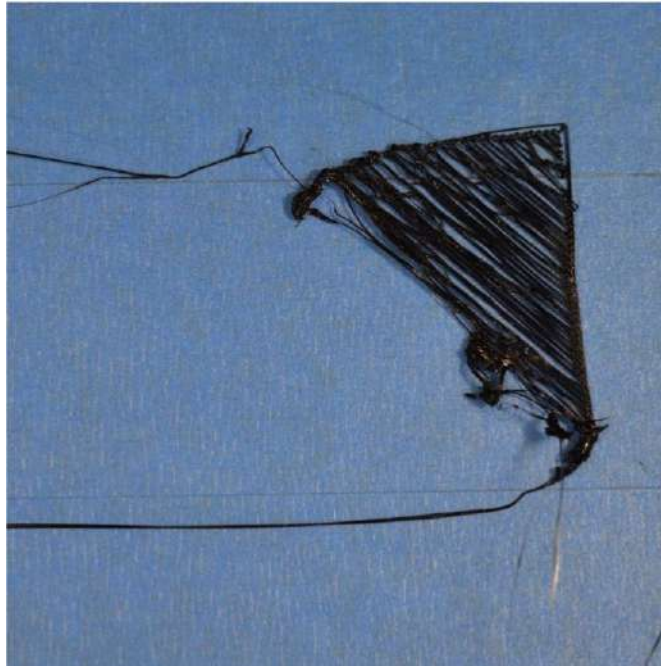
Not Extruding at Start of Print



Common Solutions

- Extruder was not primed before beginning the print.
- Nozzle was too close to the bed.
- The filament has striped against the drive gear.
- The extruder is clogged.

Not Sticking to the Bed



Common Solutions

- Build platform is not level.
- Nozzle starts too far away.
- First layer is printing too fast.
- Temperature or cooling settings.
- The build platform surface (tape, glues, and materials).
- When all else fails: Brims and Rafts.

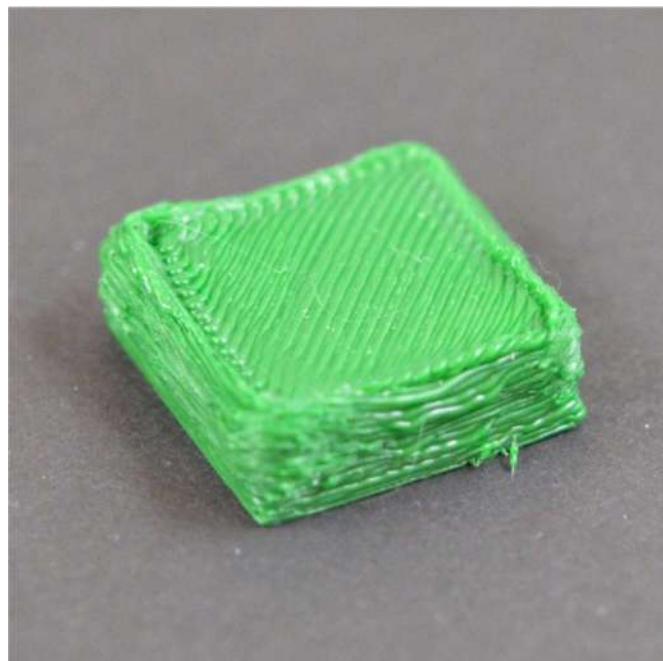
Under-Extrusion



Common Solutions

- Incorrect filament diameter (too big)
- Increase the extrusion multiplier

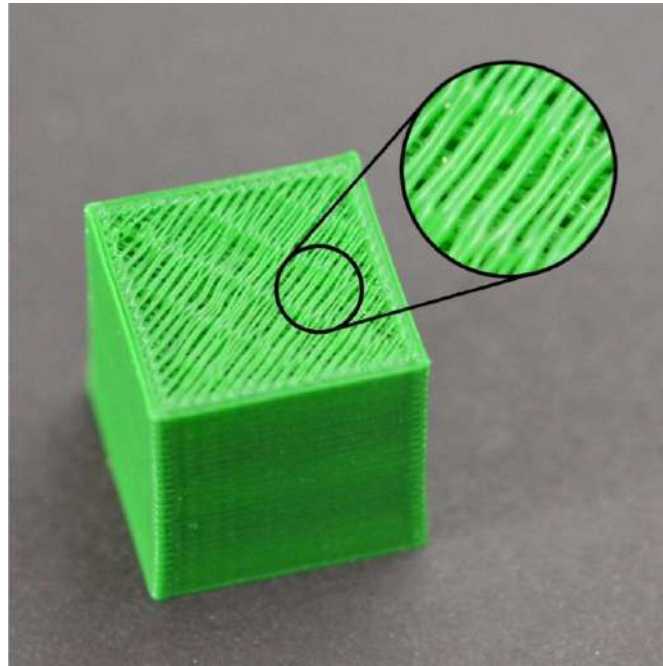
Over-Extrusion



Common Solutions

- Incorrect filament diameter (too small)
- Decrease the Extrusion multiplier

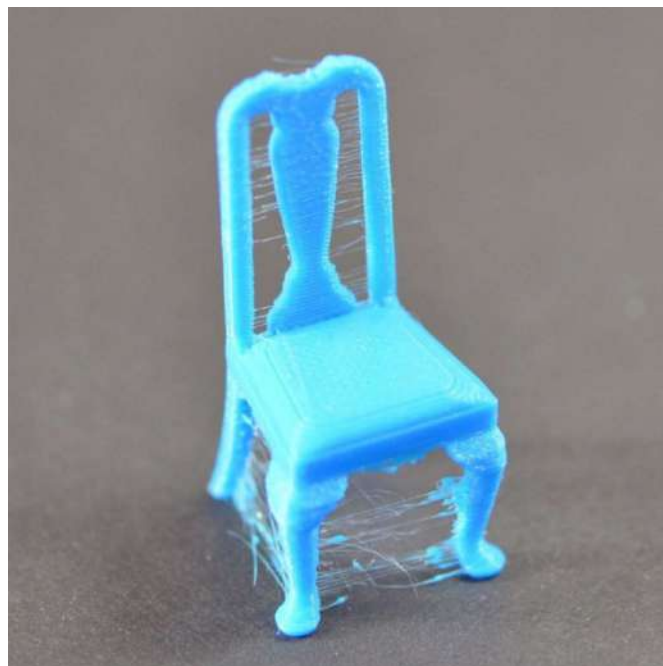
Gaps in Top Layers



Common Solutions

- Not enough top solid layers.
- Infill percentage is too low.
- Under-Extrusion

Stringing or Oozing

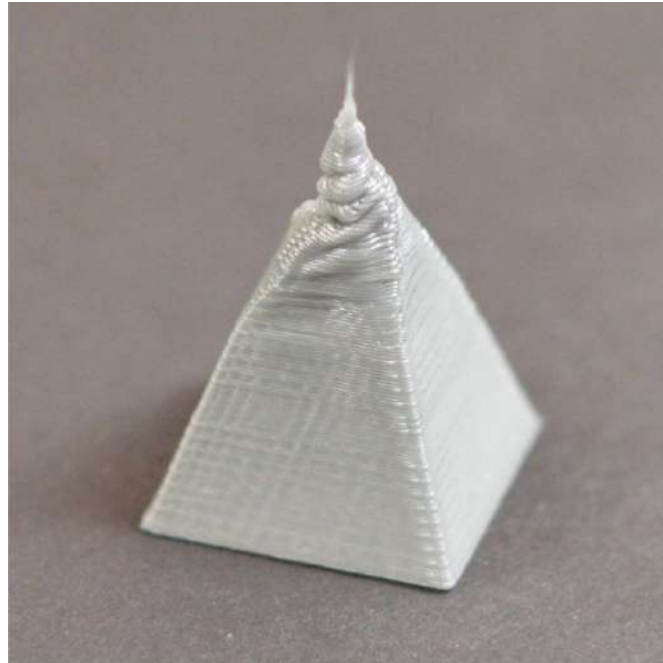


Common Solutions

- Retraction distance.

- Retraction speed.
- Temperature is too high.
- Long movements over open spaces.
- Movement Speed.

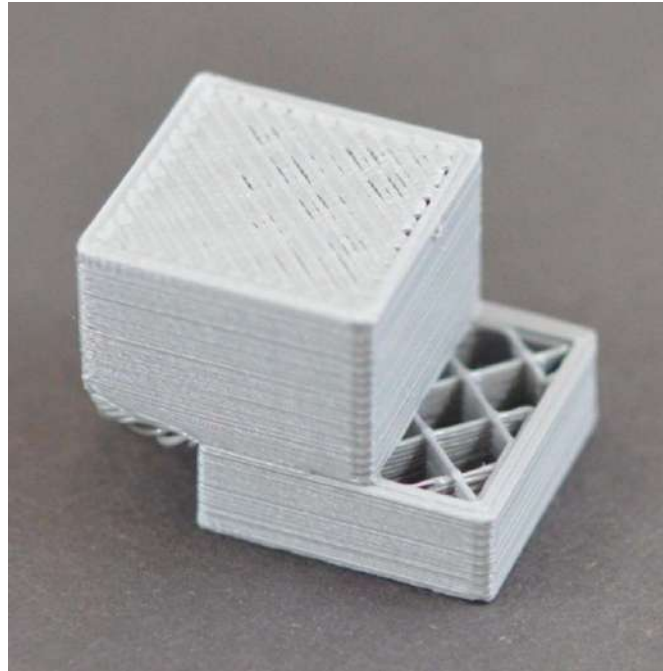
Overheating



Common Solutions

- insufficient cooling.
- Printing at too high of a temperature.
- Printing too fast.
- When all else fails: Try printing multiple parts at once.

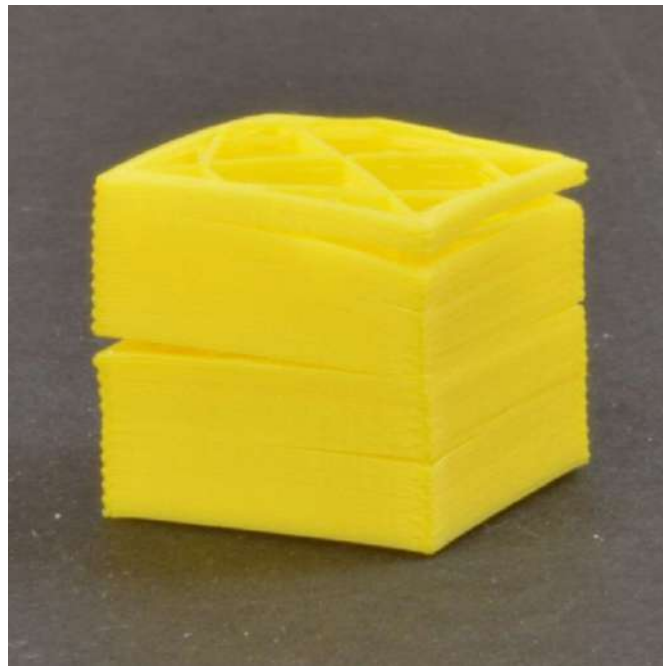
Layer Shifting



Common Solutions

- Toolhead is moving too fast.
- Mechanical or electrical issues.

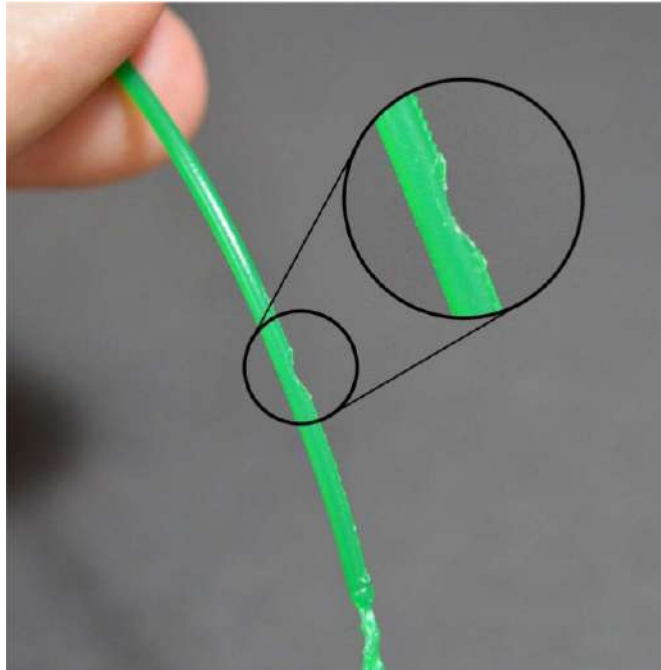
Layer Separation and Splitting



Common Solutions

- Layer height is too large.
- Print temperature is too low.

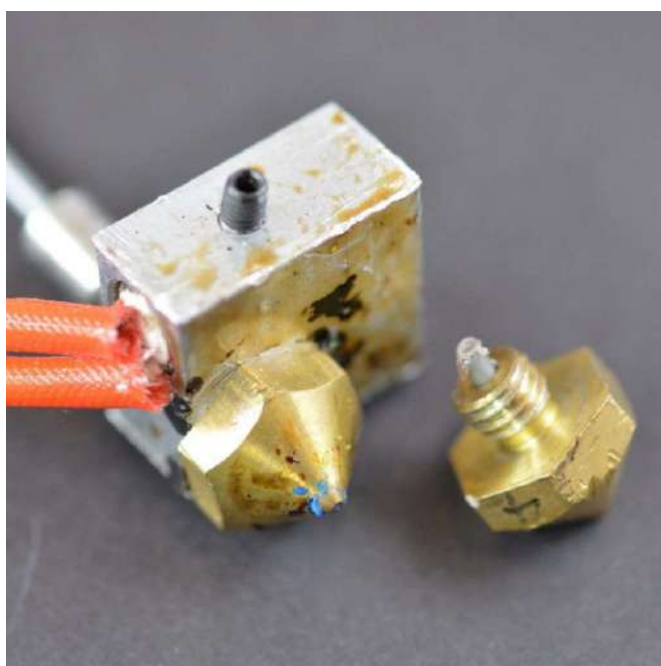
Grinding Filament



Common Solutions

- Aggressive retraction settings.
- Increase extruder temperature.
- Printing too fast.
- Check for a nozzle clog.

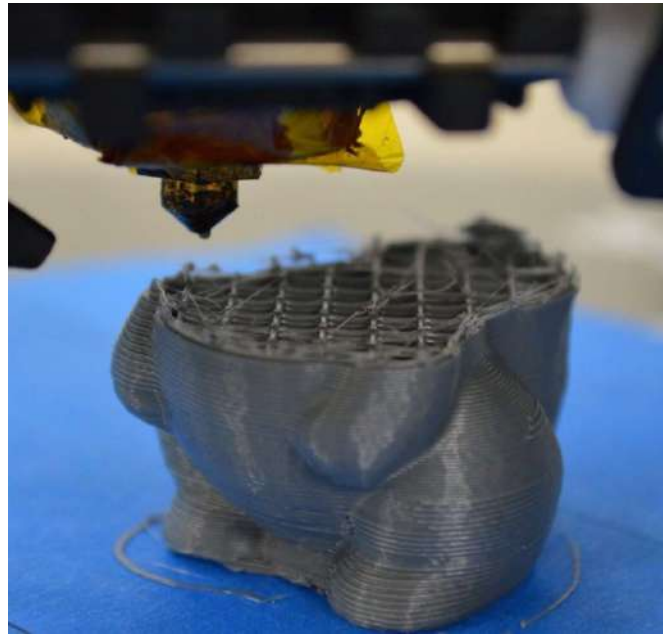
Clogged Extruder



Common Solutions

- Manually push the filament into the extruder.
- Reload the filament.
- Clean the nozzle.

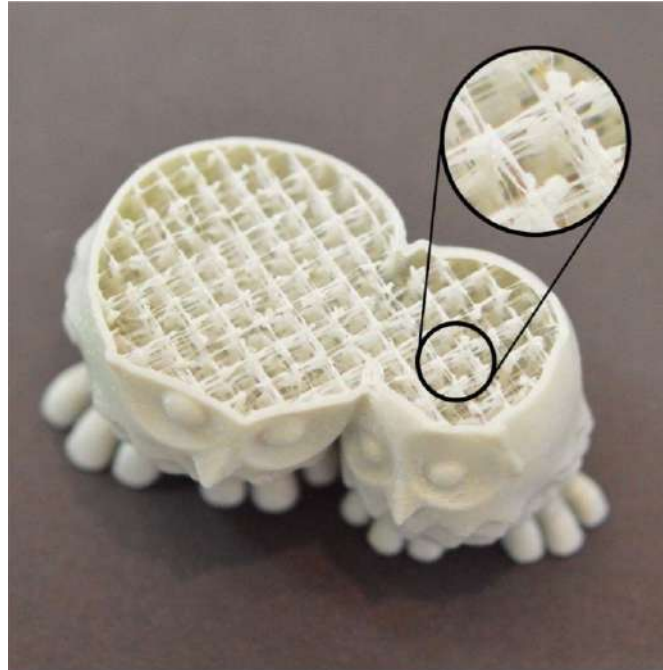
Stops Extruding Mid Print



Common Solutions

- Out of filament.
- The filament has stripped against the drive gear.
- The extruder is clogged.
- Overheated extruder motor driver.

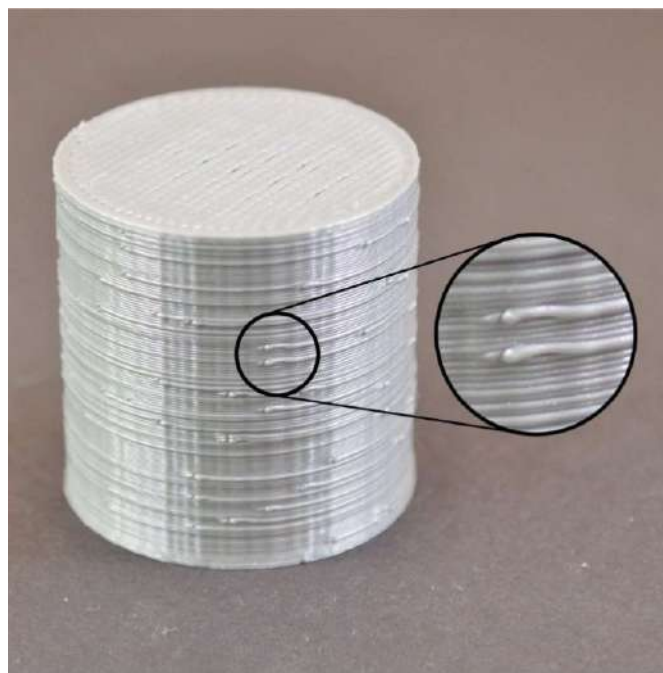
Weak Infill



Common Solutions

- Try alternate infill patterns.
- Lower the print speed.
- Increase the infill extrusion width.

Blobs and Zits

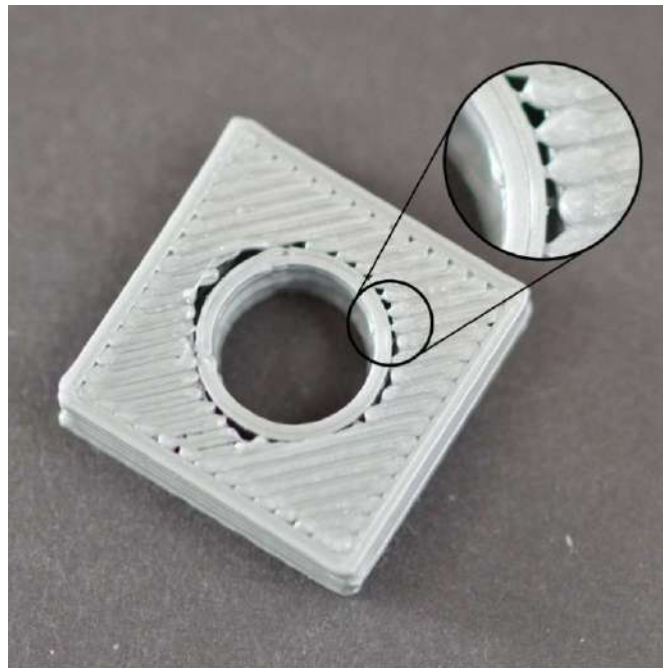


Common Solutions

- Retraction and coasting settings.

- Choose the location of your start points.

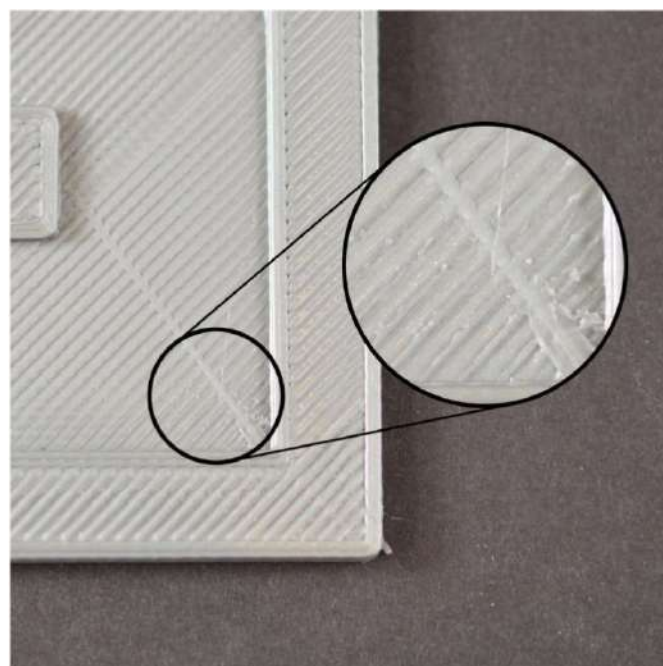
Gaps between infill and Outline



Common Solutions

- Not enough outline overlap.
- Printing too fast.

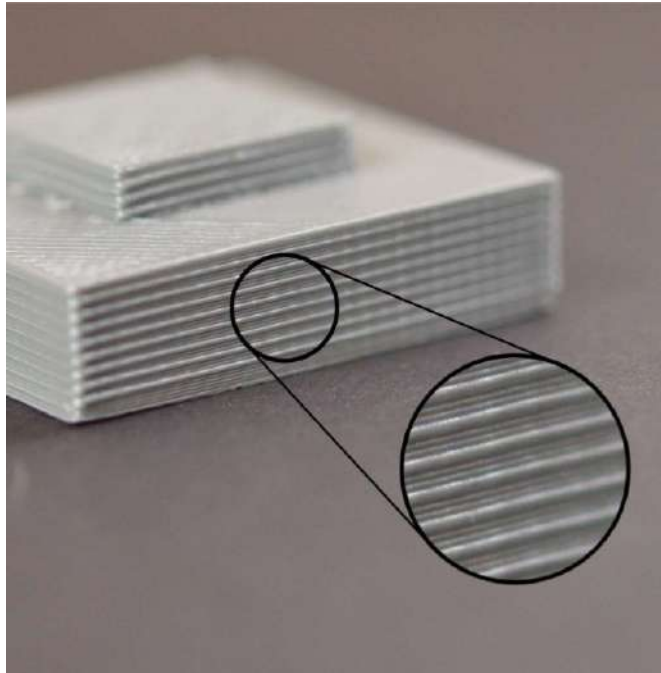
Scars on Top Surface



Common Solutions

- Extruding too much plastic.
- Vertical lift (Z-hop).

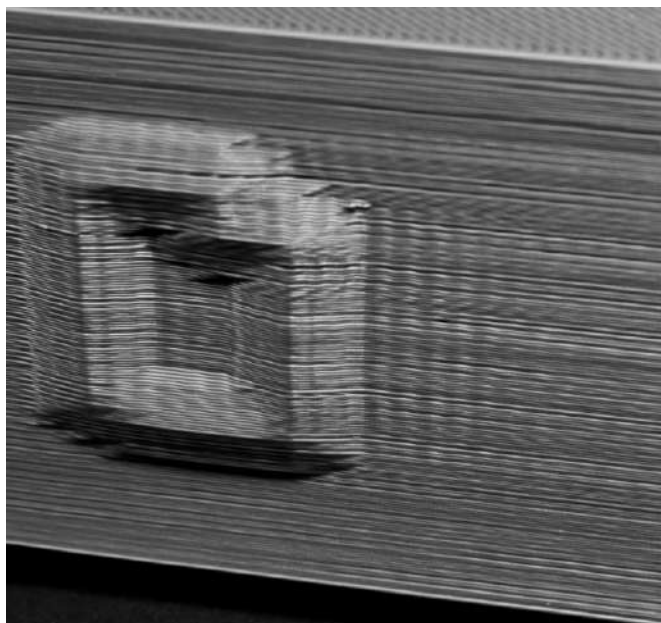
Lines on the side of the Print



Common Solutions

- Inconsistent extrusion.
- Temperature variation.
- Mechanical issues.

Vibrations and Ringing



Common Solutions

- Printing too fast.
- Mechanical issues.

Gaps in Thin Walls



Common Solutions

- Adjust the thin wall behavior.
- Change the extrusion width to fit better.

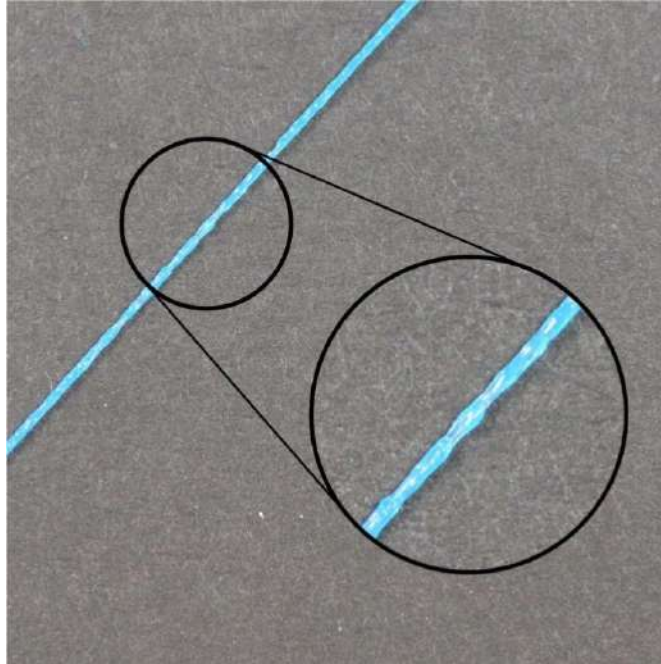
Small Features not printed



Common Solutions

- Enable single extrusion walls.
- Redesign the part to have thicker features.
- Install a nozzle with a smaller tip size.

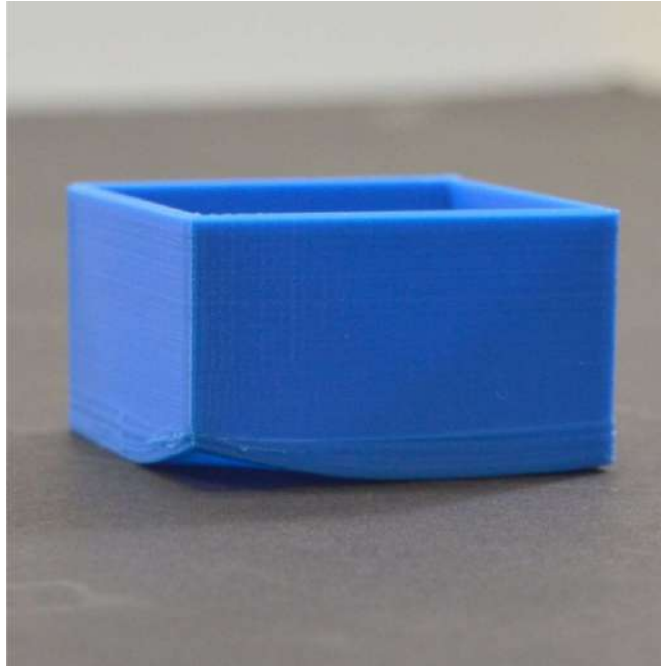
Inconsistent Extrusion



Common Solutions

- Filament is getting stuck or tangled.
- Clogged extruder.
- Very low layer height.

Warping



Common Solutions

- Use a heated bed.
- Disable Fan cooling.
- Use a Heated Enclosure.
- Brims and Rafts.

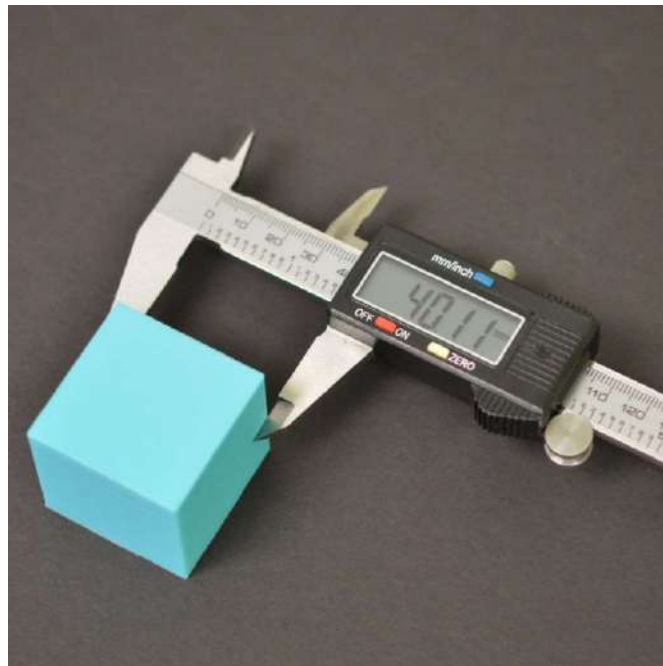
Poor surface above supports



Common Solutions

- Lower the layer height.
- Support infill percentage.

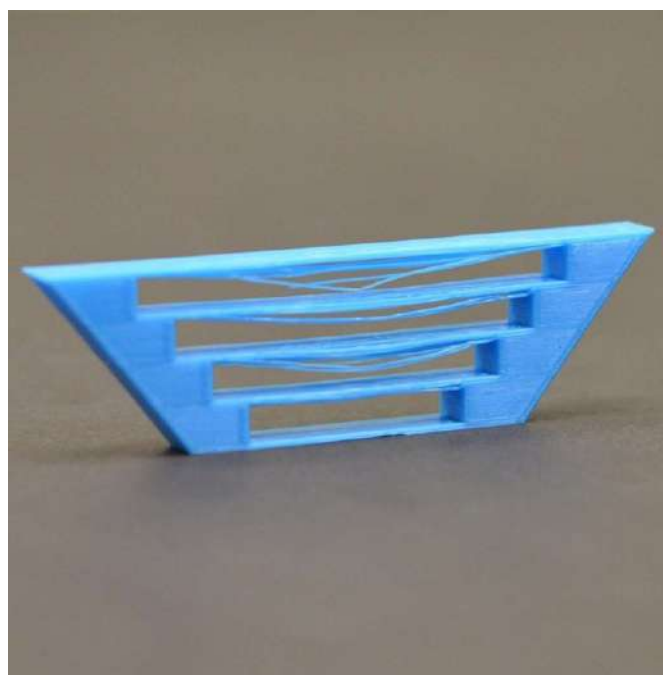
Dimensional accuracy



Common Solutions

- First layer impact.
- Under or over extrusion

Poor Bridging



Common Solutions

- Check the angle used for bridging infilling.
- Adjust settings for optimal performance.
- Use supports for longer bridges.

1.2 G-code

- FDM printer language
- Expresses each motion of the printing head with a single line
- G1, a linear motion to X, Y, Z position.
- Also control auxiliary elements.
- Makes up 95% of G-code files

```

10.1 G0 & G1: Move
10.2 G2 & G3: Controlled Arc Move
10.3 G4: Dwell
10.4 G6: Direct Stepper Move
10.5 G10: Set tool Offset and/or workplace coordinates and/or tool temperatures
10.6 G10: Retract
10.7 G11: Unretract
10.8 G12: Clean Tool
10.9 G17..19: Plane Selection (CNC specific)
10.10 G20: Set Units to Inches
10.11 G21: Set Units to Millimeters
10.12 G22: Firmware Retract
10.13 G23: Firmware Recover
10.14 G26: Mesh Validation Pattern
10.15 G28: Move to Origin (Home)
10.16 G29: Detailed Z-Probe
    10.16.1 G29 Auto Bed Leveling (Marlin - MK4duo)
    10.16.2 G29 Unified Bed Leveling (Marlin - MK4duo)
    10.16.3 G29 Manual Bed Leveling (Marlin - MK4duo)
    10.16.4 G29 Auto Bed Leveling (Repetier-Firmware)
    10.16.5 G29 Mesh Bed Compensation (RepRapFirmware)
10.17 G29.1: Set Z probe head offset
10.18 G29.2: Set Z probe head offset calculated from toolhead position

```

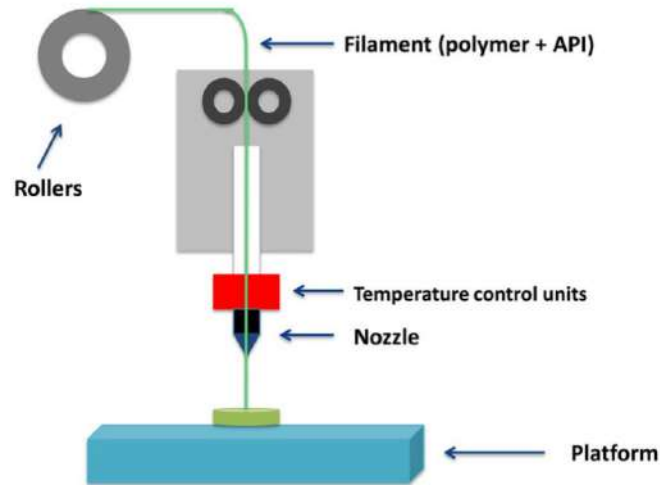
Example

G1 linear move:

G1 X0 Y0 F2400 ; move to the X=0 Y=0 position on the bed at a speed of 2400 mm/min

G1 Z10 F1200 ; move the Z-axis to Z=10mm at a slower speed of 1200 mm/min

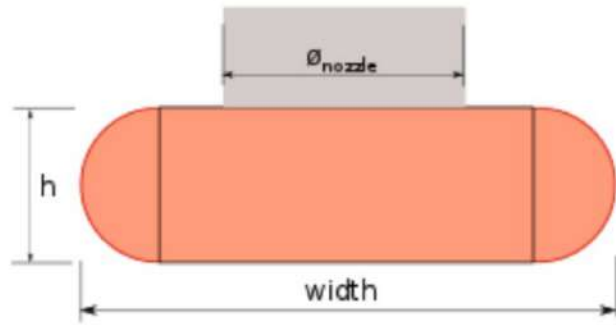
G1 X30 E10 F1800 ; push 10mm of filament into the nozzle while moving to the X=30 position a

E parameter

- The length of filament to feed into the extruder between the start and end point.
- Volume conservation,

$$E \cdot Af = A \cdot L$$

Where, Af is the cross sectional area of the filament, A is the cross sectional area of the strand and L is the print length



$$A = \text{rectangle} + \text{circle} = (w - h) \cdot h + \pi \cdot (h/2)^2$$

Dimensional Accuracy

To verify the quality of the assumptions used in the slicer a model was developed that provides information about the dimensional accuracy.

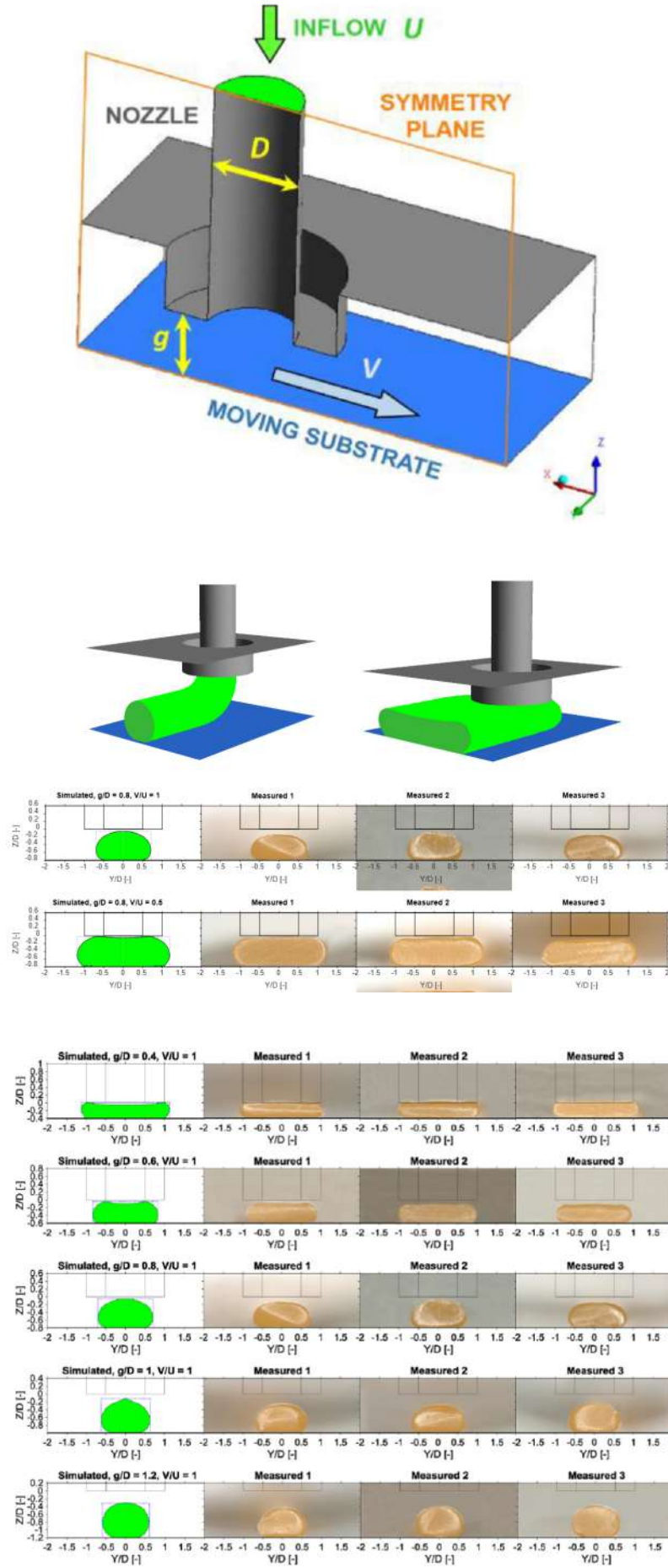


Figure 5: As can be seen from the bottom set of graphs, they follow the assumption made by the slicer, compared to the top set of graphs that do not.

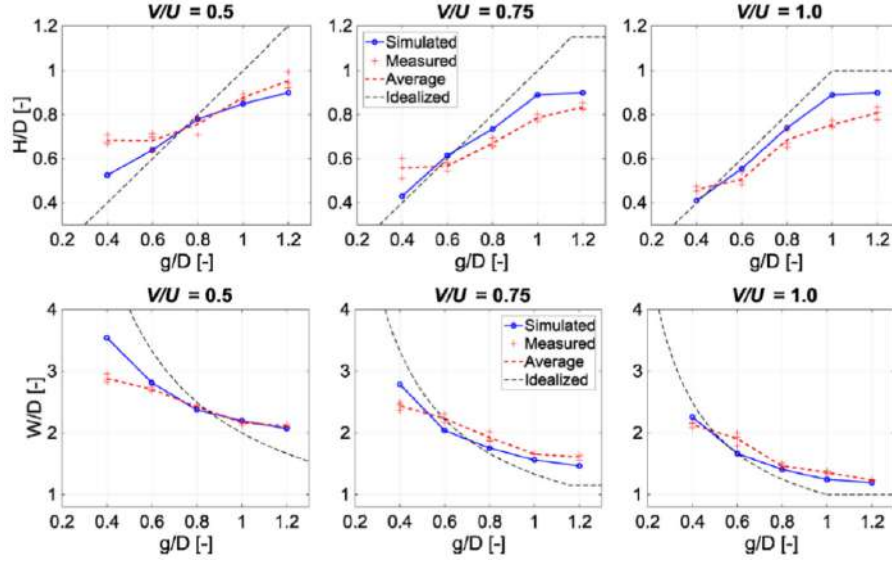
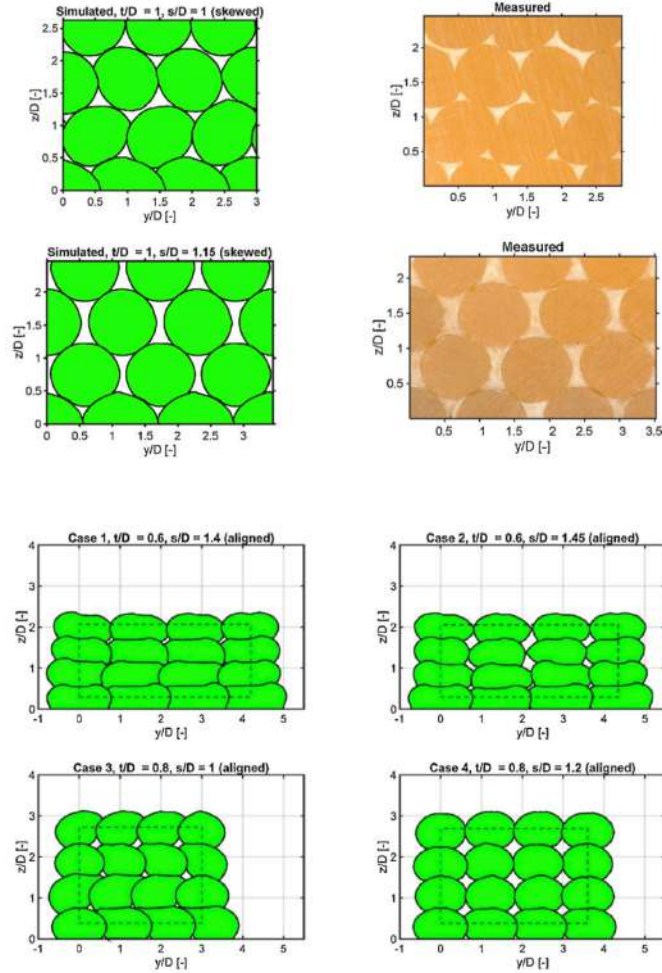


Figure 6: Various simulations and measured data, found to be very coherent, proving the model is very accurate



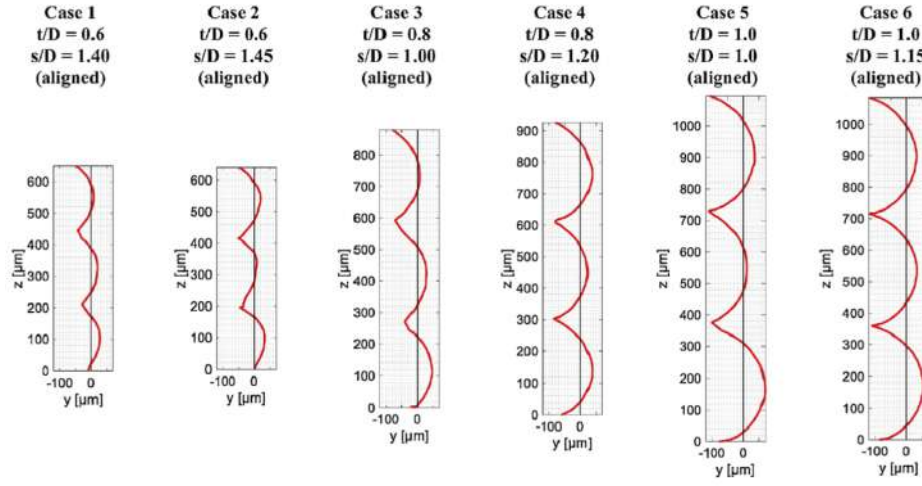


Figure 7: Even though in case 1 it can be seen that the initial layers printed well with no porosities, the following layers seem to have porosities and the following cases, continue showing more porosities as we go to higher layers

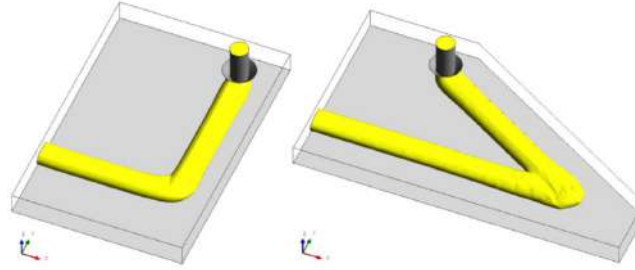


Figure 8: Effect of sharp corners in the XY -plane on dimensiona accuracy

1.3 5 and 6 degrees of freedom (DOF) printing

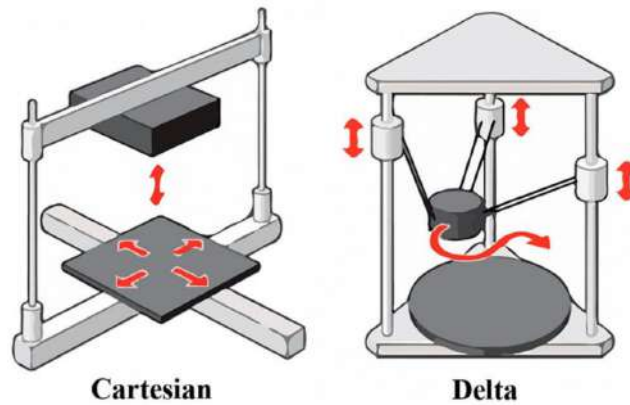


Figure 9: 3 DOF printer (Cartesian) and 5 DOF printer (Delta)

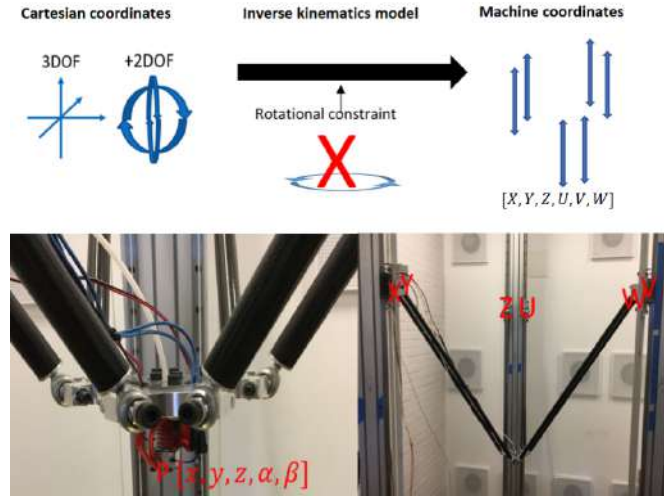
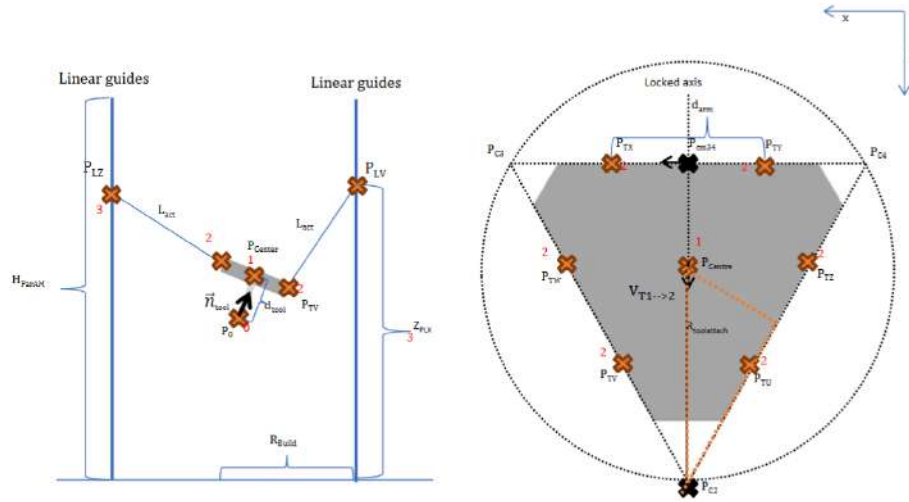


Figure 10: 5 DOF printing



Needed information for 5 DOF prints are position on the linear guide.

Pseudo code:

1. Position vector of the tip $P_0(x, y, z, \alpha, \beta)$
2. With the P_0 vector and the d_{tool} we find the P_{center}
3. With P_{center} , locked axis, orthogonality we can find the vector
4. From P_{center} vector, and distances we get the P_{c2} P_{cm34}
5. P_{cm34} , locked axis, and orthogonality we can find the vector
6. From P_{cm34} vector and the distances we can get P_{c3} , P_{c4} , P_{TX} and P_{TW}
7. P_{c2} , P_{c3} and P_{c4} we can find the vectors
8. From P_{c2} , vectors and distances we get P_{TU} , P_{TZ} , P_{TV} and P_{TW}
9. From the linear guides position we can get the line in space
10. Using P_{TV} and distances we can get the circle in space
11. From the circle and line intersection we finally get P_{LV}

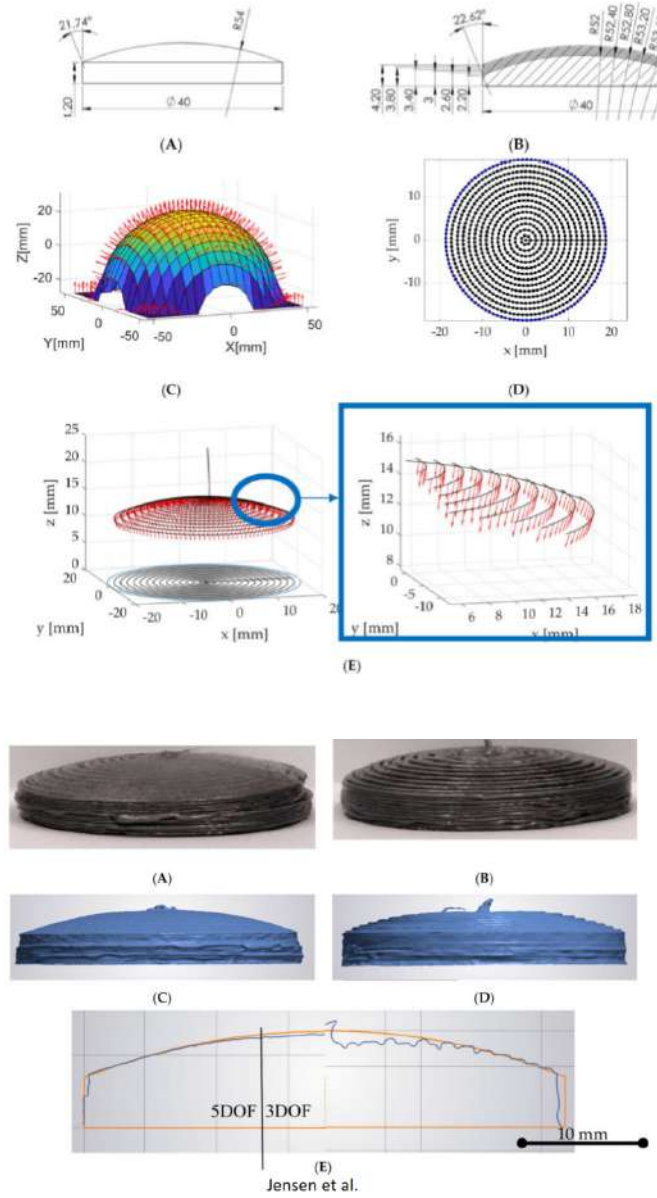


Figure 11: Using the 5 DOF printer and the 3 DOF we see how the 5 DOF shows much higher accuracy in overhangs compared to the 3 DOF

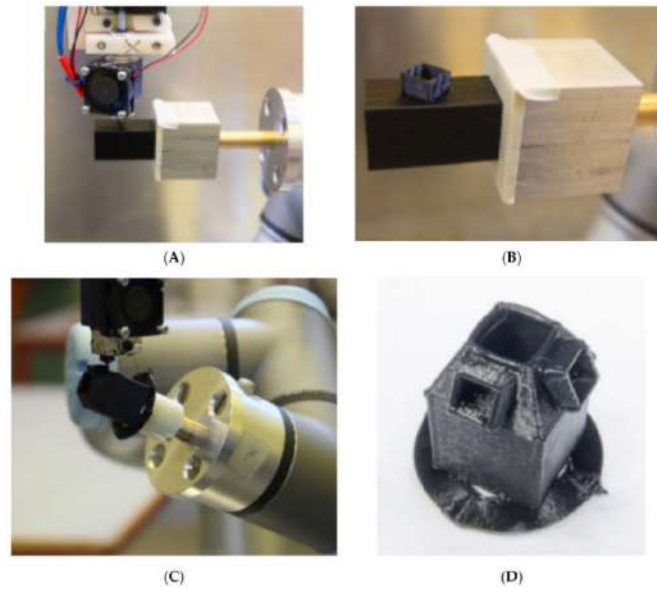


Figure 12: Example of 6 DOF printer

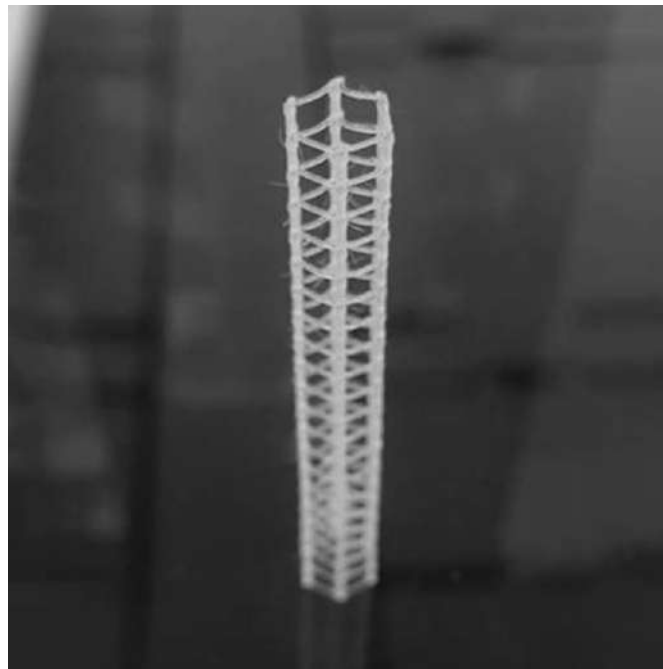


Figure 13: Example of non-planar 3D printing (Aerial printing)

2 Heat tranfer mechanisms

Thermal conduction Fourier's Law

$$\frac{\dot{q}}{A} = -k \frac{\partial T}{\partial x}, \quad \frac{\dot{q}}{A} \text{ Heat flux}$$

Variable	Symbol	Unit (SI)
Heat transfer	$\dot{q}$	W
Area	A	m ²
Thermal conductivity	k	W/(Km)
Temperature	T	K
length	x	m

Thermal convection Newton's Law of cooling

$$\frac{\dot{q}}{A} = -h(T_{\infty} - T_s)$$

Variable	Symbol	Unit (SI)
Heat transfer coefficient	h	W/(m ² K)
Temperature in fluid	T_{∞}	K
Temperature on surface	T_s	K

Irradiation Stefan Boltzmann's Law

$$\frac{\dot{q}}{A} = \sigma T^4$$

This will not be used in the FDM part.

Instationary heat conduction equation

$$T(x, y, z, t)$$

Deriving in 1D,

Heat balance,

$$\dot{Q}_{\text{inside}} = \dot{q}_{\text{in}} - \dot{q}_{\text{out}} + \dot{Q}_{\text{gen}}$$

Heat in the volume,

$$Q_{\text{inside}} = CT = Mc_p T = V\rho c_p T = A\Delta x\rho c_p T$$

Heat change per time,

$$\dot{Q}_{\text{inside}} = A\Delta x\rho c_p \frac{\partial T}{\partial t}$$

Variable	Symbol	Unit (SI)
Heat capacity	C	J/K
Volume	V	m ³
Specific heat	c_p	J/(kgK)
Density	ρ	kg ³ /m ³

Heat balance

$$A\Delta x\rho c_p \frac{\partial T}{\partial t} = \dot{q}_{\text{in}} - \dot{q}_{\text{out}} + \dot{Q}_{\text{gen}} \quad (1)$$

$$\rho c_p \frac{\partial T}{\partial t} = \frac{\frac{\dot{q}_{\text{in}}}{A} - \frac{\dot{q}_{\text{out}}}{A}}{\Delta x} + \dot{Q}_{\text{gen}}''', \quad \dot{Q}_{\text{gen}}''' = \frac{\dot{Q}_{\text{gen}}}{V} \quad (2)$$

$$(3)$$

$$\text{as } \Delta x \rightarrow 0 \quad (4)$$

$$(5)$$

$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(-\frac{\dot{q}}{A} \right) + \dot{Q}_{\text{gen}}''' \quad (6)$$

Stationary heat conduction

$$0 = \frac{\partial}{\partial x} \left(-\frac{\dot{q}}{A} \right) + \dot{Q}_{\text{gen}}'''$$

$$0 = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \dot{Q}_{\text{gen}}'''$$

Bio t

Bi describes the temperature difference in the solid relative to the temperature difference in the cooling medium

$$\frac{\dot{q}_{\text{cond.}}}{A} = -k \frac{\partial T}{\partial x} = -k \frac{\Delta T}{\Delta x} = -k \frac{T_s - T_i}{L} \quad (7)$$

$$\frac{\dot{q}_{\text{conv.}}}{A} = -h(T_{\infty} - T_s) \quad (8)$$

Heat balance

$$\frac{\dot{q}_{\text{cond.}}}{A} = \frac{\dot{q}_{\text{conv.}}}{A} \Rightarrow \frac{\Delta T_{\text{solid}}}{\Delta T_{\text{fluid}}} = \frac{T_s - T_i}{T_{\infty} - T_i} = \frac{hL}{k} \equiv Bi$$

If $Bi < 0.1$ then one can use lumped analysis.

Characteristic length

$$L = \frac{V}{A_s} = \frac{\text{Volume}}{\text{Cooling area}}$$

for a plate,

$$L = \frac{dA}{2A} = \frac{d}{2}$$

Lumped analysis

- Use it if $Bi < 0.1$
- Gives you a temperature in the solid as a function of time
- It assumes that the temperature is constant in the solid for a given time

Change of heat per time

$$V \rho c_p \frac{dT}{dt}$$

Heat balance,

$$V \rho c_p \frac{dT}{dt} - h A_s (T_\infty - T_s) = 0$$

$$V \rho c_p \frac{dT}{dt} = -h A_s (T_s - T_\infty)$$

Defining $T_s = T(t)$:

$$V \rho c_p \frac{dT}{dt} = -h A_s (T - T_\infty)$$

$$\int \frac{dT}{T - T_\infty} = \int -\frac{h A_s}{V \rho c_p} dt$$

$$\ln |T - T_\infty| = -\frac{h A_s}{V \rho c_p} t + C$$

From $T > T_\infty$:

$$T - T_\infty = \exp\left(-\frac{h A_s}{V \rho c_p} t + C\right)$$

$$T - T_\infty = \exp(C) \exp\left(-\frac{h A_s}{V \rho c_p} t\right)$$

$$T - T_\infty = C^* \exp\left(-\frac{h A_s}{V \rho c_p} t\right)$$

Using $T(t = 0) = T_i$:

$$T_i - T_\infty = C^* \exp(0)$$

$$C^* = T_i - T_\infty$$

Inserting C^* into the equation:

$$T(t) - T_\infty = (T_i - T_\infty) \exp\left(-\frac{h A_s}{V \rho c_p} t\right)$$

Printing speed

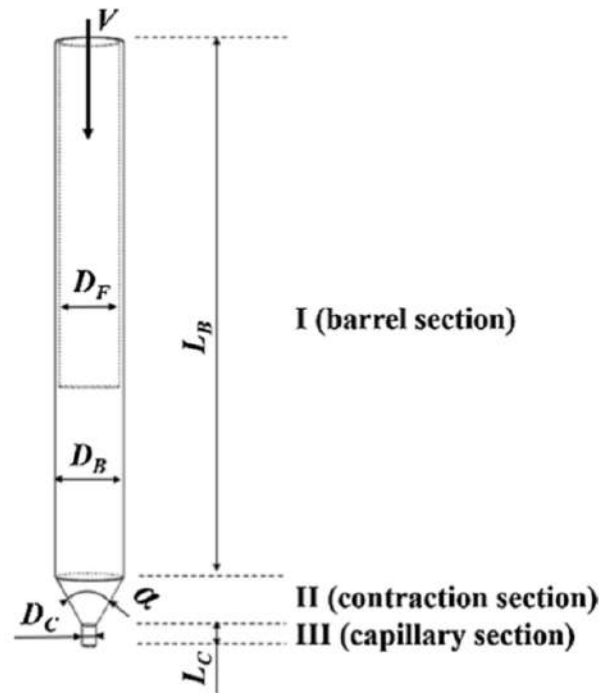


Figure 14: Hot-end channel

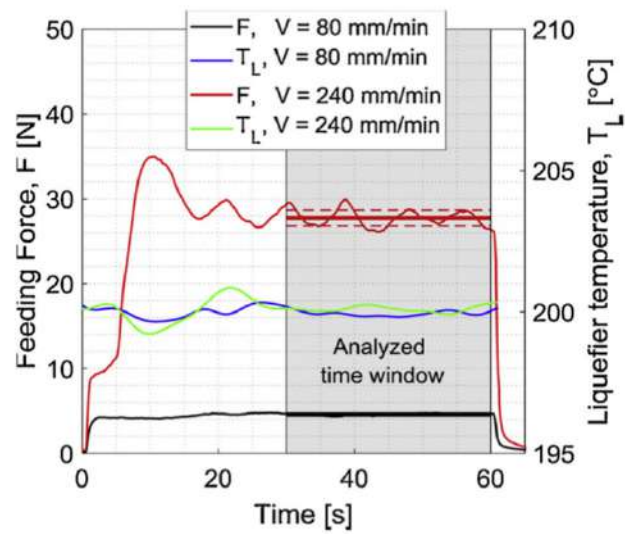


Figure 15: Filament force analysis

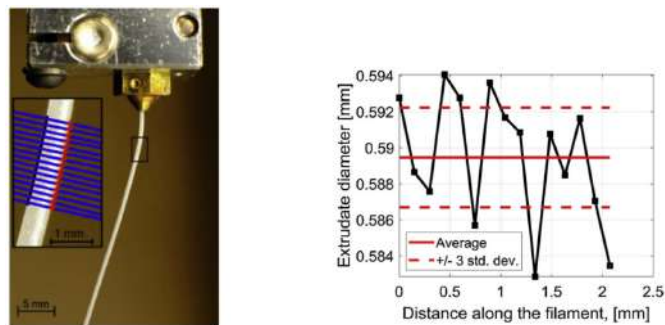


Figure 16: Extrudate swell (When the filaments diameter expands after passing through the tool head)

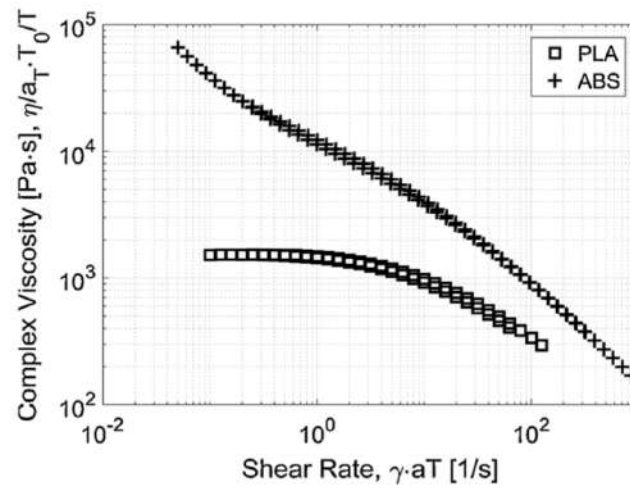


Figure 17: Rheology (study of deformation and flow of matter)

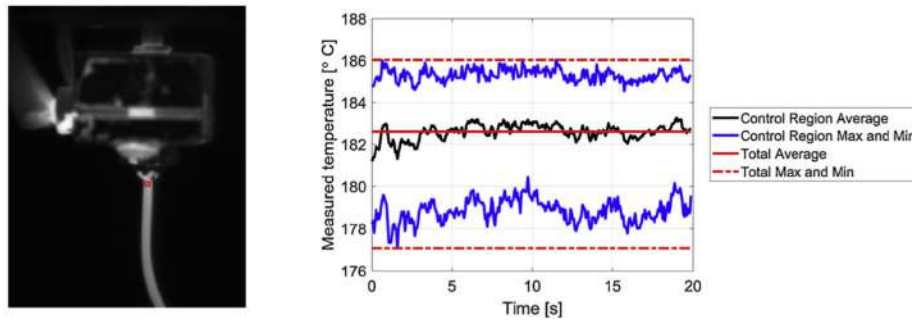


Figure 18: Temperature measurements of the filament measured as soon as it exists the toolhead

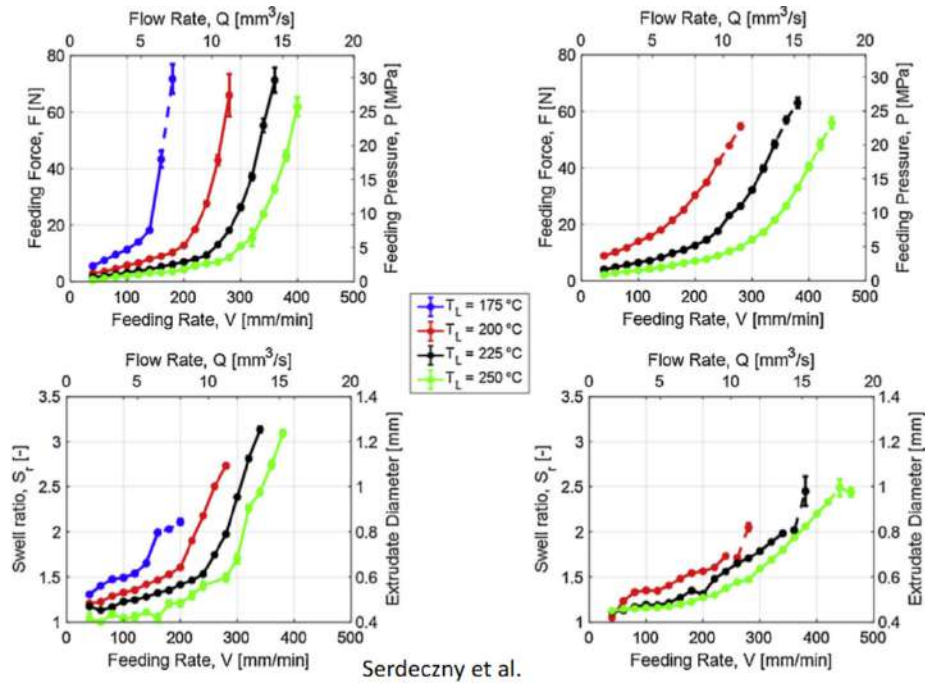


Figure 19: Feeding force and swell



Figure 20: Examples of stable and unstable extrudates

Instability	Main Causes	Main Fixes
Sharkskin	High extrusion speed Small tool head / die diameter Low melt strength Low temperature (high viscosity) Strong wall adhesion (no slip)	Reduce extrusion speed Increase die diameter Use higher melt strength polymer Increase temperature Add processing aids / induce wall slip
Melt Fracture	Very high extrusion speed Small tool head / short die High molecular weight High elastic stresses (high Wi) Sharp die exit	Lower extrusion speed Use longer / smoother die Reduce molecular weight Increase temperature Modify die exit geometry

Table 1: Main causes and mitigation strategies for sharkskin and melt fracture instabilities in polymer extrusion

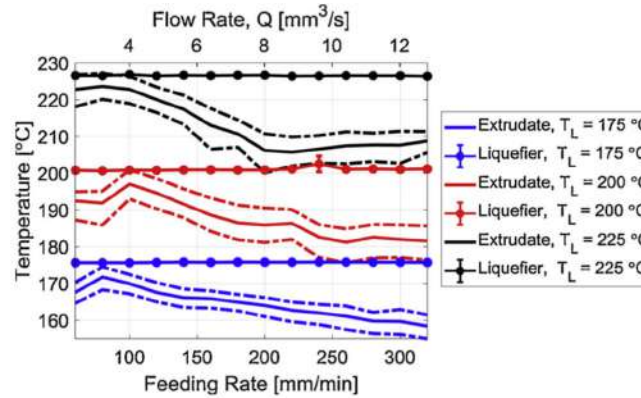


Figure 21: Extrudate temperature analysis

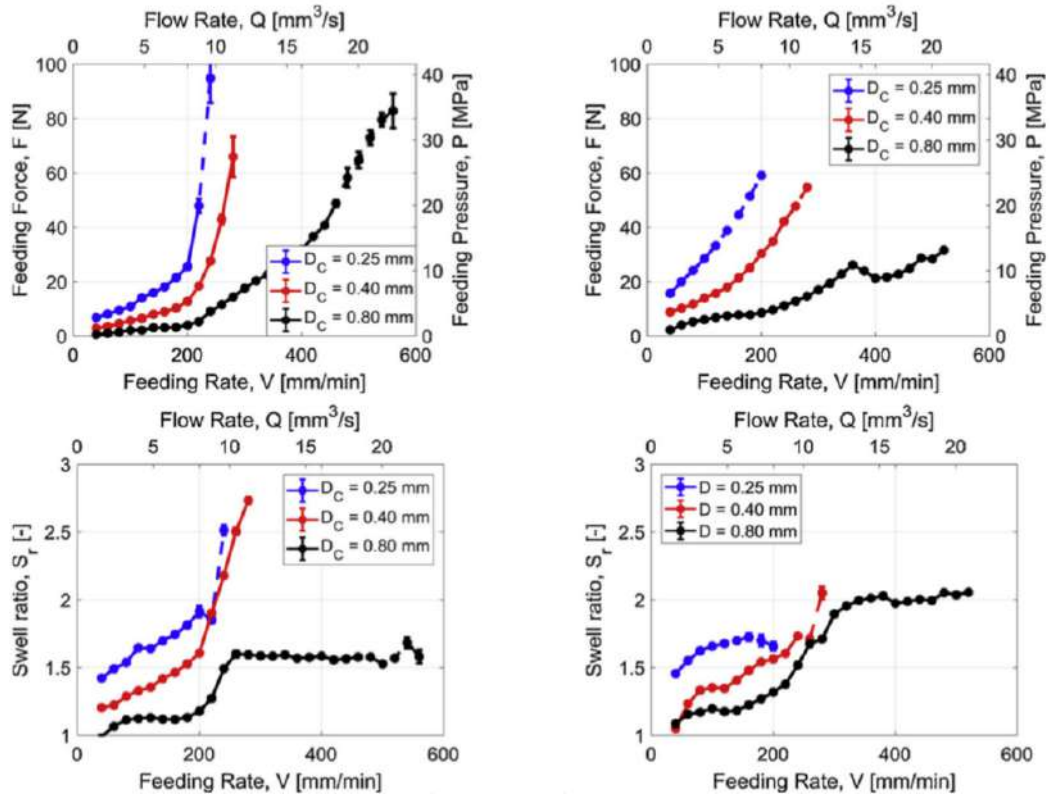


Figure 22: Different nozzle diameter analysis

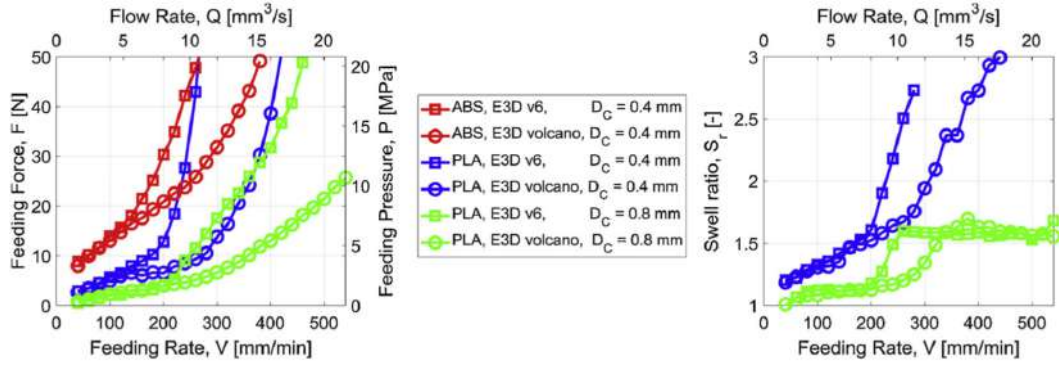


Figure 23: Liquefier length analysis (tube in which the filament is heated before exiting the toolhead)

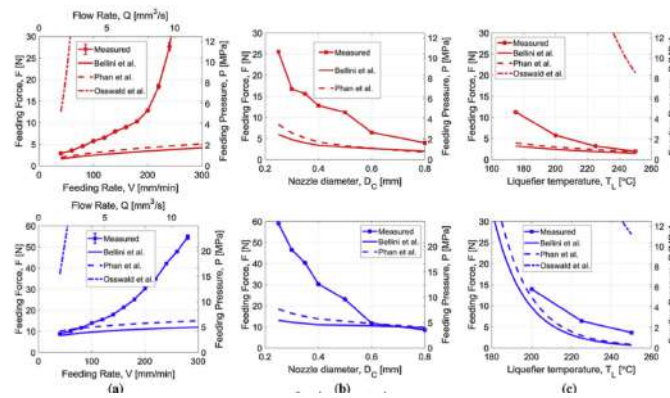


Figure 24: Models from literature and measured analysis

New model

$$\dot{Q}_H = h\Delta T\pi D_B L_B = \dot{m}(\lambda + C_m(T_{out} - T_{in}))$$

$$\Delta T = \frac{(T_L - T_{in}) - T_L - T_{out}}{\ln\left(\frac{T_L - T_{in}}{T_L - T_{out}}\right)}$$

$$\dot{m} = \rho_m = V_{\max} A_F$$

$$V_{\max} = \frac{h\Delta T\pi D_B L_B}{(\lambda + C_m(T_{out} - T_{in}))\rho_m A_F}$$

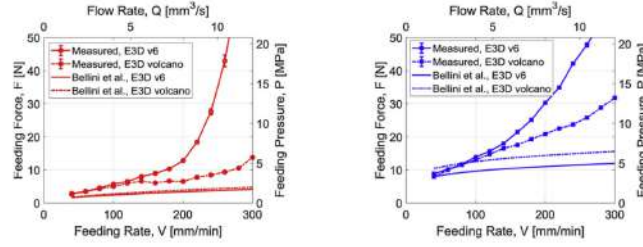
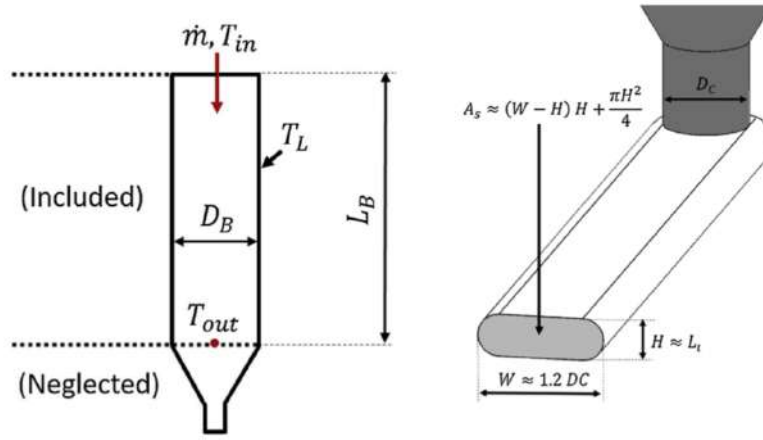


Figure 25: Models from literature

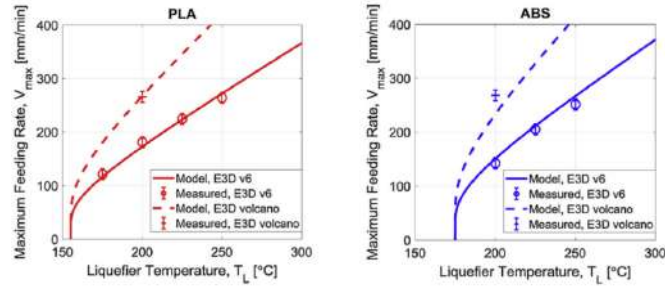


Figure 26: New model

3 Extrusion based additive manufacturing with other materials

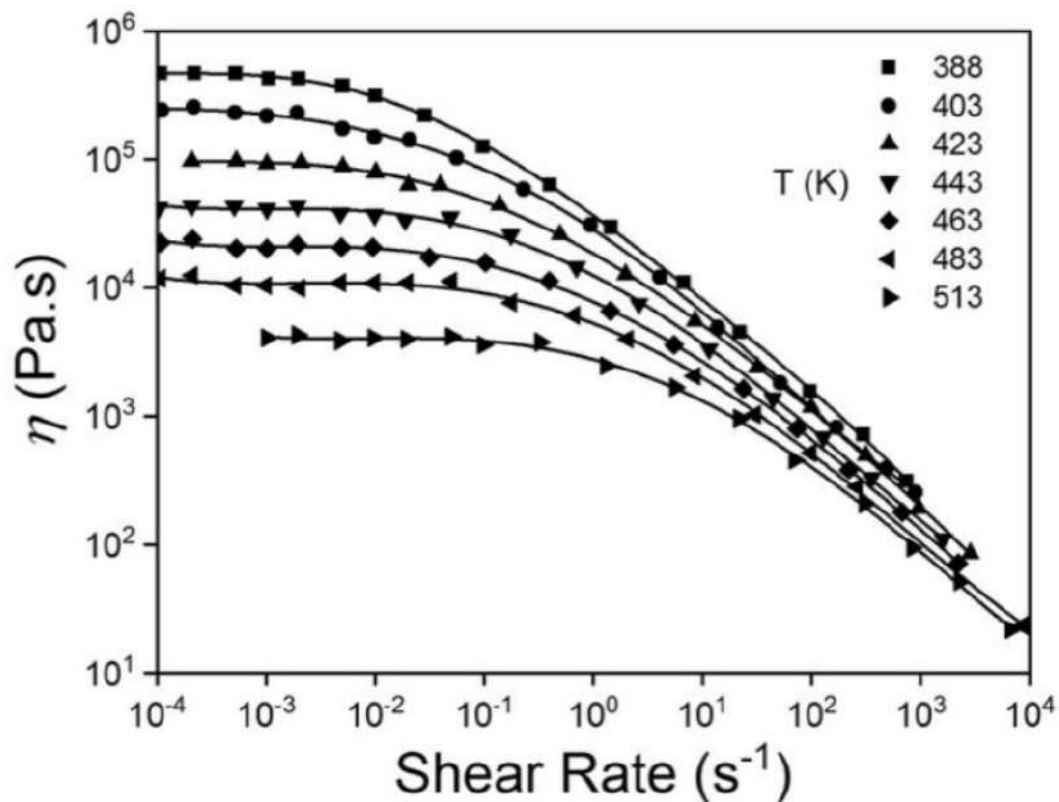


Figure 27: Rheological behavior of thermoplastic polymers

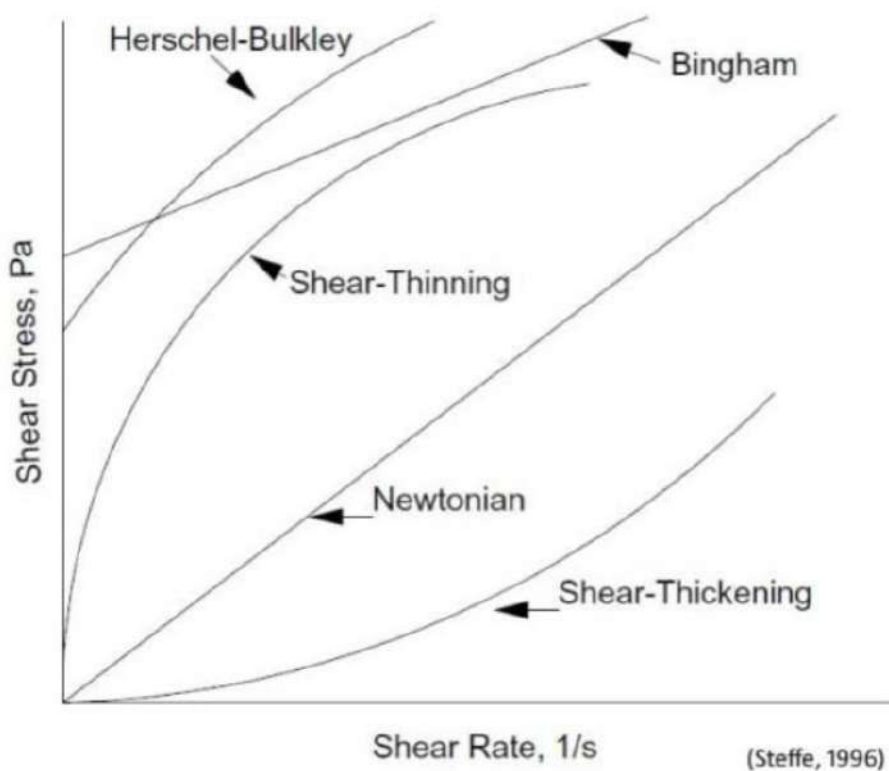


Figure 28: Different types of fluids

Additive manufacturing of composites

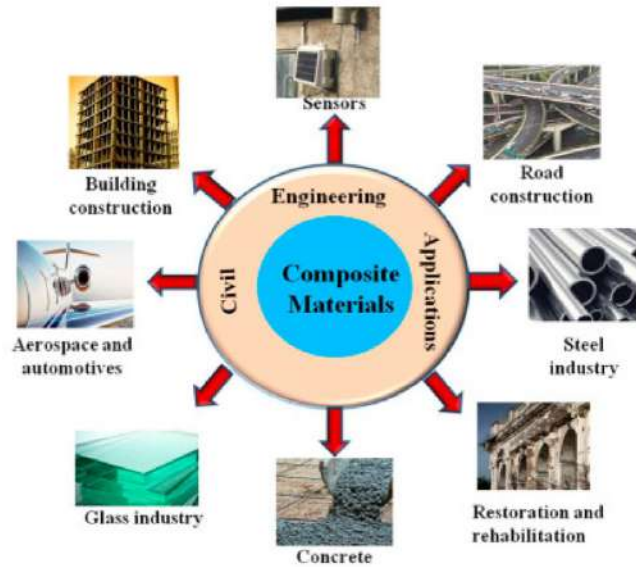


Figure 29: Composite applications

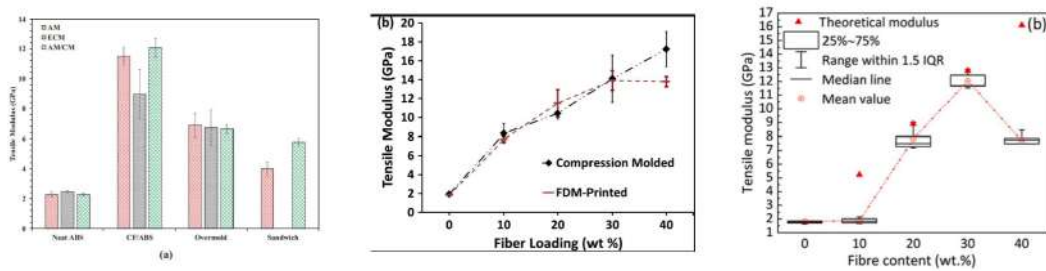


Figure 30: Effects of composites with fiber load



Figure 31: Example of produced concrete columns with formwork

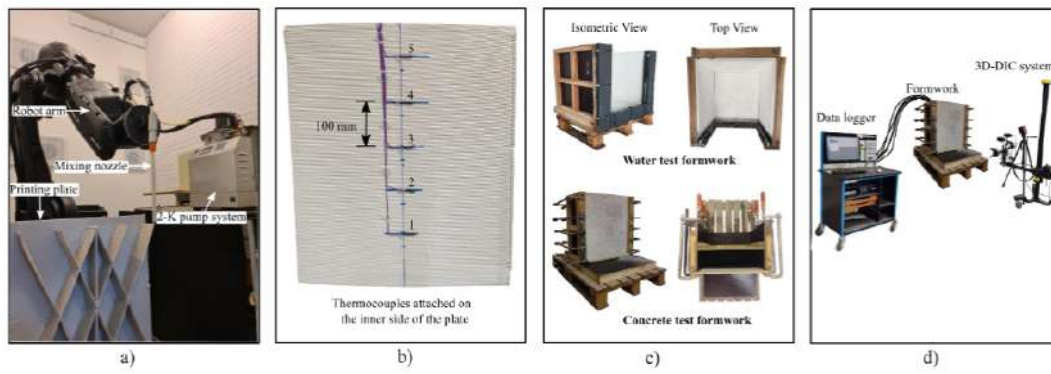
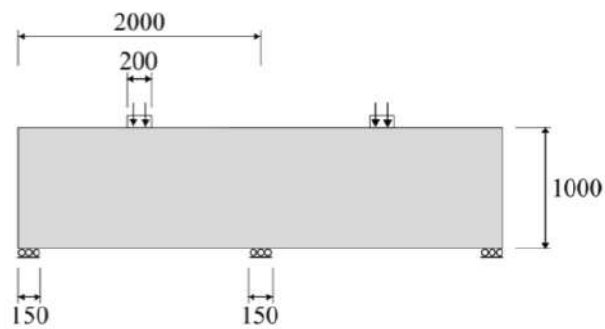


Figure 32: Printing and testing ribs-reinforced 3D printing plate formwork



Optimized design used variable thickness only, reducing concrete volume by ~66% vs. uniform beam.

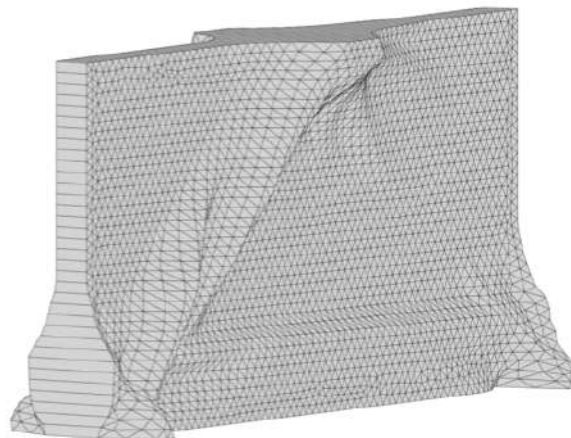


Figure 33: Complex-shaped beam, an optimized design reduces concrete volume drastically without losing weight bearing properties

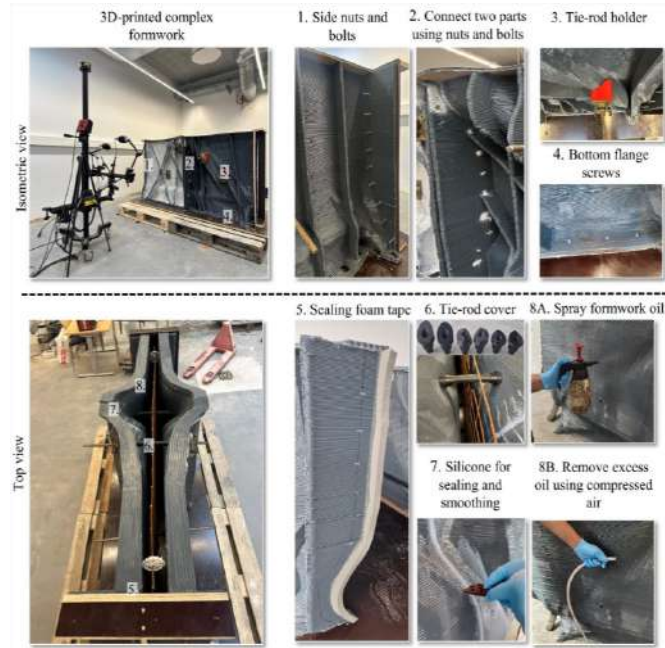


Figure 34: Experimental setup used during the printing of complex-shaped beam

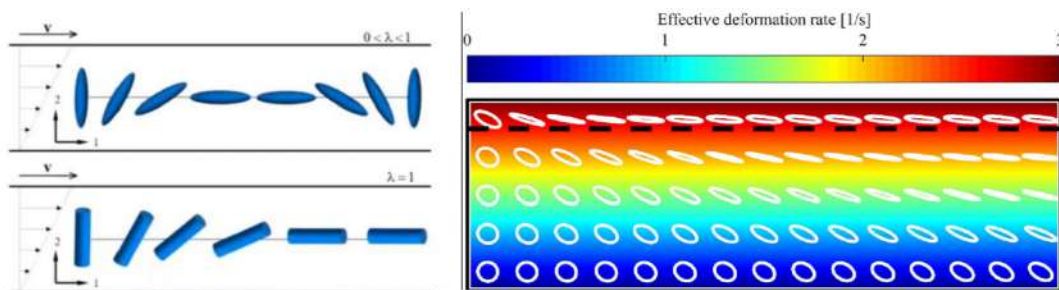
Composite formwork had 4 main steps,

1. 3D printing of the complex-shaped beam mold parts
2. Assembly of the mold
3. Casting the beam in the mold
4. Demolding of the beam

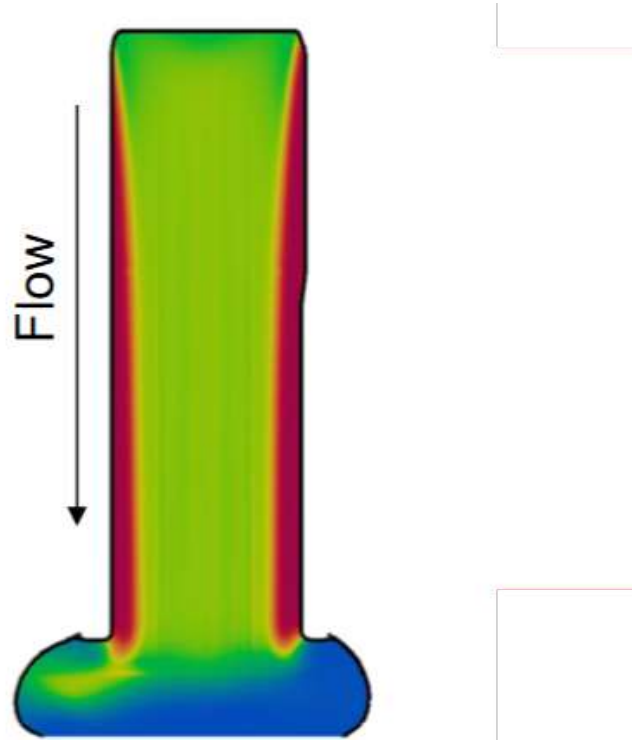
Particles in a simple shear flow

Particles are very common additions in composite manufacturing. However, one of the main factors in their effect in optimizing is their direction, and when used in injection molding it is impossible to accurately have the particles follow one direction.

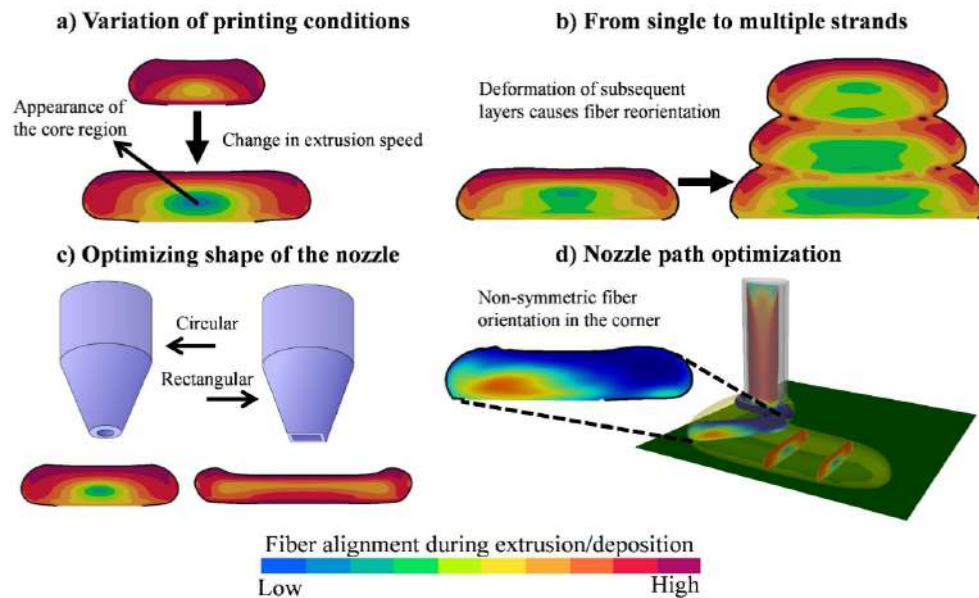
Using 3D printing the direction of these particles are much easier to control, although no where perfect.



As can be seen above it is still not easy to accurately align particles throughout the whole strand, however it is clear that more shear flow leads to better aligned particles.



As can be shown above since Polymers and other typical materials in 3D printing often show Non-Newtonian behavior. Bingham plastic is a common model to describe its behavior. In the central zone of the strand, there is little to none velocity gradient which can plug flow and lead to no re-orientation.



Orientation of the fiber inside the strands

The 3D simulations confirmed the existence of core/skin areas where low/high alignment can be found in the printing direction.

The existence of those regions is partially due to inherited microstructure from the nozzle, but there is additional re-orientation once the strands are deposited.

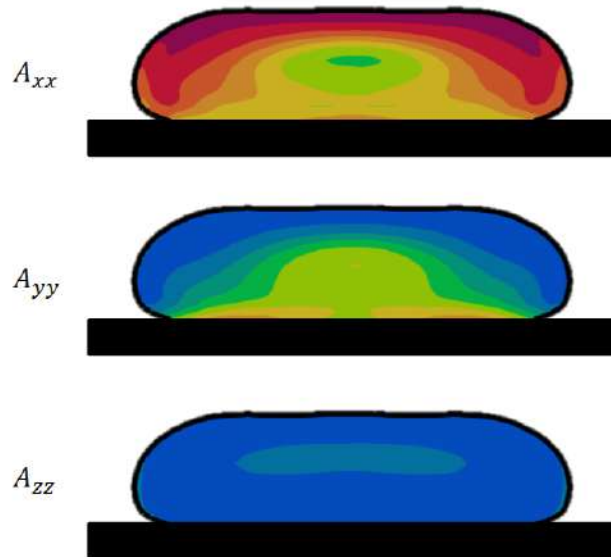


Figure 35: Orientation of fibers inside strands in the xx , yy , zz orientation

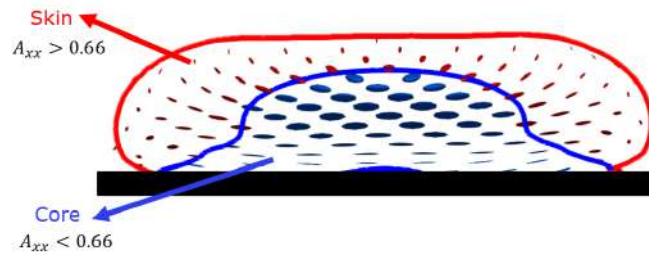


Figure 36: Difference in orientation in the Skin and Core of the strand



Opposed to engineering thought the fork in trees is supported using circular fiber alignment opposed to "U" shaped alignment.

Engineers at DTU were able to make the tool head of a printer rotate in the xy plane such that they would be able to control and vary the fiber alignment in the strand through out the layer-by-layer printing. Leaving the possibility for us to replicate the biological structure of the fork in trees.

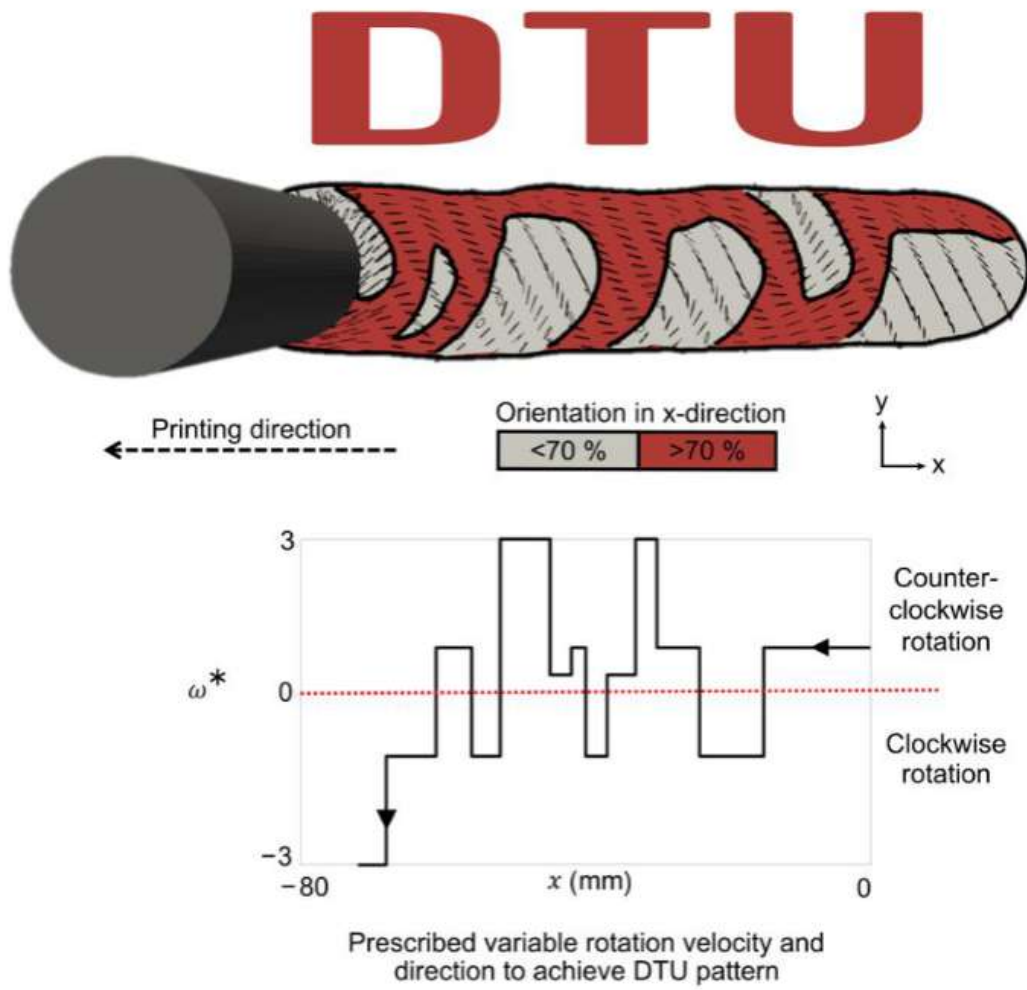


Figure 37: Digital design of the material

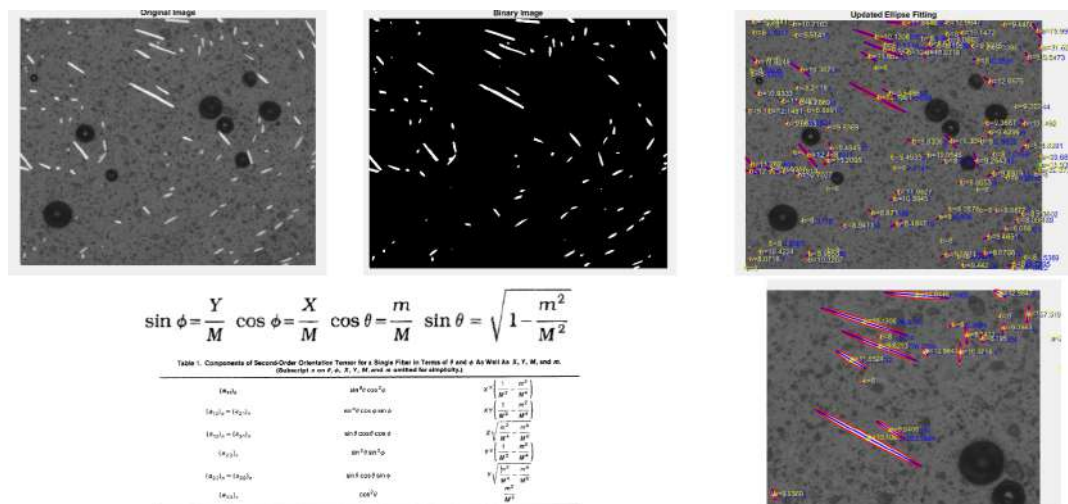


Figure 38: Taking the cross-sectional view of the strand and using computational program to fit ellipses on the strand they were able to calculate their angle of alignment

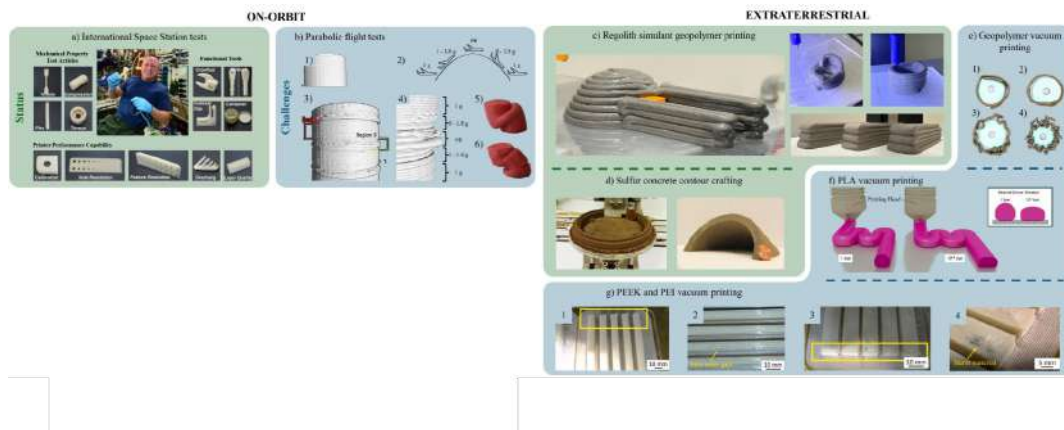
3D Concrete printing

8% of all global emissions of CO₂ emissions stem from the use of concrete and 80% of those emissions stem from cement manufacturing.

The Concrete conundrum is based off the need to increase concrete consumption to fulfill societal housing demands and the need to decrease CO₂ emissions from concrete consumption.

The idea of Large-area additive manufacturing has existed for a long time but the issue is still finding an optimal material that can be used like concrete with a chemical formula that, doesn't emit as much CO₂, has optimal characteristics for 3D printing and is not so expensive that the possibility of reducing amount of material used while keeping structural integrity is pricier than normal concrete.

In-space manufacturing and 3D printing of habitats using regolith



NASA plans to establish permanent residence on the moon by 2030 (likely too ambitious) and as such the following characteristics must be taken into account,

- Need to provide grounds for laboratories with almost total vacuum, low gravity, fundamental sciences and further extraterrestrial exploration.
- Construction issues, such as meteorites, low pressure (vacuum), galactic cosmic rays, large temperature variation and large costs

In the moon environment, regolith is very abundant and would therefore reduce cost. The use of 3D printing requires little human presence.

Therefore the suggested construction materials are lunar concrete, polymer concrete, sulfur concrete, geopolymer.

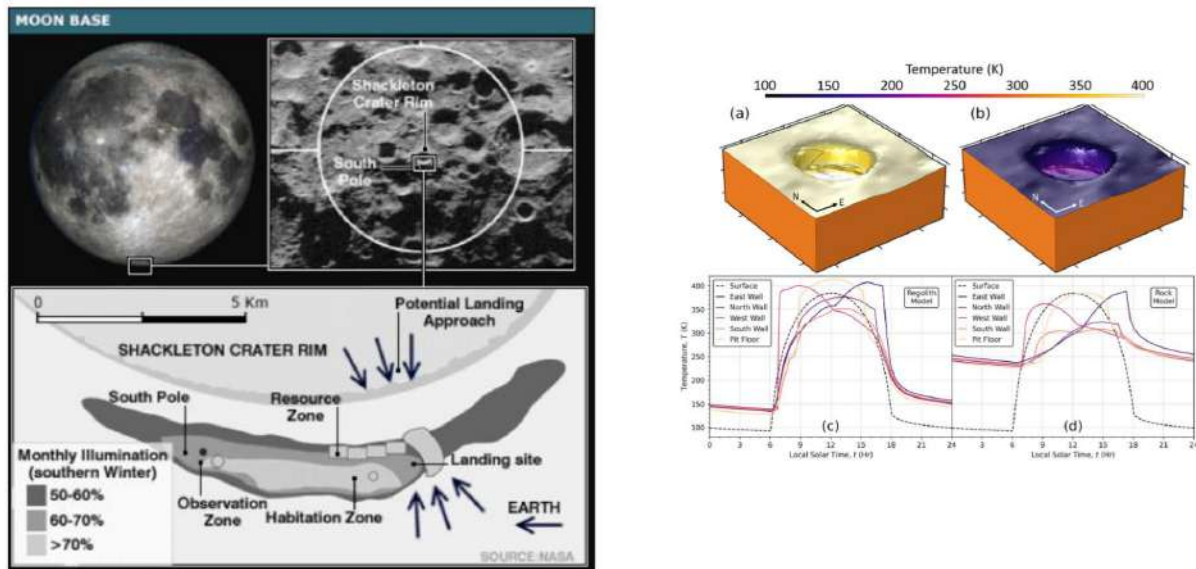


Figure 39: The most prized location on the moon is its southern pole where there less temperature variation

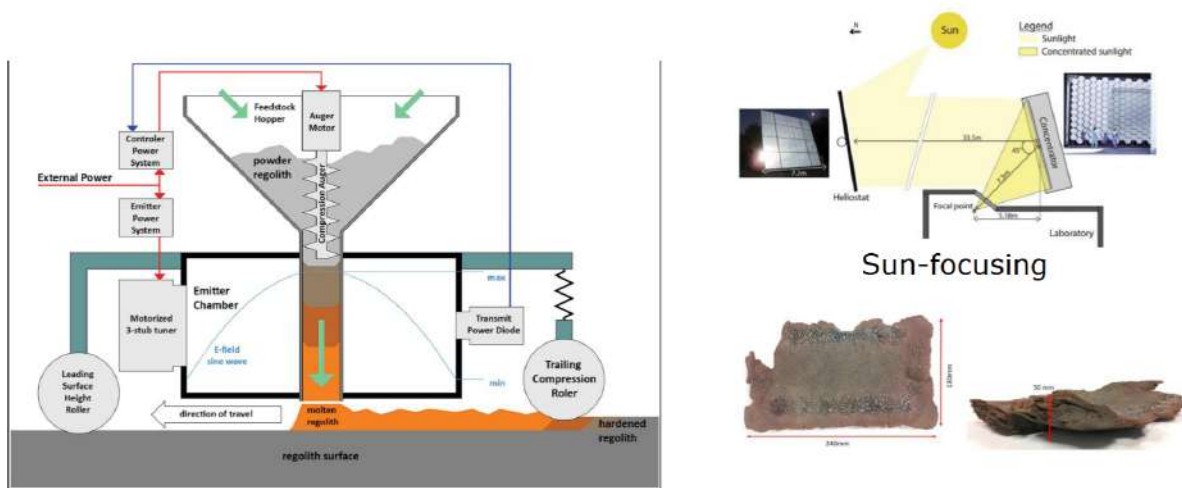


Figure 40: Melting of regolith in the lunar surface has been thought to be possible with the use of solar rays

Sulphur concrete

One of the first solutions proposed by professor Khoshnevis,

Pros:

- Sulphur present on the moon, can be extracted relatively simple«y
- Good binder, sulphur-concrete is commercially available on Earth
- Strong enough to withstand lunar gravity

Cons:

- Low melting temperature, close to daily lunar temperature
- Hard processing, very tight working window in terms of temperature
- Possibly toxic, especially once it starts sublimating/evaporating



- Sublimate quickly at high temperature, slowly at medium/low temperatures
- Shows poor behavior during temperature cycling

Lunar regolith simulant and activation solution (geopolymer)

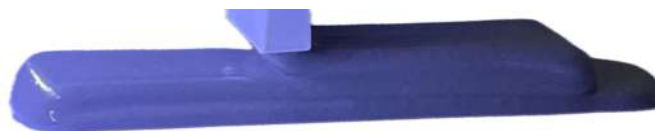
- Only a very limited amount of lunar regolith was brought back
- Lunar regolith simulants: chemical composition, particle size, different soils on the moon
- Not available lunar regolith on earth JSC-1, EAC-1 and CAS-1. Available, LMS-1, LHS-1
- Particle size distribution varies from $0.04 \mu m$ – $1000 \mu m$ Activation solution similar to flyask studies

Initial printing and adjusting the compositions for LHS-1, where 0.39 means 0.39 unit activator to 1 unit lunar regolith.

- Initial prints: 0.39*, 0.38, 0.38 cured

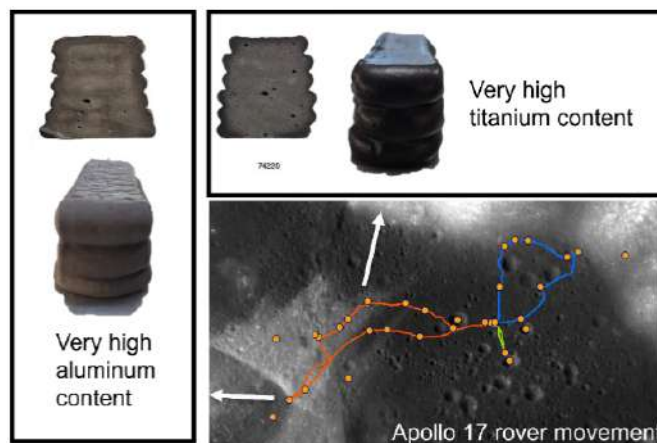


- Beam: 0.38 changed to 0.37

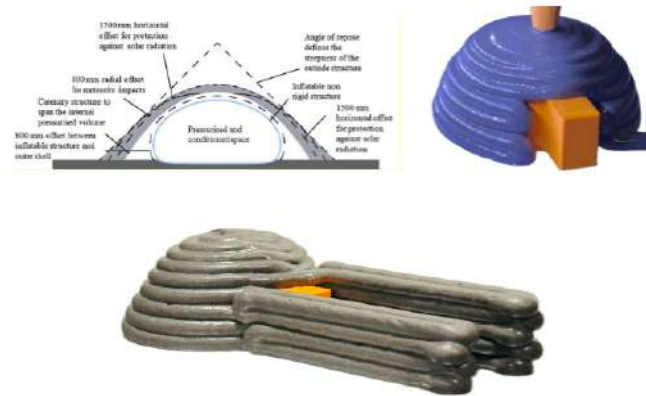


- 3 domes with 0.3, last with 0.36

3D printing with regolith



"No scoop of the lunar soil is the same", If the possibility of printing on the moon arises we can not be surprised by local material behavior.



In situ resource utilization, usage of regolith through geopolymerization. An inflatable core, regolith shell, ESA design.

The 0.36 dome habitat shown above cracked. Potential causes could be Low temperature → slow curing → shrinkage, or uneven air exposure.

To mitigate the issues an increase in curing temperature and addition of filler (unmilled regolith and fibers).



Figure 41: Three-point bending test of LHS-1 with 3 beams

	Beam 1	Beam 2	Beam 3	Mean	Std
Flexural strength MPa	28.6	12.4	8.9	16.6	8.6

The flexural strength is higher than that of sulphur concrete of 4.8 MPa. all 3 beams with 0.37 mass ratio cured. However, standard problems such shape (uneven load), fractured at angle, porosities.



Figure 42: Compression test of LHS-1

Sample	1	2	3	4	5	Mean	Std
Compressive strength, initial, MPa	70.8	67.4	70.4	109.1		79.4	17.2
Compression strength, casted, MPa	103.6	62.3	87.3	76.6	87.4	83.4	13.6

Tests were made with the remains of beams. Standard deviation is large due to the surface being uneven. Casted sample had smoother surface, which eliminate porosities. The compressive strength was also higher than that of sulphur concrete of approximately 26.0 MPa.

4 Introduction to Precision Metal Additive Manufacturing

4.1 Additive Manufacturing in Industry

Additive Manufacturing (AM), commonly referred to as 3D printing, is a manufacturing process in which parts are produced layer-by-layer from a digital 3D model. While often perceived as a new technology, its origins date back to the 1980s, with the first commercial systems emerging in 1986.

Initially, AM was primarily used for **rapid prototyping**, enabling fast and automated production of design models. However, over time, improvements in:

- precision,
- repeatability,
- and material capabilities

have enabled its transition toward **industrial production applications**.

Despite this progress, AM is still evolving and has not yet fully reached the performance level of conventional manufacturing processes.

4.2 Advantages of Additive Manufacturing

The main advantages of AM stem from its **tool-less manufacturing approach**:

- No need for expensive tooling
- Ability to manufacture highly complex geometries
- Easy design modifications without additional cost
- Possibility for mass customization

A key industrial example is the redesign of fuel injector nozzles in aerospace, where AM enables complex internal geometries that improve performance and lifetime.

4.3 Limitations and Industrial Challenges

Although AM shows significant potential, its large-scale industrial adoption is still limited. The main challenges include:

- Limited precision compared to conventional machining
- Need for extensive post-processing
- Integration with existing manufacturing workflows
- Qualification and reliability requirements

Therefore, AM must not only produce parts economically, but also integrate with:

- design processes,
- material supply chains,
- and post-processing operations.

4.4 Precision Engineering in Additive Manufacturing

Precision engineering is a key requirement for the advancement of AM technologies. It focuses on achieving high levels of:

- accuracy,
- repeatability,
- and reliability.

A common way to define precision is through the ratio:

$$\text{Precision} \sim \frac{\text{Error}}{\text{System Range}} \quad (9)$$

A system is typically considered **precision engineering** when this ratio approaches 10^{-6} .

For example:

- Precision machine: repeatability $\sim 100 \text{ nm}$ over $100 \text{ mm} \Rightarrow 10^{-6}$
- Metal AM machine: repeatability $\sim 100 \text{ }\mu\text{m}$ over $100 \text{ mm} \Rightarrow 10^{-3}$

This highlights that current AM systems are still far from true precision engineering standards.

4.5 Sources of Lack of Precision

The main factors limiting precision in AM include:

- Thermal effects
- Mechanical vibrations
- Loose joints and structural compliance
- Limited sensor and actuator resolution
- Friction and contamination

Improving AM precision requires identifying and minimizing these sources of error.

4.6 Towards Precision Additive Manufacturing

Achieving high precision in AM requires a **holistic approach**, where precision principles are applied throughout the entire process chain:

- Design
- Machine construction and control
- Process monitoring
- Post-processing
- Metrology and validation

Rather than relying on trial-and-error and heavy post-processing, future AM systems aim to achieve **deterministic and predictable manufacturing outcomes**.

Non-destructive evaluation Inspection of AM parts is challenging due to complex geometries. Common methods include:

- X-ray computed tomography (CT)
- Ultrasonic testing

CT is particularly useful for detecting internal defects.

4.7 Development of Precision Additive Manufacturing Processes

This topic focuses on how precision in metal additive manufacturing (AM) can be improved, mainly for laser powder bed fusion (L-PBF), which is one of the most important processes for manufacturing metal AM parts. A major limitation of metal AM is that precision is still not high enough for many demanding applications. Lack of precision can arise from:

- variation in material properties,
- strong interactions between process parameters,
- limited machine and equipment accuracy,
- insufficient in-process monitoring and control,
- and the strongly thermal nature of the process.

This is especially important in applications such as aerospace, automotive, marine and medical devices, where geometric accuracy, repeatability and reliability are critical.

4.7.1 Why precision in metal AM matters

In metal AM, small geometric deviations can accumulate from layer to layer. Since each new layer is deposited on top of the previous one, any small error may propagate through the build. Precision process development therefore aims to make the process:

- repeatable,
- predictable,
- defect-minimising,
- and suitable for industrial production.

4.7.2 L-PBF process sequence

In laser powder bed fusion, a laser selectively melts metallic powder in a powder bed to build a part layer by layer. The process sequence is:

1. A CAD model of the part is created.
2. The CAD file is converted to STL.
3. Process parameters are set in AM pre-processing software.
4. The geometry is digitally sliced into layers.
5. A laser selectively melts powder on the build platform.
6. The molten pool solidifies after cooling.
7. The build platform is lowered by one layer thickness.
8. A new powder layer is spread.
9. The next layer is scanned and melted.
10. The sequence repeats until the full part is built.
11. The build platform is removed for heat treatment and finishing.
12. Excess powder is removed, sieved and possibly reused.

L-PBF offers:

- high geometric flexibility,
- near-net-shape production,
- reduced material waste,
- and very good buy-to-fly ratios compared with conventional subtractive manufacturing.

4.7.3 Setting priorities in AM process development

Process improvement in AM should be considered at two levels:

1. **Process level:** improving the AM process itself, for example by optimising process parameters, dimensional accuracy, energy efficiency and material usage.
2. **System level:** analysing how AM fits into the full manufacturing chain, including lead time, logistics, cost, qualification and integration with existing industrial workflows.

Therefore, precision AM process development must be holistic: improving one part of the process should also be evaluated in terms of its effect on the complete production system.

4.7.4 Significant process parameters

The quality of L-PBF parts depends on many interacting process parameters. These are commonly grouped into four categories.

Laser-related parameters

- **Laser power** P : optical power delivered by the laser.
- **Spot size** d : diameter of the focussed laser beam.
- **Focal offset distance**: distance between the focal plane and the printing surface.

Scan-related parameters

- **Scanning speed** v : translational speed of the laser beam.
- **Hatch spacing** h : distance between adjacent scan tracks.
- **Scan pattern**: path followed by the laser on the powder bed.

Common scan strategies include:

- hexagonal cells,
- rectangular cells,
- and stripe patterns.

Powder-related parameters

- **Particle size and size distribution**: affect packing density, flowability and laser absorption.
- **Powder bed density**: influences porosity, thermal conductivity and final surface quality.
- **Layer thickness** t : thickness of each deposited layer.

Smaller layers generally improve detail and strength, while thicker layers increase build speed but reduce resolution.

Build chamber-related parameters

- **Build chamber temperature**: affects thermal gradients, residual stresses and mechanical properties.
- **Atmosphere**: inert gases such as argon or helium are used to reduce oxidation and porosity.

4.7.5 Combined processing parameters

Since many parameters interact, researchers often use combined energy-density measures.

Volumetric energy density

$$E_V = \frac{P}{vht} \quad (10)$$

Linear input energy density

$$E_I = \frac{4P}{v\pi d^2} \quad (11)$$

Surface energy density

$$E_S = \frac{P}{vd} \quad (12)$$

Linear energy density

$$E_L = \frac{P}{v} \quad (13)$$

These quantities are useful because they relate multiple process parameters to the heat input delivered to the material.

Energy density regimes

- Low energy density $\rightarrow$ lack of fusion
- Optimal energy density $\rightarrow$ dense parts
- High energy density $\rightarrow$ keyhole formation and porosity

Melt pool dynamics The stability of the melt pool strongly influences part quality:

- Unstable melt pools can cause balling and porosity
- Spatter (ejection of molten material) can redeposit particles and create defects

4.7.6 Microstructure formation in L-PBF

Due to extremely high cooling rates, L-PBF produces non-equilibrium microstructures:

- Columnar grains form near melt pool boundaries
- Equiaxed grains form near the center of the melt pool

Microstructure evolution is governed by thermal gradients and solidification conditions.

4.7.7 Additive manufacturing performance indicators

To improve precision, suitable output measures are needed. Important AM performance indicators include the following.

Mechanical properties These include:

- tensile strength,
- yield strength,
- hardness,
- and other structural performance metrics.

Dimensional accuracy Dimensional accuracy describes how close the manufactured part is to the intended CAD geometry. In L-PBF, it is strongly affected by:

- thermal gradients,
- residual stresses,
- part orientation,
- support structures,
- and post-processing.

Residual stresses are particularly important. During printing, upper hot layers expand while colder underlying layers restrain them. After cooling, this leads to tensile residual stresses, which can cause:

- warping,
- cracking,
- and geometric distortion.

Origin of residual stresses Residual stresses originate from thermal gradients during the process:

- Newly deposited layers are hot and expand
- Previously solidified layers are cooler and constrain deformation
- Cooling leads to tensile stresses in the material

These stresses can lead to warping, cracking and delamination.

Effect of part orientation Because L-PBF is a layer-based process, part orientation strongly affects quality. Features such as holes and cylinders are usually printed most accurately when their axis is aligned with the build direction, reducing staircase effects.

Support structures Supports are often required for down-facing regions. They:

- provide mechanical support,
- improve heat conduction to the base plate,
- reduce deformation,
- but increase build time and post-processing effort.

Support removal may also introduce burrs and extra surface defects.

Post-processing Stress-relief heat treatment is commonly applied before removing the part from the build platform. This reduces residual stresses and helps avoid severe distortion or failure.

Surface texture Surface texture is another key quality indicator. In AM it is influenced by:

- layer-wise fabrication,
- powder morphology,
- scan strategy,
- and overhang formation.

Part density Internal defects such as:

- pores,
- cracks,
- lack-of-fusion regions,
- and inclusions

reduce the density of the final part and often degrade both mechanical performance and dimensional quality.

A “fully dense” part is often taken as one with density greater than 99%, although no single universal definition exists.

Total build time Build time is a major limitation of metal AM in industrial use. Reducing build time improves productivity and reduces lead time.

Energy consumption Energy consumption is important both economically and environmentally. Since AM is already attractive because of its good material efficiency, improving energy efficiency can further strengthen its sustainability advantage.

System-wide performance indicators In industrial production, AM must also be evaluated using broader manufacturing KPIs, such as:

- on-time delivery,
- cycle time,
- changeover time,
- yield,
- throughput,
- overall equipment efficiency (OEE),
- and manufacturing cost per unit.

A common expression for overall equipment efficiency is:

$$\text{OEE} = \text{Availability} \times \text{Performance} \times \text{Quality} \quad (14)$$

These indicators are important because AM is increasingly being integrated into real industrial process chains rather than being used only in research environments.

4.7.8 Data-driven process improvement

Modern AM process development is increasingly data-driven. Process improvement typically follows five steps:

1. data collection,
2. data sharing,
3. data analysis,

4. optimisation,
5. and feedback to the machine or operator.

Data can come from:

- in-situ sensors,
- cameras,
- machine logs,
- and off-line inspection or testing.

Since this data is often heterogeneous, it must be digitised, cleaned and normalised before it can be used for modelling and optimisation.

Design of Experiments (DoE) DoE methods are used to systematically study the relationship between process inputs and output responses with as few experiments as possible.

A DoE uses:

- **factors:** input variables,
- **levels:** values assigned to each factor,
- **responses:** measured outputs.

Full factorial design If all factor-level combinations are tested, the total number of experiments is:

$$n_r = n_l^{n_f} \quad (15)$$

where:

- n_r is the number of runs,
- n_f is the number of factors,
- n_l is the number of levels.

Because this grows very quickly, fractional factorial methods are often preferred.

Central composite design (CCD) CCD combines:

- factorial points,
- axial points,
- and replicated centre points.

It is especially useful when quadratic trends are expected.

Orthogonal arrays (Taguchi designs) Orthogonal arrays reduce the number of trials while preserving balanced combinations of factor levels. They are often written as:

$$L_n(s^m)$$

where:

- n is the number of experiments,
- m is the number of factors,
- s is the number of levels.

Modelling of process performance Once data has been collected, input-output relationships can be modelled. This helps predict process behaviour and identify the most influential parameters.

Regression and statistical analysis Regression models are useful when the process-response relationship can be reasonably approximated with an analytical expression. They are often combined with analysis of variance to determine which factors are statistically significant.

Artificial neural networks (ANNs) ANNs are especially useful when the process is highly nonlinear and difficult to model explicitly. In metal AM, ANNs can be used to predict:

- density,
- surface quality,
- dimensional deviation,
- and mechanical performance

from process parameters and sensor data.

Their main advantage is that they can capture complex nonlinear relationships, although they usually require a sufficiently large dataset.

Process optimisation After a process model has been built, optimisation methods can be used to determine parameter sets that best satisfy the chosen objective, for example:

- maximising density,
- minimising surface roughness,
- minimising geometric deviation,
- or reducing cost and build time.

This step is essential because AM processes typically involve multiple interacting variables, and manual tuning is inefficient.

4.7.9 Precision AM in the context of Industry 4.0

The development of precision AM is strongly linked to Industry 4.0. In this context, AM systems are increasingly connected, sensor-rich and data-driven.

Real-time monitoring of AM processes Real-time monitoring aims to detect process instabilities during the build itself. This can involve:

- melt-pool monitoring,
- optical imaging,
- thermal sensing,
- and machine-state monitoring.

The goal is to detect anomalies such as:

- lack of fusion,

- overheating,
- powder spreading defects,
- and geometric distortion

before they lead to severe part defects.

Artificial intelligence and decision-making systems AI-based systems can support digital quality control by:

- interpreting monitoring data,
- recognising defect signatures,
- predicting quality in real time,
- and suggesting or applying corrective actions.

This moves AM from passive monitoring toward intelligent process control.

4.7.10 Future perspectives for precision AM processes

Future development of precision AM is expected to focus on:

- better understanding of complex parameter interactions,
- improved process models,
- wider use of machine learning and AI,
- more effective in-situ monitoring and control,
- stronger integration with Industry 4.0 systems,
- and improved standardisation and qualification methods.

The long-term goal is to move from trial-and-error AM toward robust, deterministic and industrially reliable processes.

4.7.11 Main takeaway

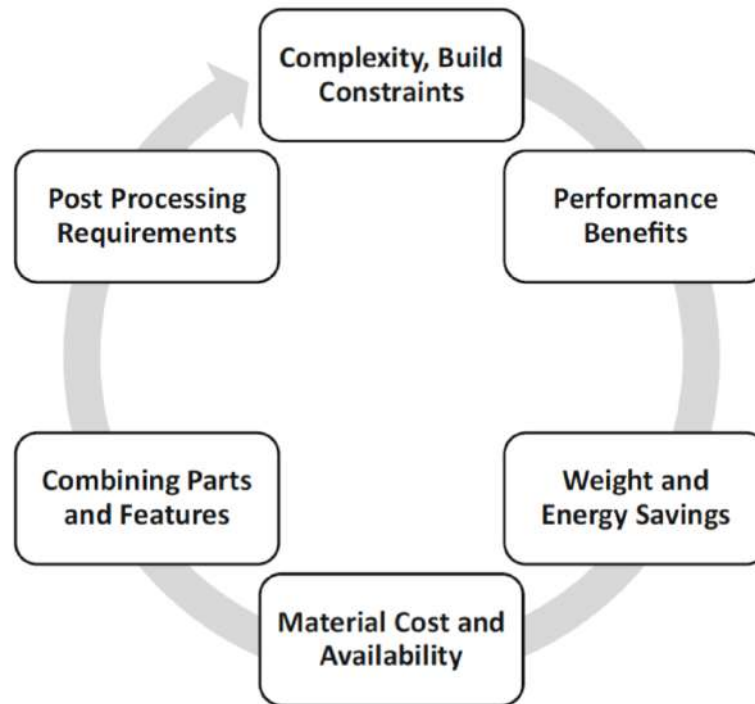
Precision in metal additive manufacturing depends on understanding and controlling the full process chain. In L-PBF this means:

- selecting suitable process parameters,
- monitoring relevant performance indicators,
- modelling the process using experimental or sensor data,
- optimising parameters systematically,
- and integrating monitoring, AI and feedback for closed-loop control.

In other words, precision AM is not achieved by a single parameter change, but by combining process understanding, data-driven modelling, optimisation and intelligent control across the full manufacturing system.

5 Metal Additive Manufacturing

When to opt for AM? It all starts with the question "what do I get out of it?"



- | | |
|---------------------|---|
| 1. Size: | Does it fit? |
| 2. Material: | Does it weld? |
| 3. Economics: | Is there Value? |
| 4. Performance: | Is it met? |
| 5. Qualification: | Is it Yours? (Do you know what is critical) |
| 6. Design: | Will it Work? |
| 7. Post Processing: | Is it Painful? |

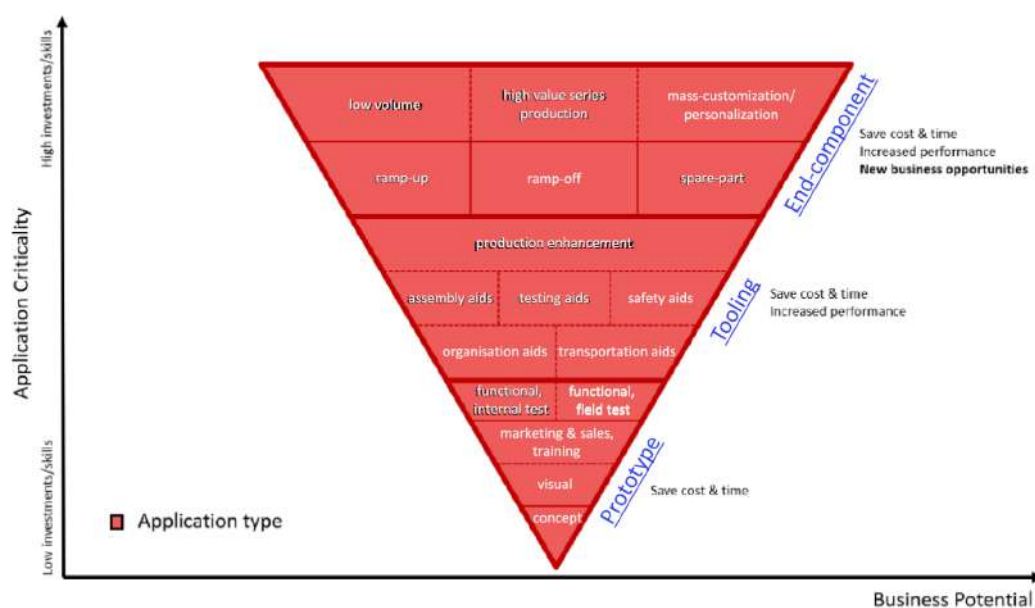
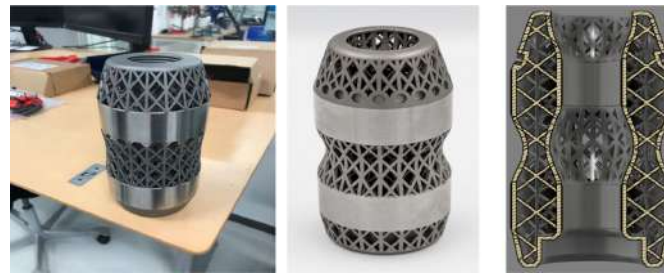


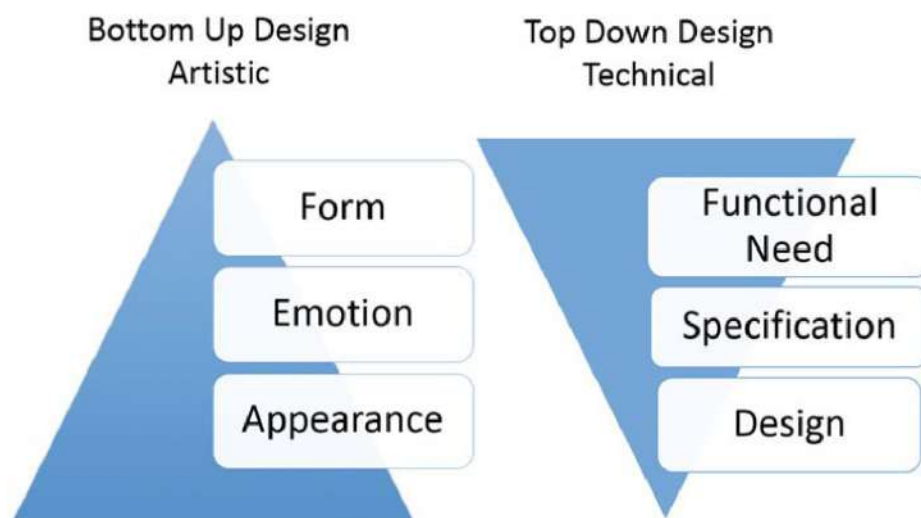
Figure 43: How to identify components: Output from AMLine project

Examples where it makes sense

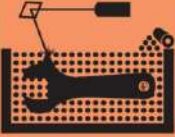
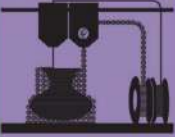


Building a rocket out of chill

5.1 Parts and design guidelines for Additive Manufacturing



First impressions on DfAM (Design for Additive Manufacturing)

7 Families of Additive Manufacturing						
According to ISO/ASTM52900-15 (formerly ASTM F2792)						
						
VAT PHOTO-POLYMERIZATION	POWDER BED FUSION (PBF)	BINDER JETTING	SHEET LAMINATION	MATERIAL JETTING	MATERIAL EXTRUSION	DIRECTED ENERGY DEPOSITION (DED)
Alternative Names: SLA™ - Stereolithography Apparatus DLP™ - Digital Light Processing 3DP™ - Scan, Spin, and Selectively Photocure CLIP™ - Continuous Liquid Interface Production	Alternative Names: SLM™ - Selective Laser Melting (aka a SLS™ - Selective Laser Sintering), DMLS™ - Direct Metal Laser Sintering (aka a Sintering), EBM™ - Electron Beam Melting, MJF™ - Multi Jet Fusion, SHS™ - Selective Heat Sintering	Alternative Names: 3DP™ - 3D Printing Binder Jetting™ is available from ExOne, Voxeljet, Desktop Metal's Production System™, and others *metal & ceramic require post-print sintering	Alternative Names: LOM - Laminated Object Manufacturing SLM - Selective Deposition Lamination UAM - Ultrasonic Additive Manufacturing	Alternative Names: PolyJet™ SGM™ - Smooth Curvature Printing MJM - Multi-Jet Modeling Project™	Alternative Names: FFF - Fused Filament Fabrication FDM™ - Fused Deposition Modeling APD™ - Augmented Polymer Deposition ADM™ - Atomic Diffusion Additive Mfg™ BMD™ - Bound Metal Deposition™	Alternative Names: LMD - Laser Metal Deposition LENS™ - Laser Engineered Net Shaping LMD - Laser Metal Deposition Laser cladding WAM - Wire-arc Additive Manufacturing
Description: A vat of liquid photopolymer resin is cured through selective exposure to light (via a laser or projector) which then initiates polymerization and converts the exposed areas to a solid part.	Description: Powdered materials are selectively consolidated by melting it together using a heat source such as a laser or electron beam. The powder surrounding the consolidated part acts as support material for overhanging features.	Description: Liquid binding agents are selectively applied onto thin layers of powdered material to build up parts layer by layer. The binders include organic and inorganic materials. Metal or ceramic powdered parts are typically fired in a furnace after they are printed.	Description: Sheets of material are stacked and laminated together to form an object. The lamination method can be adhesive or chemical (paper/paste), ultrasonic welding, or bonding (metal). Unneeded regions are cut out layer by layer and removed after the object is built.	Description: Droplets of material are deposited layer by layer to make parts. Common variants include jetting a photosensitive resin and curing it with UV light, as well as jetting thermally sensitive materials that then solidify in ambient temperatures.	Description: Material is extruded through a nozzle or orifice in tracks or beads, which are then combined into multi-layer models. Common variants include heated thermoplastic extrusion (similar to a hot glue gun) and syringe dispensing.	Description: Powder or wire is fed into a melt pool which has been generated on the surface of the part where it adheres to the underlying part or layers by using an energy source such as a laser or electron beam. This is essentially a form of automated build-up welding.
Strengths: • High level of accuracy and complexity • Smooth surface finish • Accommodates large build sizes	Strengths: • High level of complexity • Powder acts as support material • Wide range of materials	Strengths: • Allows for full color printing • High productivity • Uses a wide range of materials	Strengths: • High volumetric build rates • Relatively low cost (non-metals) • Allows for combinations of materials, including embedding components	Strengths: • High level of accuracy • Allows for full color parts • Enables multiple materials in a single part	Strengths: • Inexpensive and economical • Allows for multiple colors • Can be used in an office environment • Parts have good structural properties	Strengths: • Not limited by direction or axis • Effective for repairs and adding features • Multiple materials in a single part • Highest single-point deposition rates
Typical Materials UV-Curable Photopolymer Resins	Typical Materials Plastics, Metal and Ceramic Powders, and Sand	Typical Materials Powdered Plastic, Metal, Ceramics, Glass, and Sand	Typical Materials Paper, Plastic Sheets, and Metal Foils/Tapes	Typical Materials Photopolymers, Polymers, Waxes	Typical Materials Thermoplastic Filaments and Pellets (FFF), Liquids, and Slurries (Syringe Type)	Typical Materials Metal Wire and Powder, with Ceramics

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DESIGN RULES FOR 3D PRINTING										
3D HUBS										
	Supported Walls	Unsupported Walls	Support	Embossed & Engraved Details	Horizontal Bridges	Holes	Connecting & Moving Parts	Escape Holes	Minimum Features	Pin Diameter
	Walls that are connected to other structures on at least two sides.	Walls that are connected to the rest of the print on only one side.	The maximum angle a wall can be printed out without requiring support.	Features on the model that are raised or recessed below the model surface.	The span a technology can print without the need for support.	The minimum diameter a technology can successfully print a hole.	The recommended clearance between moving or connecting parts.	The minimum diameter of escape holes to allow for the removal of build material.	The recommended minimum size of a feature to ensure it will not fail to print.	The minimum diameter a pin can be printed at.
										
Fused Filament Fabrication	0.8mm	0.8mm	45°	0.6mm wide & 2mm high	10mm	Ø2mm	0.5mm		2mm	3mm
Stereolithography	1mm	0.5mm	support always required	0.1mm embossed & 0.4mm for engraved details		Ø0.5mm	0.5mm for moving parts	4mm	0.2mm	0.5mm
Selective Laser Sintering	0.7mm	NA as power offers support		1mm wide & high		Ø1.5mm	0.3mm for moving parts & 0.1mm for connections	3.5mm	0.8mm	1mm
Material Jetting	1mm	1mm	support always required	0.5mm wide & high		Ø0.5mm	0.2mm for moving parts		0.5mm	0.5mm
Binder Jetting	2mm	3mm		0.5mm wide & high	20mm	Ø1mm		5mm	2mm	2mm
Direct Metal Laser Sintering	0.4mm	0.5mm	support always required	1mm wide & high	2mm	Ø1.5mm		3mm	0.6mm	1mm

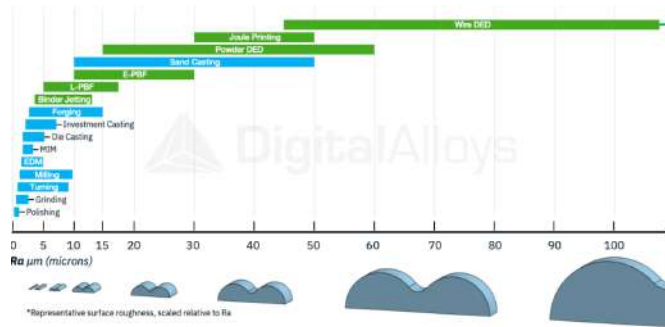


Figure 44: Typical Surface Roughness of Metal AM and Conventional Processes

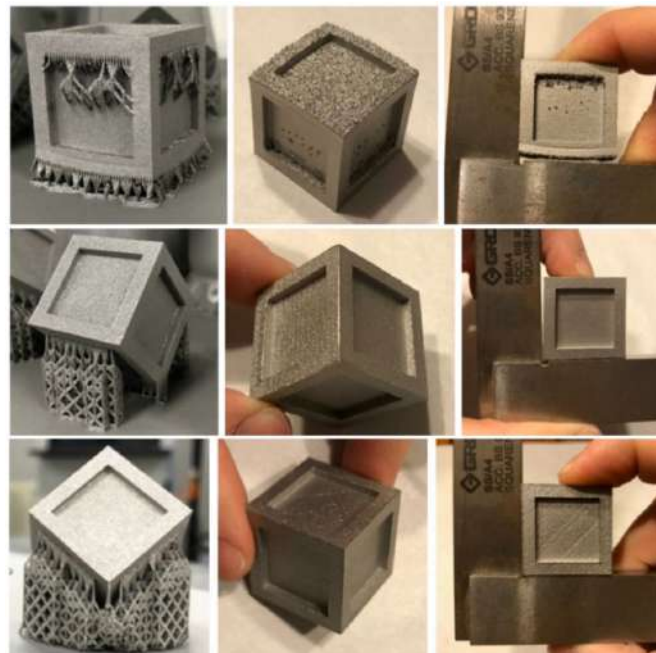
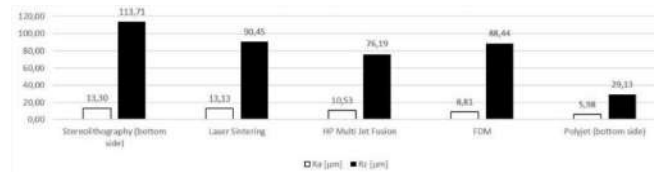


Figure 45: Design the orientation in which the part is printed

Effect of part design choices for Case 1

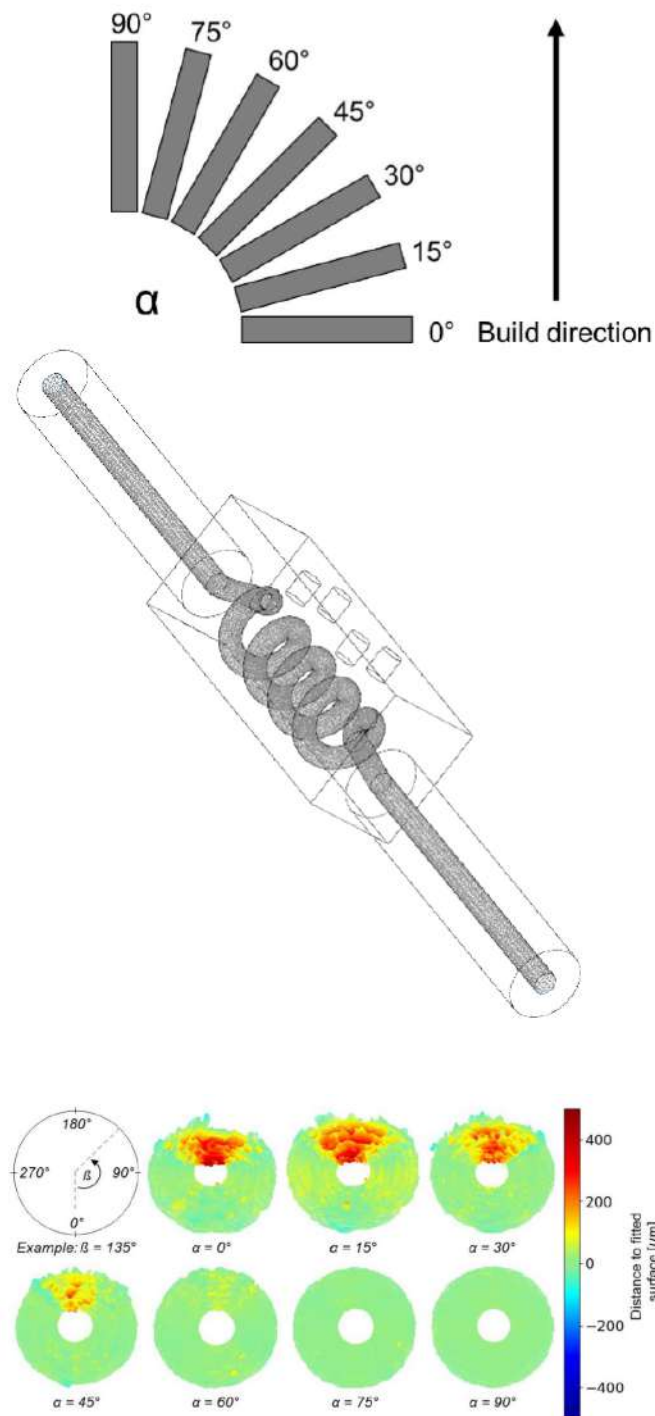


Figure 46: through view of the channels, visualizing the roughness dependency on the location inside a channel (local orientation β) and build direction (global orientation α)

Principle	Description
Part orientation	Some shapes print better in certain orientations, though sometimes there is no optimal direction. Orientation can affect the surface finish of the part, especially when dealing with rounded features. The orientation can influence the strength between horizontal and vertical components, since layers are added in the vertical (z) direction.
Removal of supports	Removing support structures significantly reduces surface finish and increases the need for post-processing. Sharp inner edges can reduce the need for support structures as the layers build.
Hollowing out parts / Manufacturing features	When functionally possible, thick walls and hollow interiors can reduce print time. While AM does not require undercuts or draft angles, many parts are prototypes that will later be manufactured using traditional processes.
Interlocking features	AM processes have a finite build space and may require large parts to be broken up and joined later using interlocking features. Joined/interlocking gap dimensions should be minimized to enable robust support removal and reduce dimensional deviations.
Reduction of part count	If the AM part is intended as a final product, the number of parts in an assembly may be reduced.
Identification marks	When producing many parts, it is easy to lose track of which model is which.
Avoid sharp edges	Removing sharp edges results in better accuracy; rounding radii correlate with outer radii of simple-curved elements.
Round inner edges	Rounding edges simplifies the removal of dispersed support structures (e.g., powder).
Blunt extreme points	Vertical points should be blunted parallel to the build plane; horizontal points should be blunted orthogonal to the build plane.
Short overhang	Ensures robust manufacturability and prevents layers from collapsing.
Low island positions	Has a significant impact on build time.

Table 2: Guidelines described by prior papers

Design for Additive Manufacturing

A quick method for reducing the number of printing and prototyping failures, by Joran Booth

Instructions: Mark one for each category for the part you plan to print. Check daggers and stars first, then scores

Mark One	Complexity	Mark One	Functionality	Mark One	Material Removal	Mark One	Unsupported Features	Sum Across Rows	Totals
†○	The part is the same shape as common stock materials, or is completely 2D	*○	Mating surfaces are bearing surfaces, or are expected to endure for 1000+ of cycles	○	The part is smaller than or the same size as the required support structure	○	There are long, unsupported features	x5 =	
*○	The part is mostly 2D and can be made in a mill or lathe without repositioning it in the clamp	*○	Mating surfaces move significantly, experience large forces, or must endure 100-1000 cycles	○	There are small gaps that will require support structures	○	There are short, unsupported features	x4 =	
○	The part can be made in a mill or lathe, but only after repositioning it in the clamp at least once	○	Mating surfaces move somewhat, experience moderate forces, or are expected to last 10-100 cycles	○	Internal cavities, channels, or holes do not have openings for removing materials	○	Overhang features have a sloped support	x3 =	
○	The part curvature is complex (splines or arcs) for a machining operation such as a mill or lathe	○	Mating surfaces will move minimally, experience low forces, or are intended to endure 2-10 cycles	○	Material can be easily removed from internal cavities, channels, or holes	○	Overhanging features have a minimum of 45deg support	x2 =	
○	There are interior features or surface curvature is too complex to be machined	○	Surfaces are purely non-functional or experience virtually no cycles	○	There are no internal cavities, channels, or holes	○	Part is oriented so there are no overhanging features	x1 =	
Mark One	Thin Features	Mark One	Stress Concentration	Mark One	Tolerances	Mark One	Geometric Exactness		
○	Thin features will almost always break	○	Interior corners must transition gradually	○	Mating parts should not be the same size	○	Large, flat areas tend to warp		
○	Some walls are less than 1/16" (1.5mm) thick	○	Interior corners have no chamfer, fillet, or rib	○	Hole or length dimensions are nominal	○	The part has large, flat surfaces or has a form that is important to be exact	x5 =	
○	Walls are between 1/16" (1.5mm) and 1/8" (3mm) thick	○	Interior corners have chamfers, fillets, and/or ribs	○	Hole or length tolerances are adjusted for shrinkage or fit	○	The part has medium-sized, flat surfaces, or forms that should be close to exact	x3 =	
○	Walls are more than 1/8" (3mm) thick	○	Interior corners have generous chamfers, fillets, and/or ribs	○	Hole and length tolerances are considered or are not important	○	The part has small or no flat surfaces, or forms that need to be exact	x1 =	
				Starred Ratings	Total Score			Overall Total	
				* Consider a different manufacturing process	33-40 Needs redesign				
				† Strongly consider a different manufacturing process	24-32 Consider redesign				
					16-23 Moderate likelihood of success				
					8-15 Higher likelihood of success				

REID Research in Engineering and Interdisciplinary Design

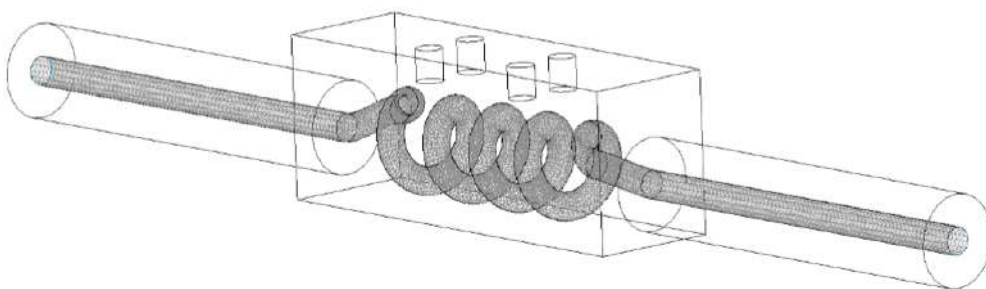
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Figure 47: Design for Additive Manufacturing (DfAM)

- Nominal means very important
- Mating surfaces are things that touch and stop touching repeatedly

Case 1: Cooling/ heating channels inside a stamp.



- A hot stamp is to be used to create a label on rubber-like material. During its operation, the stamp heats up the material and causes it to char / mark, after which the stamp is removed. The stamp then needs to be heated up rapidly to its preferred starting temperature before the next label is marked. The current heating setup is sought to be replaced by a new design incorporating heating channels within the stamp.
- Evaluate the incorporation of the channels make sense.

For this we will need to know lumped analysis. Assuming an object has a constant temperature at a given time, the lumped analysis gives the temperature of the object as a function of time:

$$T(t) - T_{inf} = (T_i - T_{inf}) \exp\left(-\frac{hA_c t}{\rho c_p V}\right)$$

Where T_{inf} is the temperature in the cooling medium.

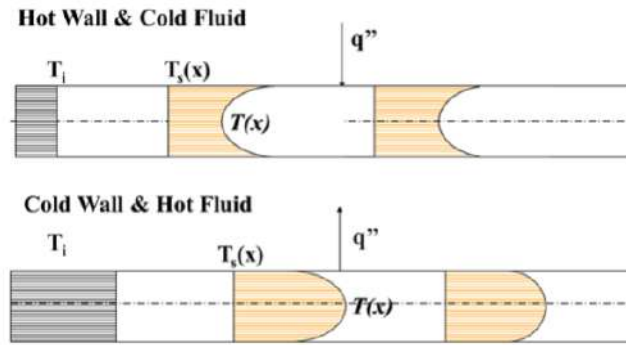
The ratio of heat removed from the channel surface by convection to the heat removed by conduction (Nusselt number for a channel or pipe uses the diameter D),

$$Nu_D(x) = \frac{h_x D}{k_{fluid}} = \frac{\frac{q_x}{A(T_{wall} - T_m(x))} D}{k_{fluid}}$$

Usually the Nusselt number is always defined as,

$$Nu = \frac{hL}{k}$$

Where h is the heat transfer coefficient, k is the thermal conductivity of the fluid and L is the characteristic length



For this specific case we can assume the thermal conductivity of water to be 0.598 W/(mK) , and the diameter of the channel to be 6 mm .

The Surface averaged Nusselt number,

$$\overline{Nu}_D = 0.3 + \frac{0.62 \text{Re}_D^{1/2} \text{Pr}^{1/3}}{(1 + (0.4/\text{Pr})^{2/3})^{1/4}} \left(1 + \left(\frac{\text{Re}_D}{282000}\right)^{5/8}\right)^{4/5} \quad \text{PrRe}_D \geq 0.2$$

- A number of tests were carried out to see how well the channels heat up the part.
- Since a hot liquid to be pushed into the channels using a pump, the only available flow setting options to the table shown below.

$$\text{Pr} = 7.56$$

Re	Nu_D
12000	129.56
14000	141.29
16000	152.42
18000	163.05
20000	173.27

- The operating temperature of the stamp is 50°C . The rubber like material reduces the temperature to 30°C -

- The current heating solution dips the stamp in a huge pool of stagnant water held at 70 °C.
 - The area in contact with water is A_{out}
 - The characteristic length is 1 m
 - The consequent Nusselt number for cooling is equal to 1
- The new heating solution flows the water within the channel at 70 °C
 - The area in contact with water is A_{in}

Assuming the stamp behaves as a lumped thermal mass, the heating time is,

$$t = \frac{\rho V c_p}{hA} \ln \left(\frac{T_{inf} - T_i}{T_{inf} - T(t)} \right) = \frac{\rho L c_p}{h} \ln \left(\frac{T_{inf} - T_i}{T_{inf} - T(t)} \right)$$

For both cases,

$$T_{inf} = 70 \text{ °C}, \quad T_i = 30 \text{ °C}, \quad T(t) = 50 \text{ °C}$$

So the logarithmic term is the same in both cases and also ρc_p is the same stamp. Therefore,

$$t \propto \frac{L}{h}$$

So the ratio of heating times is,

$$\frac{t_{new}}{t_{old}} = \frac{L_{new} h_{old}}{L_{old} h_{new}}$$

$$h_{old} = \frac{Nu_{old} k}{L_{old}}$$

for the old solution, it is given,

$$Nu = 1, \quad k = 0.598 \text{ W/(m} \cdot \text{K)}, \quad L = 1 \text{ m}$$

$$h_{old} = \frac{1 \cdot 0.598}{1} = 0.598 \text{ W/(m}^2 \text{K)}$$

For the new solution, for the channel,

$$L_{new} = D = 6 \text{ mm} = 0.006 \text{ m}$$

Therefore the ratio,

$$\frac{t_{new}}{t_{old}} = \frac{L_{new} h_{old}}{L_{old} h_{new}} = \frac{0.006 \cdot 0.598}{1 \cdot h_{new}} = \frac{0.003588}{h_{new}}$$

In practice this won't be entirely correct since the pipe won't always be circular throughout the whole path as seen in [46](#).

6 Metal AM Current major systems

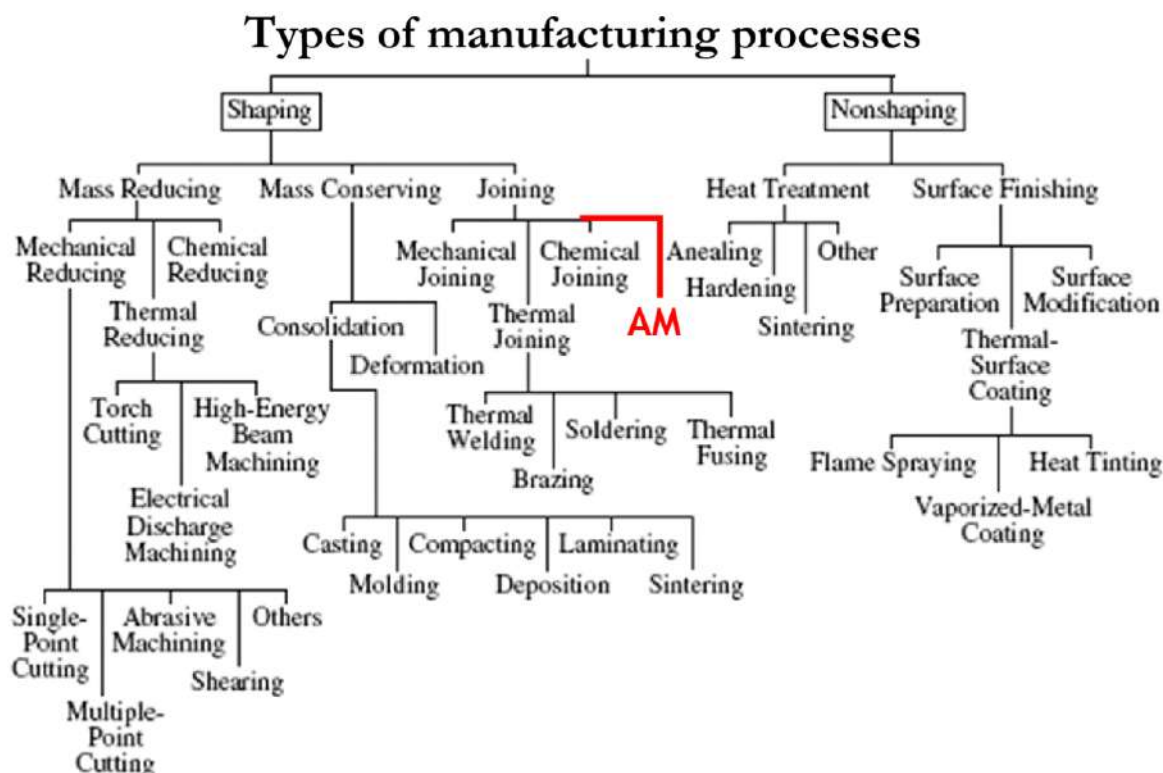


Figure 48: Types of manufacturing processes

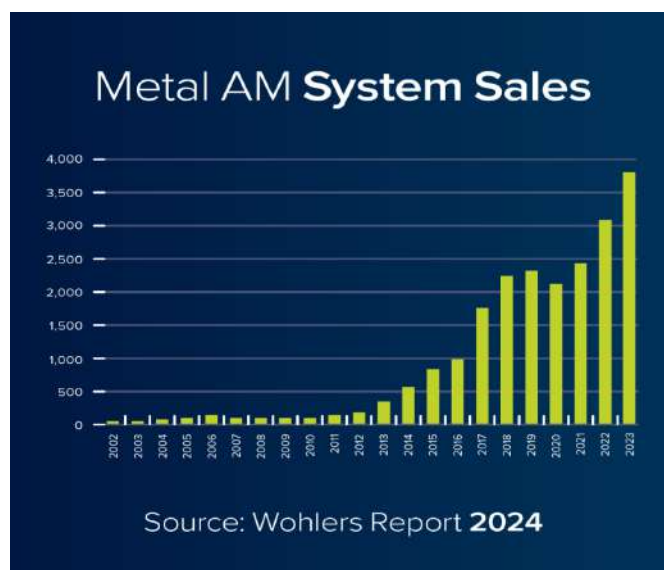


Figure 49: AM system sales and material revenue for total AM market and metal-based AM

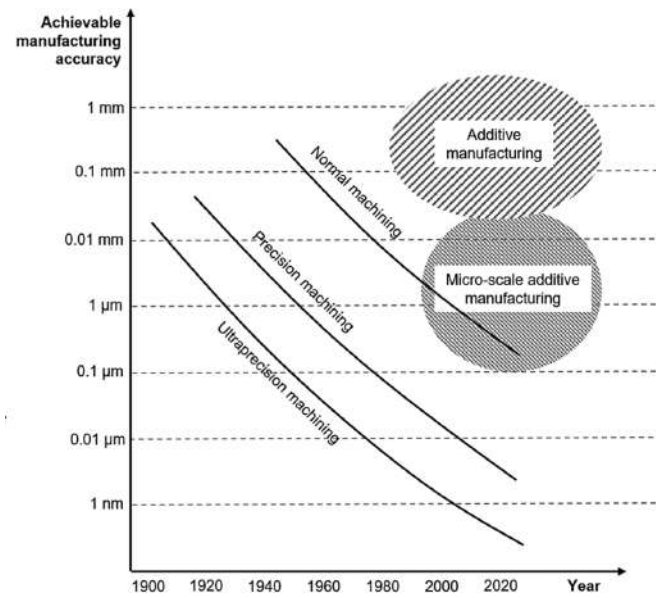


Figure 50: Taniguchi plot indicating the development of precision manufacturing capability and AM as function of time

Process category (ISO/ASTM 52900)		Single / Multi	Metal-based Additive Manufacturing Process	Feedstock	No. suppliers
	Powder Bed Fusion	Single	Laser-based powder bed fusion (L-PBF)	Powder	36
		Single	Electron beam powder bed fusion (EBM)	Powder	4
		Multi	Metal SLS	Powder	1
	Directed Energy Deposition	Single	Powder feed laser energy deposition	Powder	18
		Single	Resistance welding	Wire	1
		Single	Plasma arc energy deposition	Wire	2
		Single	Wire arc energy deposition	Wire	7
		Single	Electron beam energy deposition	Wire	2
		Single	Wire feed laser energy deposition	Wire	2
	Extrusion	Multi	Metal fused deposition modeling	Filament	7
		Multi	MIM fused deposition modeling	Granulate	2
	Binder Jetting	Multi	Binder jetting	Powder	4
		Multi	Hybrid binder jetting	Powder	1
	Sheet Lamination	Single	Friction welding	Sheet	1
	Process category not defined	Single	Cold spray	Powder	4
		Single	Friction stir welding	Rods	1
		Single	Liquid metal printing	Rods	1
		Multi	Nano-particle jetting	Dispersion	1

Table 3: Overview of metal additive manufacturing processes categorized according to ISO/ASTM 52900.

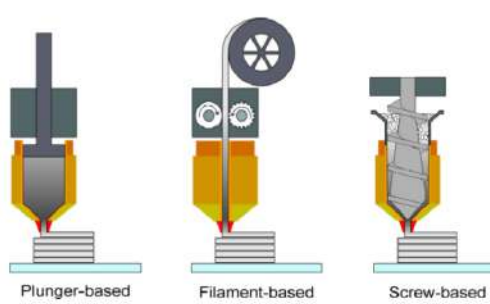


Figure 1. Different types and approaches for extrusion-based additive manufacturing.

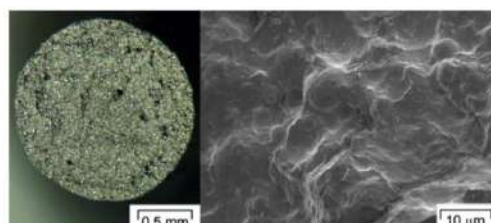


Figure 4. Optical microscopy and SEM images of a filament filled with steel particles [62,73].

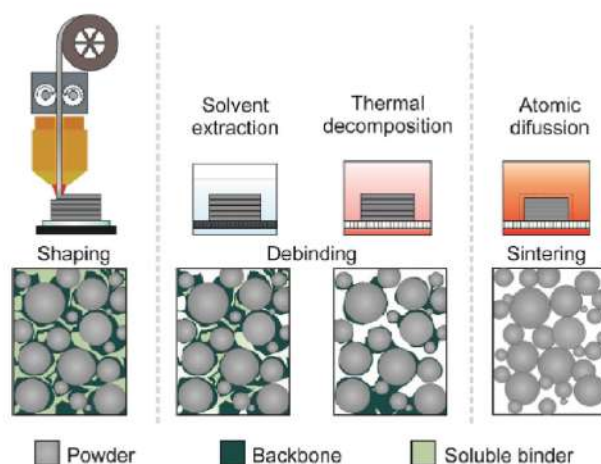


Figure 2. Schematic representation of the shaping, debinding, and sintering (SDS) process and respective morphology of the parts for the fabrication of metal, ceramic, or metal-ceramic components.

Source: *Additive Manufacturing of Metallic and Ceramic Components by the Material Extrusion of Highly-Filled Polymers: A Review and Future Perspectives*

Figure 51: Material extrusion

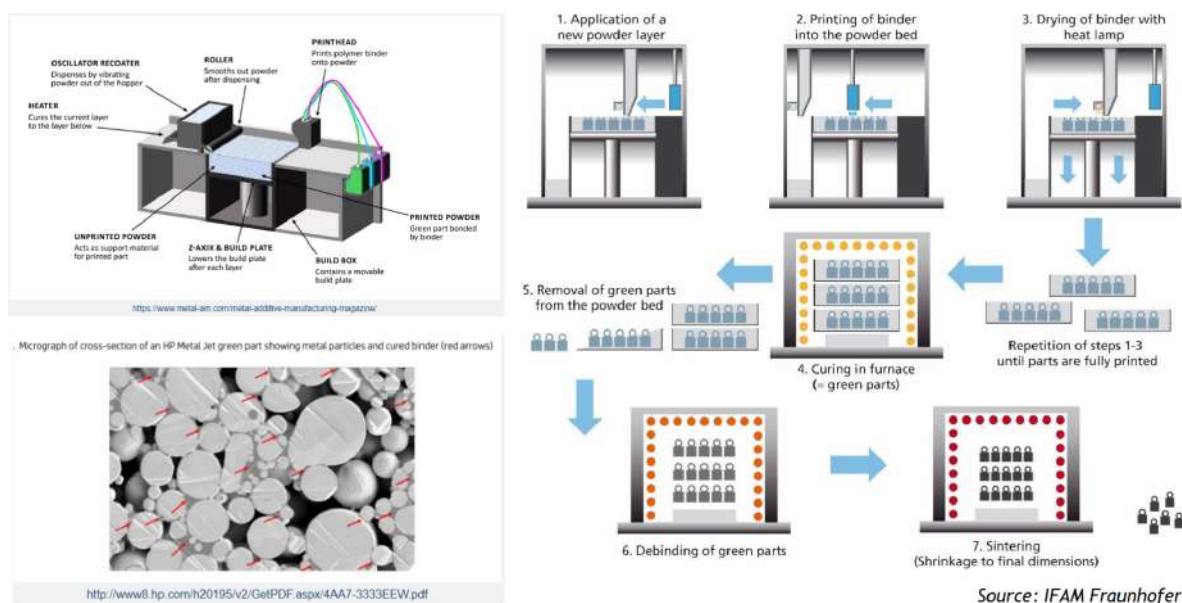


Figure 52: Binder jetting

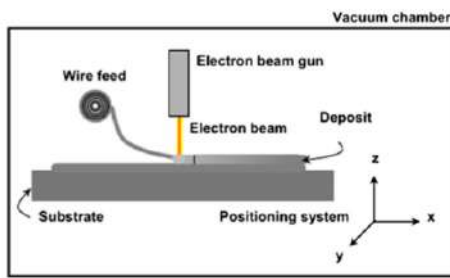


Fig. 7 Schematic of electron beam freeform fabrication (EBF) system components [13]

Fig. 8 Schematic diagram of the a GMAW, b GTAW, and c PAW process

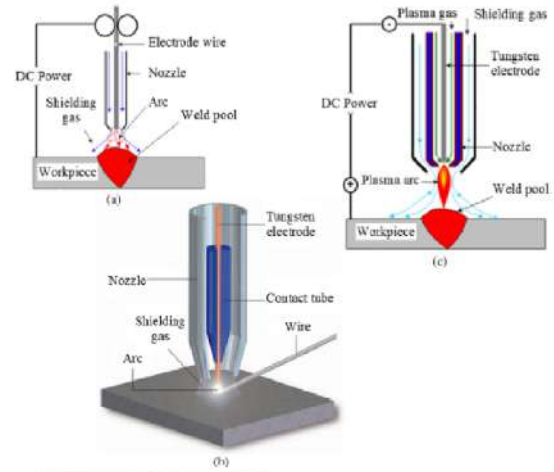


Fig. 22 a As deposited sample made by wire-feed LAM (AeroMet), b thick wall structure produced by hybrid layer manufacturing (HLM), c near-net shape of the die pair and d finished die pair produced by HLM [29]

Source: IFAM Fraunh

Figure 53: Wire-based DED

6.1 Comparison of Metal AM Processes (LPBF vs EBM)

- LPBF: high precision, high residual stress, inert gas atmosphere
- EBM: lower precision, lower residual stress, vacuum + preheating
- LPBF: finer powders (10–45 μm), better surface finish
- EBM: coarser powders (45–105 μm), faster build rates
- Trade-off: accuracy vs speed and residual stress

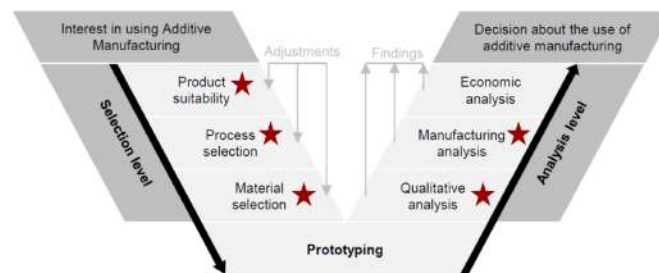


Figure 54: Systematical evaluation of AM processes for SME

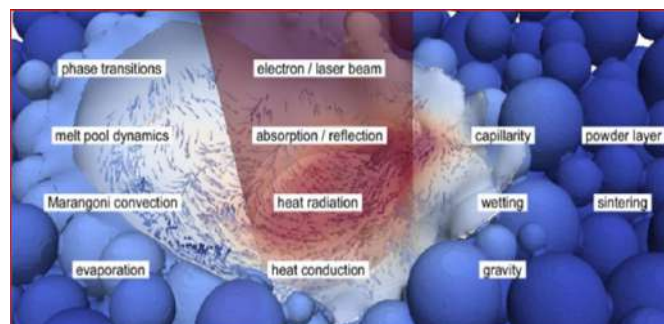


Figure 55: Laser Powder Bed Fusion

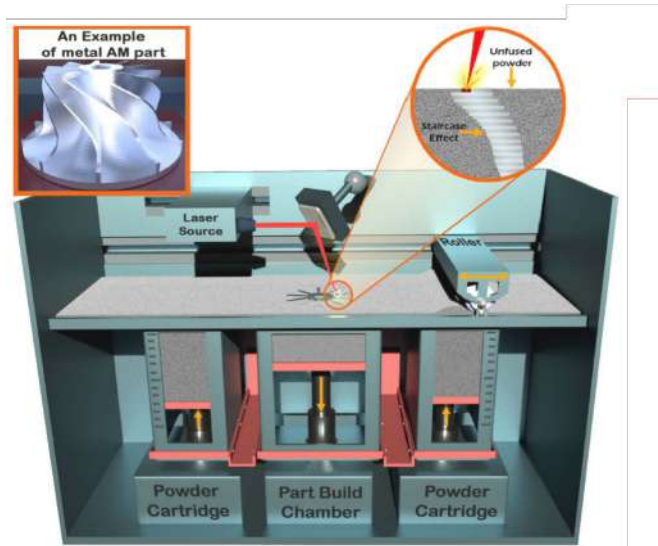


Figure 56: Laser Powder Bed Fusion

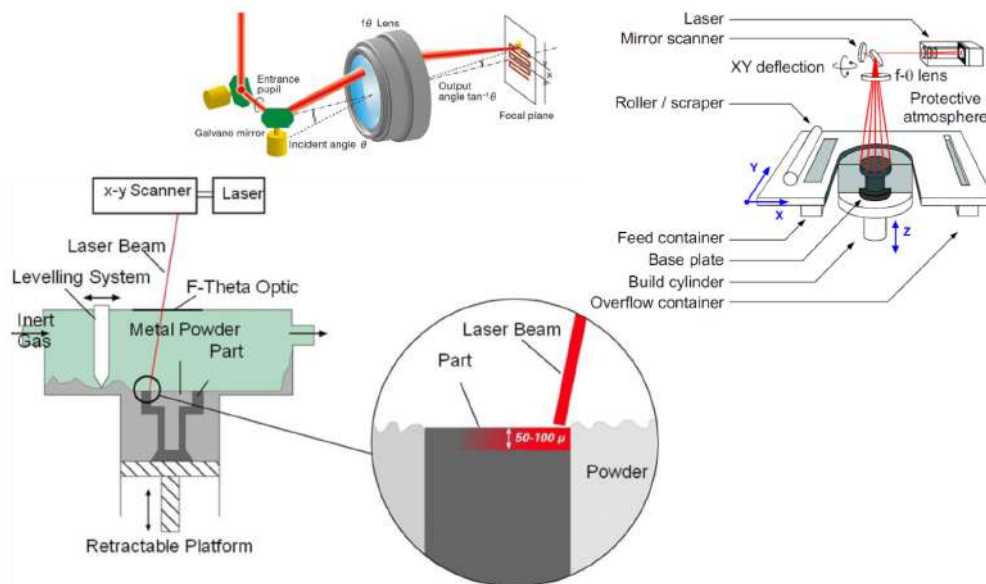


Figure 57: Process fundamentals for powder bed fusion

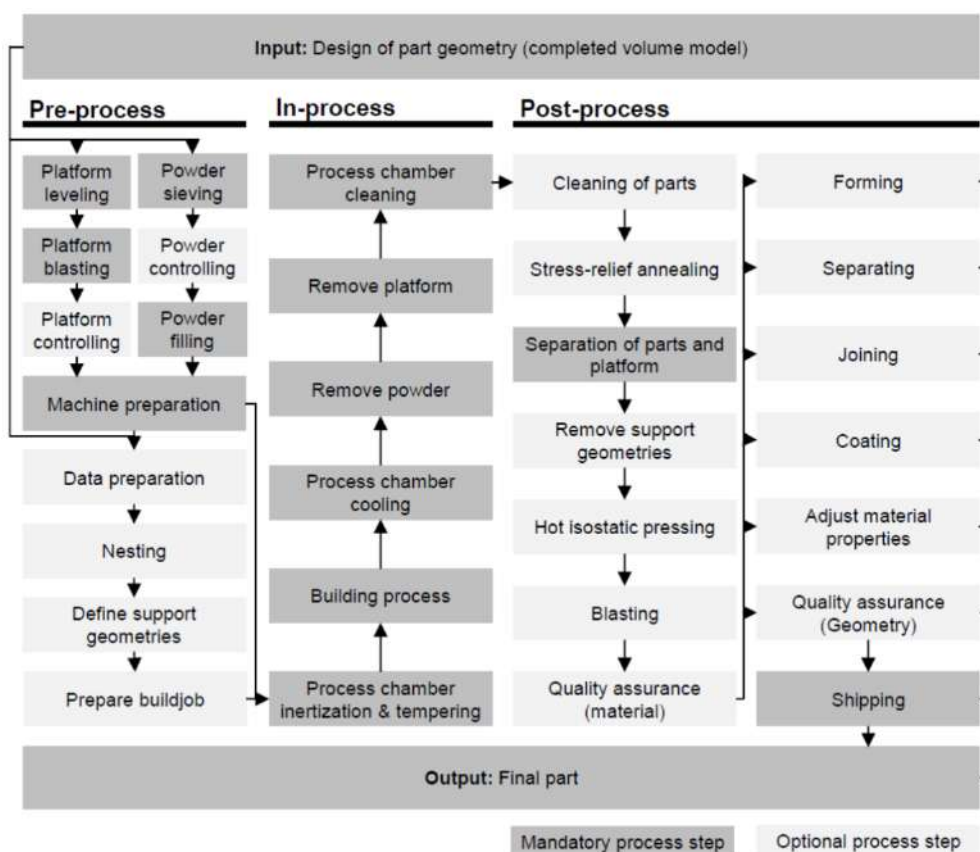


Figure 58: L-PBF Process chain

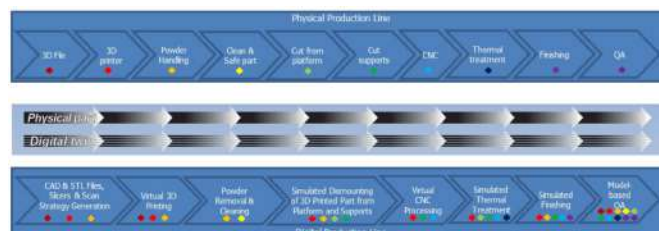


Figure 59: Typical PBF production line

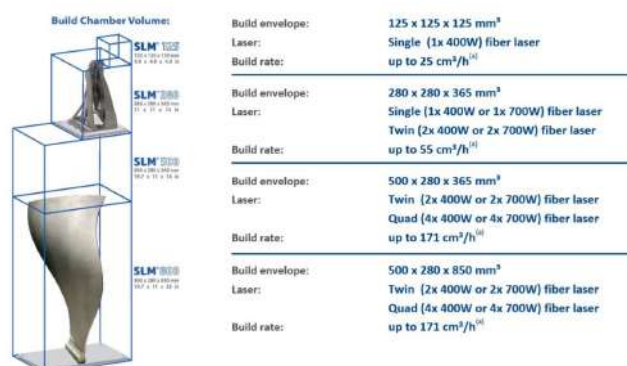


Figure 60: Build chamber aspects

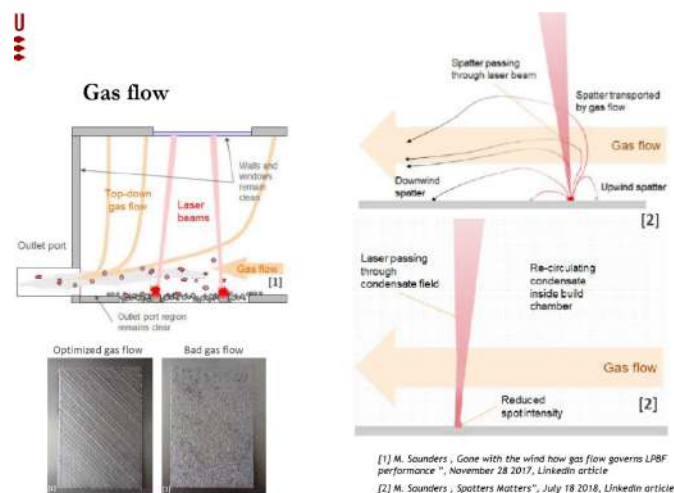


Figure 61: Gas flow



Figure 62: Particle shapes used in DEM simulations, each size and shape class are represented by a color circle, as shape varied with size, six blue circles (15-25 μm), eight purple circles (25-35 μm), five red circles (35-45 μm), and five green circles (45-55 μm)

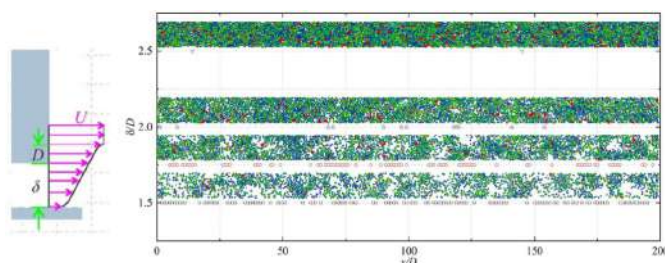


Figure 63: Powder morphology and powder bed formation



Figure 64: Roller/scrapper DEM simulations

Main process parameters

- Power (laser/electron beam)
- Scan speed
- Layer thickness
- Hatch distance
- Scan pattern
- Build chamber temperature

Relevant process variables

- Build job file repeatability
- Powder material quality
- Inert gas purity
- Ambient temperature stability
- Ambient humidity stability
- Recoating quality
- Build chamber temperature stability
- Build plate temperature stability
- Build plate thickness
- Inert gas temperature stability
- Continuity of inert gas flow
- Stability of inert gas level stability (build chamber)
- Stability of oxygen level (build chamber)
- Scanning stability
- Laser power stability (ramp-up/down)
- Machine firmware
- Machine cool-down stability
- Repeatability of powder removal
- Repeatability of stress relief
- Repeatability of part removal from the platform
- Repeatability of support removal
- Repeatability of blasting

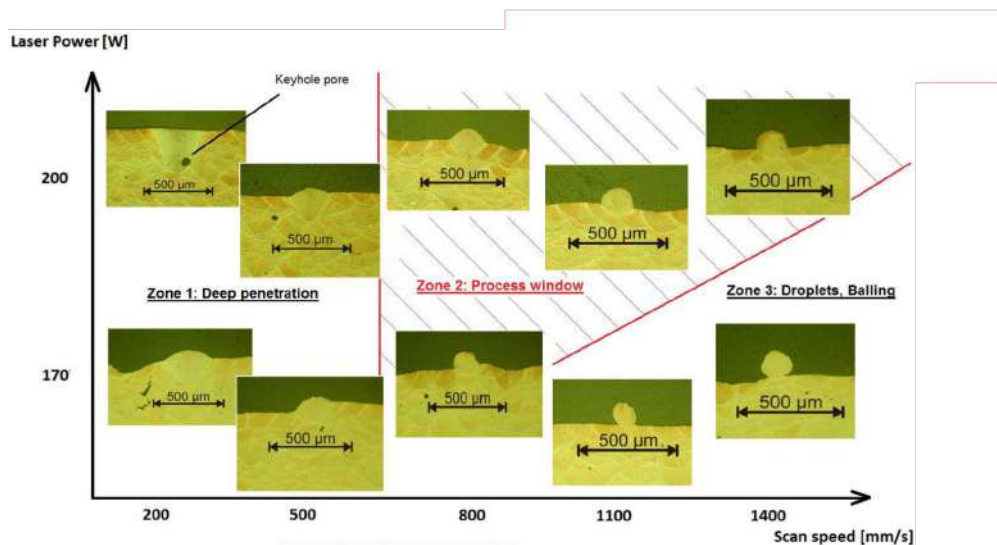


Figure 65: Process windows

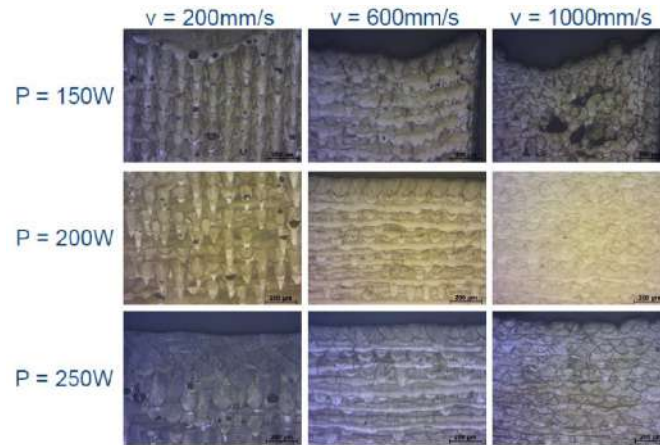


Figure 66: Process windows

	Application name	Hinge of engine hood
	Branch & Company	Automotive; EDAG Engineering, Voestalpine, Simufact
	AM benefits	Lightweight design (50% mass reduction)
	Material	Metal
	Part dimension	< 400x400x400 mm ³
	Part mass	600 g (estimated)
	Application type	☐ Serial application ☒ Prototype
	Application name	Mold segment for tires
	Branch & Company	Automotive; EOS
	AM benefits	Complex design only producible by AM
	Material	Steel
	Part dimension	< 400x400x400 mm ³
	Part mass	500 g (estimated)
	Application type	☒ Serial application ☐ Prototype

Figure 67: Example of parts produced with L-PBF

	Application name	Piston
	Branch & Company	Automotive; HardMarque Future Factories, Altair
	AM benefits	Lightweight design, Topology optimization
	Material	Titanium
	Part dimension	< 100x100x100 mm ³
	Part mass	300 g (estimated)
	Application type	☒ Serial application ☐ Prototype
	Application name	Reactor vessel
	Branch & Company	Industrial application; Thaitec GmbH, Jurec
	AM benefits	Cooling channels, Thin walls
	Material	Steel
	Part dimension (bounding box) in mm ³	D80 x 300 mm ³
	Part mass	4.7 kg
	Application type	☐ Serial application ☒ Prototype

Figure 68: Example of parts produced with L-PBF

6.2 Ishikawa (Fishbone) Diagrams in AM

Ishikawa diagrams are used to analyse the causal factors influencing a given outcome, particularly in quality control.

- Developed by Kaoru Ishikawa in the 1960s
- Represent cause–effect relationships
- Each causal factor is a source of variation and potential defect
- Used to identify and structure sources of process variability

In additive manufacturing, Ishikawa diagrams are used to:

- identify process parameters affecting part quality
- structure complex parameter interactions
- support process optimisation and control

Typical structure in L-PBF Main branches:

- Part design (geometry-related factors)
- Process parameters (machine and process-related factors)

6.3 Classification of L-PBF Parameters

Process parameters in L-PBF can be classified into:

- **Controllable parameters:** directly adjustable during the process
- **Predefined parameters:** material or system-dependent properties

These parameters influence:

- **Process signatures** (observable and derived quantities)
- **Product qualities:**
 - Geometric
 - Mechanical
 - Physical

6.4 Typical Application Requirements

- **Mechanical:** tensile strength, impact strength, hardness, ductility, fatigue, corrosion resistance
- **Thermal:** thermal conductivity, coefficient of thermal expansion (CTE)
- **Electrical:** electrical conductivity
- **Chemical:** chemical resistance
- **Surface:** surface roughness, tribology
- **Dimensional:** dimensional accuracy

6.5 L-PBF Process Parameters

6.5.1 Laser and Scanning Parameters

- Laser power P
- Peak power and pulse characteristics
- Wavelength λ and polarization
- Beam quality (M^2)
- Spot size (d_x, d_y)

- Scan velocity v
- Scan spacing s
- Scan strategy (hatching, zig-zag, etc.)

6.5.2 Material Properties

- Density
- Thermal conductivity
- Heat capacity
- Latent heat of fusion
- Melting and boiling temperature
- Melt pool viscosity
- Surface energy

6.5.3 Powder Bed and Recoating Parameters

- Powder bed density
- Absorptivity and emissivity
- Particle size distribution and morphology
- Layer thickness
- Recoater parameters (velocity, pressure, type)
- Powder bed temperature

6.5.4 Build Environment Parameters

- Shield gas type (Ar, N₂, He)
- Oxygen level (critical for oxidation)
- Gas flow velocity
- Chamber pressure
- Ambient temperature
- Convective heat transfer

6.6 Defects and Process Parameter Correlations

Defects in L-PBF are strongly linked to process parameters.

Defect formation mechanisms Defects are not independent but often occur as a chain of events. For example:

- Low energy density leads to lack of fusion
- Lack of fusion results in unmolten powder regions
- Poor wetting promotes balling
- Balling increases surface roughness and porosity

6.6.1 Predefined Parameters

- Powder shape → flowability, packing density, porosity
- Powder size → balling, density, oxidation, strength
- Powder size distribution → packing density
- Powder condition → heterogeneity, porosity, oxidation
- Powder density → porosity

6.6.2 Environment Parameters

- Gas flow → dimensional accuracy, bonding between layers
- Oxygen content → oxidation, balling
- Chamber temperature → shrinkage, deformation

6.6.3 Controllable Parameters

- Laser power → porosity, balling, warping, melt pool geometry
- Scan speed → roughness, porosity, cracks, layer distortion
- Hatch spacing → density, roughness, heterogeneity
- Scan pattern → anisotropy, porosity
- Layer thickness → staircase effect, bonding, accuracy
- Substrate temperature → distortion, warping

6.7 Defects in parts produced by powder bed fusion

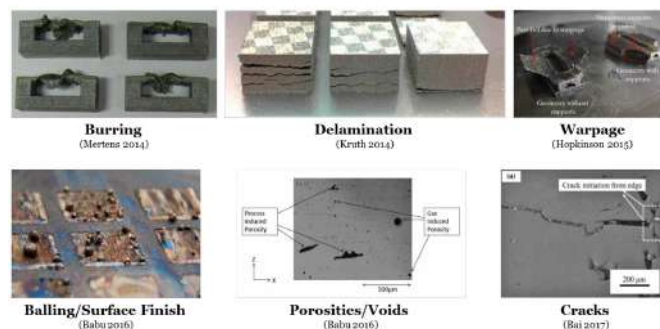


Figure 69: Challenges with metal additive manufacturing

Types of defects

- Powder bed defects
- Consolidation defects
- Microstructural defects
- (Macro-)structural defects

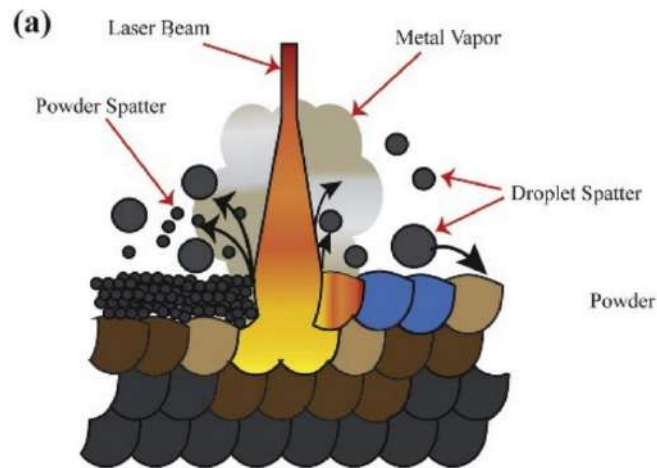


Figure 70: Spatter formation in selective laser melting process using multi laser technology

Defect classification Defects in L-PBF can be grouped as:

- Powder bed defects
- Consolidation defects
- Microstructural defects
- Macrostructural defects

Based on characteristics/type defects can be divided into four major categories,

- **Geometry**
 - Geometry inaccuracy (from dimensional deviations)
 - Dimensional inaccuracy (size dimensional deviation)
- **Surface quality (finishing)**
 - Surface roughness and surface morphology
 - Balling
 - Surface deformation
 - Surface oxidation
- **Microstructure**
 - Anisotropy
 - Heterogeneity
 - Porosity (poor density)

- **Mechanical properties**
 - Fracture/cracks/holes
 - Bonding between layers
 - Porosity
 - Low strength

Types of porosity

- Gas porosity (spherical): caused by trapped gas or excessive energy density
- Lack-of-fusion porosity (irregular): caused by insufficient melting

Based on when they appear they could then be,

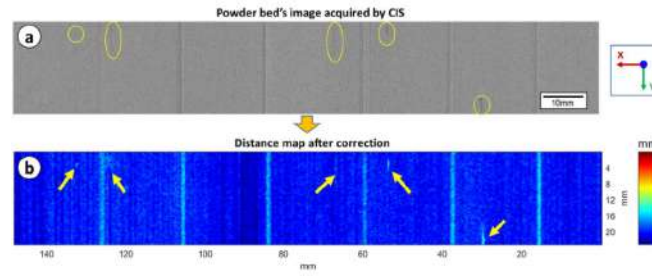
- Powder bed defects
- Consolidation defects
- Microstructural defects
- Macrostructural

No.	Process parameter	Affected items
First category: predefined parameters		
1	Powder shape	Surface contamination, flowability (smooth deposition), powder packing density, apparent density, porosity, powder fluidity
2	Powder size	Flowability of bulk powder, balling, packing density, apparent density, material strength, oxidation, layer thickness
3	Powder size distribution	Powder packing density and apparent density
4	Powder condition / drying treatment	Balling, heterogeneity, porosity, oxidation
5	Powder packing density / apparent density	Porosity, density
6	Gas flow (rate and direction)	Dimensional tolerance, bonding between layers
7	Building direction (layer orientation)	Dimensional inaccuracy, anisotropy, porosity, mechanical properties
8	Spot diameter and beam radius	Dimensional inaccuracy, surface roughness, porosity, penetration depth, energy density, energy absorption
9	Laser mode (pulse / continuous)	Porosity (density), balling (wetting angle)
10	Laser pulse length / pulse ratio	Surface roughness, bonding between layers
Second category: predefined parameters to be monitored		
11	Chamber temperature and powder bed temperature (workpiece temperature)	Shrinkage, heterogeneity, surface deformation, absorptivity
12	Gas flow rate	Dimensional inaccuracy, porosity, mechanical properties, bonding between layers
13	Contained gas of chamber	Balling, oxidation
14	Percolation of contained gas through powder layer	Surface oxidation
Third category: parameters to be monitored and controlled		
15	Laser power	Shrinkage, balling (energy density), warping, layer distortion, porosity, melt pool size and morphology, penetration depth
16	Scanning speed	Shrinkage, surface roughness, balling (energy density, poor wetting), layer distortion, porosity, melt pool size and morphology, fractures/cracks/holes
17	Laser positioning error	Geometry inaccuracy
18	Hatch spacing	Shrinkage, surface roughness, energy density, heterogeneity, porosity, melt pool size and morphology, dimensional accuracy (microstructural waviness), surface deformation
19	Scan length	Shrinkage, warping, fractures/cracks/holes
20	Scan pattern / scan direction	Surface roughness, porosity/density, heterogeneity, anisotropy
21	Platform movement error	Geometry inaccuracy
22	Layer thickness / resolution	Staircase effect, shrinkage, energy density, porosity/density, bonding between layers/penetration depth, construction speed
23	Blade edge or surface of rolling	Pit surface, powder deposition/distribution
24	Substrate temperature	Balling, distortion and warpage, poor wetting, melt pool shape

Table 4: Correlation between process parameters and defects in L-PBF across predefined, monitored, and controlled categories.

Typical powder bed defects

- Lines due to blade/roller defects
- Empty patches due to poor powder spreading behavior
- Empty spots due to "obstructions"
- Non-uniformity due to random effects

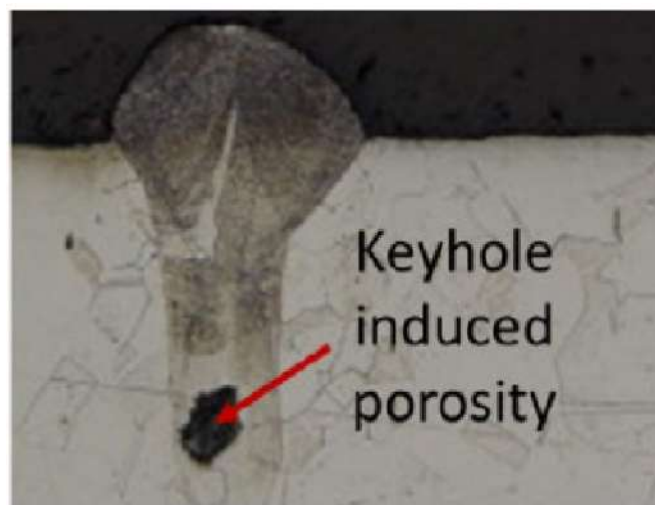
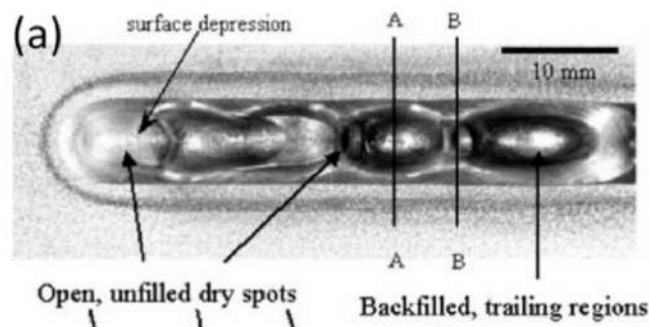


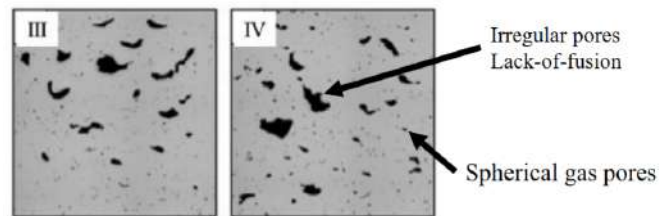
$$\Delta V_{ED} = \frac{P}{s\sigma\Delta t}$$

Where ΔV_{ED} represents the variation of the volumetric energy density (in J/m^3), P (in W) the power of the high energy beam, s (in m/s) the beam scanning speed, σ (in m) the hatch spacing, and Δt (in m) the non-uniformity of the powder layer thickness caused by PBDs.

Consolidation defects

- Porosities
- Poor surface roughness from balling effect (escalated by subsequent powder bed defects)
- Poor internal structure roughness from dross formation at overhanging surfaces





Build speed Vs Part density: Effect of scan speed and hatch spacing

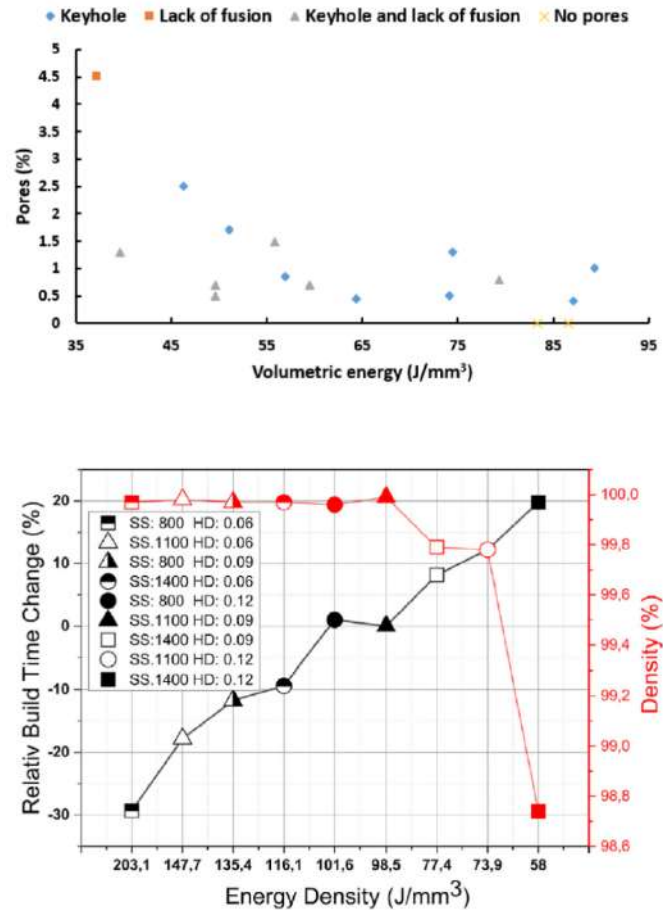


Figure 71: graph of relative build time and density as function of energy density. The process parameters are provided in the legend as a combination of scan speeds (SS in mm/s) and hatch distance (HD in mm)

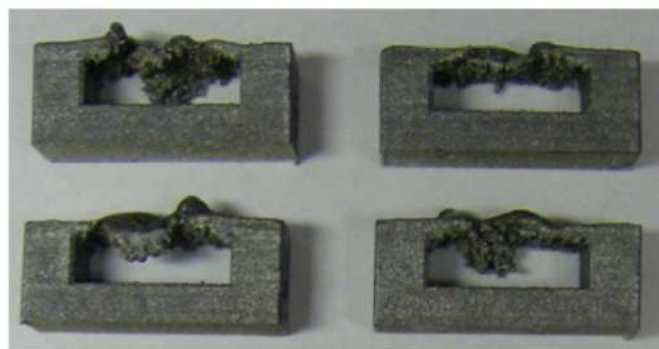


Figure 72: Poor internal structure surface roughness

Microstructural defects

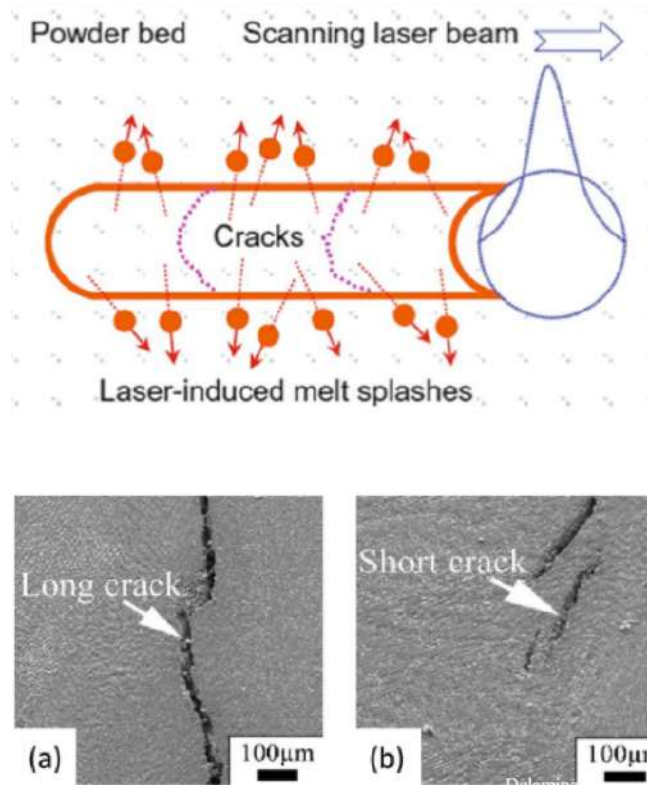


Figure 73: Micro-mechanical defects

These can be caused by,

- Hot tearing/Hot cracking
- Hot spots/Stress cracks
- Localized oxidation/inclusion

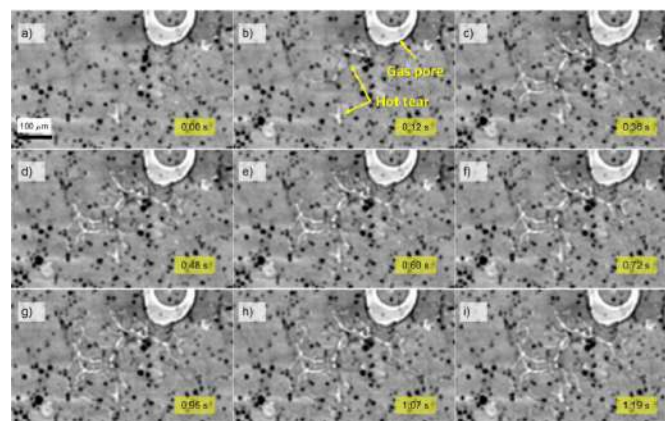
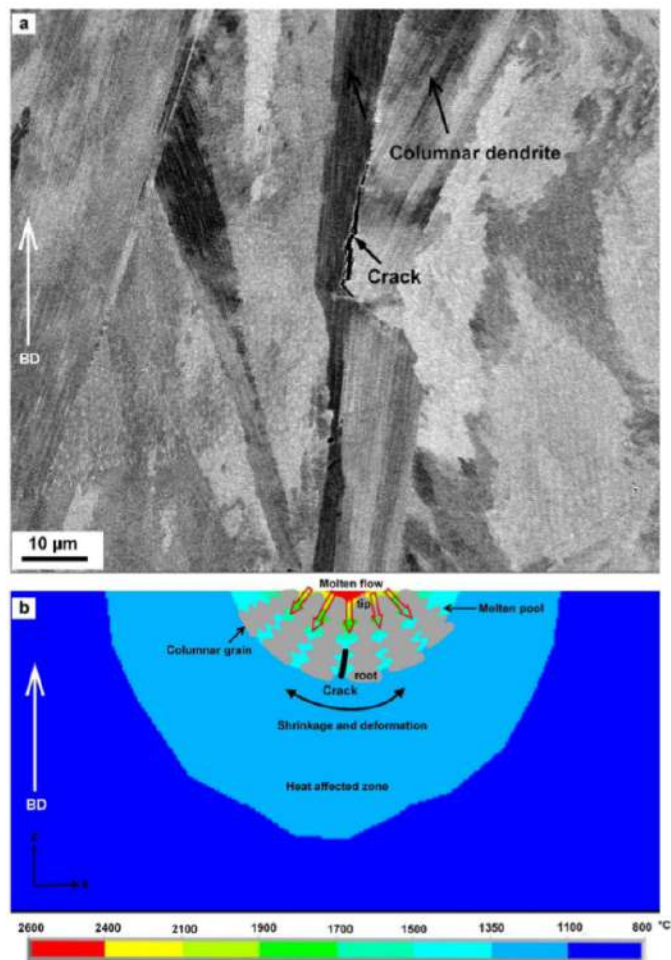
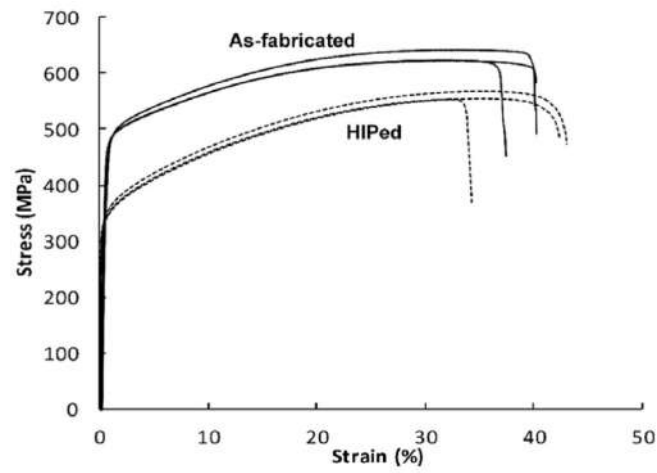


Figure 74: Hot tearing

Hot cracking during metal AM



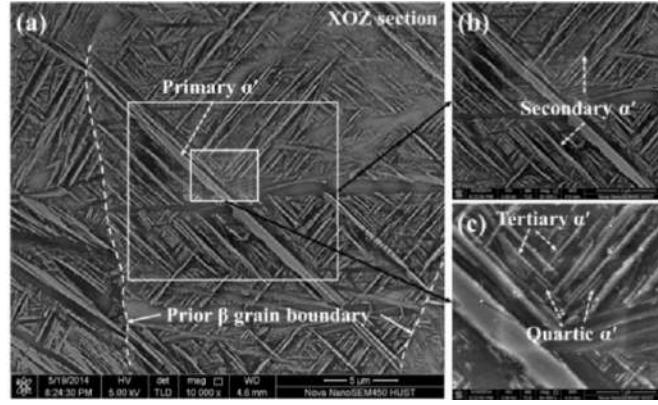


Figure 75: Metallurgical defects

Defect Criteria in SLM / L-PBF

Phenomenon	Criteria	Notes
Microcracks	STR minimisation	Reduces formation of low melting point eutectics
	PI maximisation	Defined as thermal shock resistance
	Presence of δ -ferrite during solidification	Alleviates detrimental effects of impurity elements (relevant for steels)
Lack of fusion	$D/t > 1.5$	Higher ratios improve bonding between layers
	$h/W < 1$	Alternative criterion: $(h/W)^2 + (t/D)^2 < 1$
Keyholes	$W/D > 2$	Lower values may still be acceptable depending on process conditions
	$\Delta H/h_s < 7$	Threshold may vary significantly with material and energy input
Balling	$L/W < 2.3$	Indicates stable melt pool geometry

Table 5: Summary of defect formation criteria in SLM / L-PBF.

Notation

- D : melt pool depth
- W : melt pool width
- L : melt pool length
- h : hatch spacing
- t : layer thickness
- STR : solidification temperature range

$$STR = T_{\text{liquidus}} - T_{\text{solidus}}$$

- PI : material performance index
- ΔH : enthalpy
- h_s : enthalpy of melting
- $\Delta H/h_s$: normalized enthalpy
- A : Powder absorptivity
- P : laser power

- d : thermal diffusivity
- v : Laser scan speed
- σ : Laser spot size (area covered by the laser)
- T_b : Boiling temperature
- T_m : Melting temperature

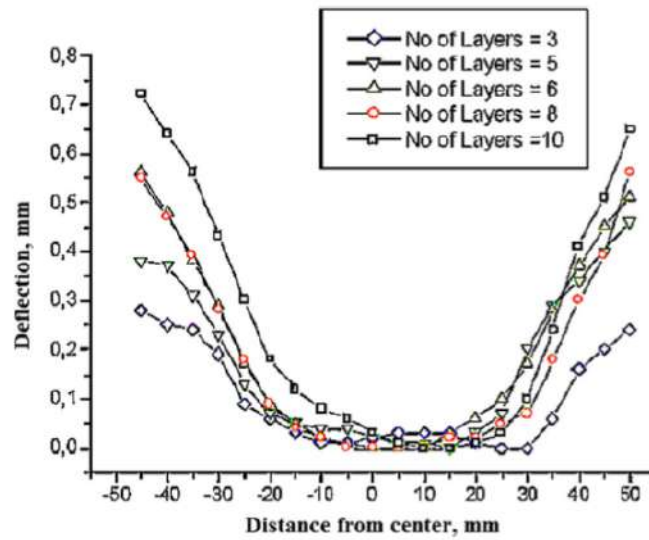
The normalized enthalpy can be calculated via the following equation,

$$\frac{\Delta H}{h_s} = \frac{AP}{h_s \sqrt{\pi d v \sigma^3}}$$

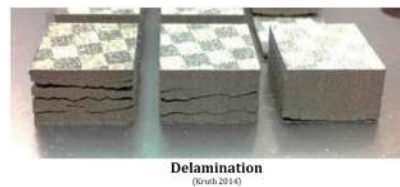
An estimate for keyhole threshold in terms of normalized enthalpy is,

$$\frac{\Delta H}{h_s} > \frac{\pi T_b}{T_m}$$

Macro-structural defects



- Delamination
- Warpage



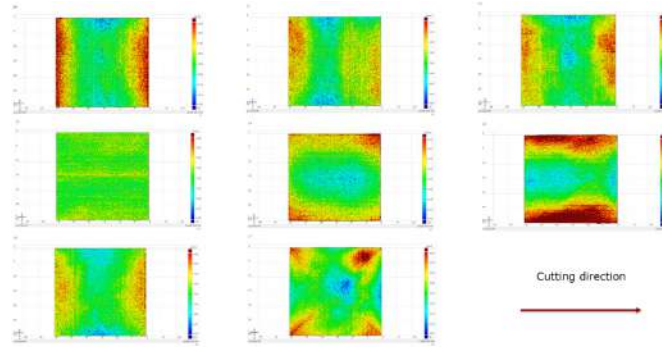


Figure 76: Different warpage in same components produced by different scanpaths

6.7.1 Typical post-processing to remove/reduce defects

- Heat treatment
- Hot isostatic pressing
- Machining, Grinding, Cutting, EDM
- Support and Powder removal
- Blasting and surface treatments
- Sanding, polishing, honing
- Etching, electro-polishing, plating

Purpose of heat treatment Heat treatment is used to:

- Reduce residual stresses
- Modify microstructure
- Improve mechanical properties

Purpose of hot isostatic pressing Hot isostatic pressing is used to:

- Remove internal porosity
- Increase density
- Improve fatigue performance

Purpose of machining, grinding, cutting, EDM These processes are used to:

- Achieve final dimensions and tolerances
- Improve surface finish
- Remove excess material or supports

Purpose of support and powder removal Support and powder removal is used to:

- Eliminate unused powder from internal cavities
- Remove support structures
- Prepare the part for further processing

Purpose of blasting and surface treatments Blasting and surface treatments are used to:

- Clean the surface
- Remove loosely attached particles
- Improve surface uniformity

Purpose of sanding, polishing, honing These processes are used to:

- Reduce surface roughness
- Improve surface quality
- Achieve tighter surface tolerances

Purpose of etching, electro-polishing, plating These processes are used to:

- Modify surface properties
- Improve corrosion resistance
- Enhance surface finish or appearance

6.8 Impact of defects on mechanical properties

Failure mechanisms Defects can lead to failure through:

- Fatigue crack initiation (most critical)
- Surface crack propagation
- Multiple failure modes originating from a single defect

Porous structures have lower strength,

$$\frac{E}{E_s} = \left(\frac{\rho}{\rho_s} \right)^2$$

Where,

- E_s is the Young's modulus of the solid (bulk) material
- E is the Young's modulus of the porous material
- ρ_s is the density of the solid (bulk) material
- ρ is the density of the porous material

However controlled/designed porosity is not a defect.

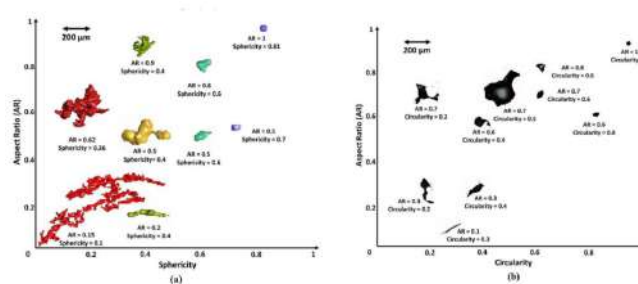


Figure 77: Shapes of Pores

The aspect ratio determines the equivalent crack depth for fatigue behavior.

Fracture mechanics is a failure theory that,

1. Determines material failure by energy criteria, possibly in conjunction with strength (or yield) criteria
2. Considers failure to be propagating throughout the structure rather than simultaneous throughout the entire failure zone or surface

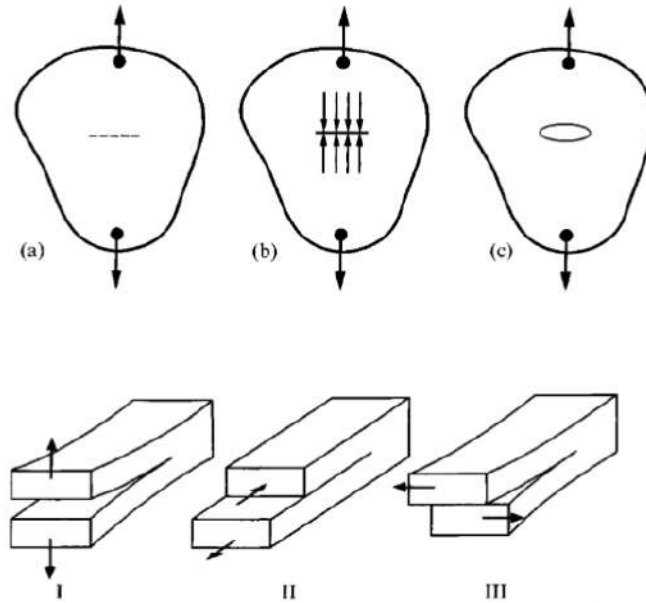


Figure 78: Enter Caption

- Linear elastic fracture mechanics (LEFM) is the basic theory of fracture, that deals with sharp cracks in elastic bodies
- Elastic fracture mechanics is the theory of ductile fracture, usually characterized by stable crack growth

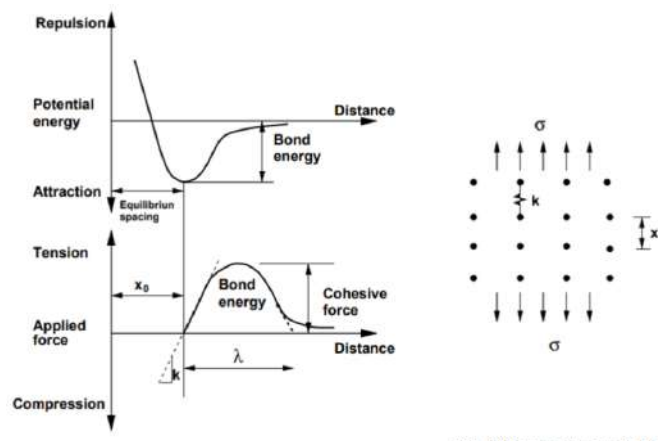


Figure 79: Crack initiation and Growth at pores

Theoretical cohesive strength,

$$\sigma_c = \frac{E}{\pi}$$

Actual cohesive strength,

$$\sigma_c = \sqrt{\frac{E\gamma_s}{a}}$$

- γ_s : Surface energy
- a : Crack length

Flaws must lower the global strength by magnifying the strength locally

6.8.1 Stress intensity factor

Stress Intensity Factor K

Stress Intensity Factor (K) = $\frac{\text{Peak stress}}{\text{Nominal stress}}$, a measure of stress concentration at a location

$$\frac{da}{dN} = C(\Delta K)^m, \quad (\Delta K_{th} < \Delta K < K_{Ic})$$

- a : Crack length
- da/dN : Fatigue crack growth for a load cycle N
- C : is an experimentally determined material constant which depend on environmental effects, stress ratio and characteristics specimen size
- m : is an experimentally determined material constant which depend on environmental effects, stress ratio and characteristics specimen size
- ΔK : is the difference between the maximum and minimum stress intensity factors for each load cycle, defined as,

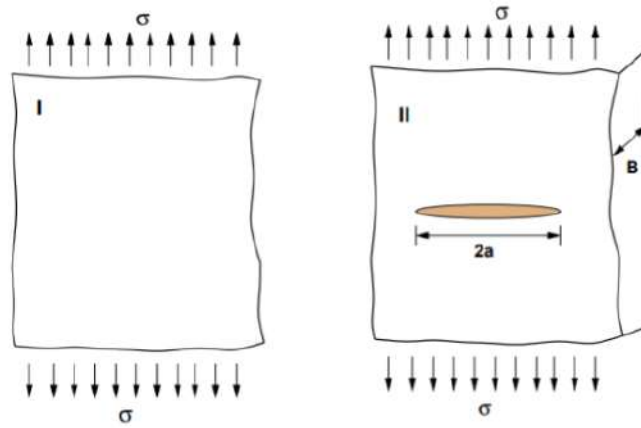
$$\Delta K = K_{\max} - K_{\min}$$

6.8.2 Griffith's criteria

The critical condition for the fracture can be defined as the point at which crack growth under equilibrium conditions,

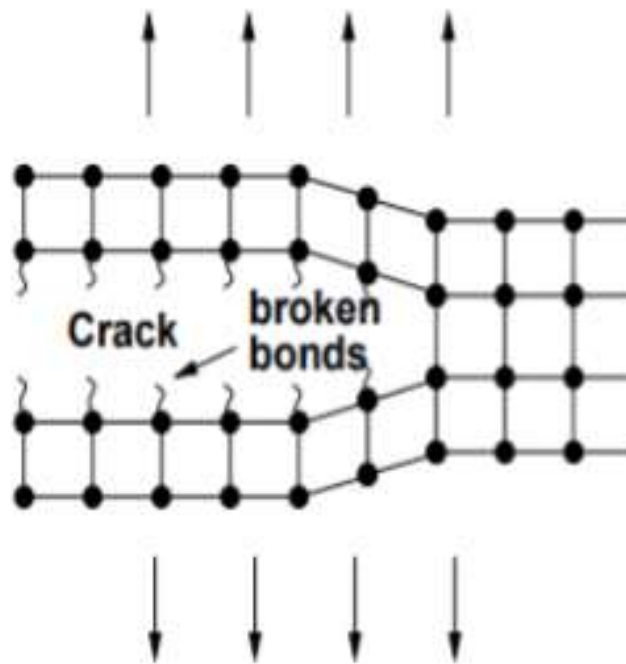
$$\frac{dE}{dA} = \frac{d\Pi}{dA} + \frac{dW_s}{dA} = 0$$

- E : Energy at crack
- A : Surface area of crack
- Π : Potential energy
- W_s : Energy from the applied force (external)



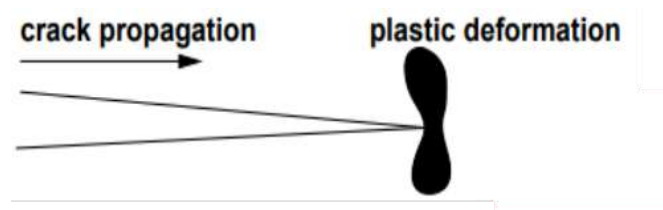
$$E - E_0 = \frac{\pi \sigma^2 a^2 B}{E} + 4aB\gamma_s$$

$$\sigma_f = \sqrt{\frac{2E\gamma_s}{\pi a}}$$



The part seen above will fail catastrophically if the local stress at the crack tip reaches σ_f , where,

$$\sigma_f = \sqrt{\frac{2Ew_f}{\pi a}}$$



Where quasi-brittle elastic-plastic material,

$$w_f = \gamma_s + \gamma_p$$

For brittle material with crack meandering and branching,

$$w_f = \gamma_s \left(\frac{\text{true area}}{\text{projected area}} \right)$$

6.8.3 Crack Initiation at Pores and Stress Intensity Factor

- Initial equivalent crack depth:

$$a_i = \sqrt{\text{area}_i} \sqrt{\frac{2}{\pi}} \quad (16)$$

- Stress intensity factor range:

$$\Delta K = Y \Delta \sigma \sqrt{\pi \sqrt{\text{area}}} \quad (17)$$

- Shape/location factor:

- $Y = 0.5$ for internal defects
- $Y = 0.65$ for surface defects

- Surface defects are more critical than internal defects of the same size.

- For equal SIF:

$$\frac{\text{area}_{\text{int}}}{\text{area}_{\text{surf}}} = \left(\frac{0.65}{0.5} \right)^4 \approx 2.86 \quad (18)$$

- Therefore, an internal defect must be about 2.9 times larger in area than a surface defect to produce the same SIF.
- More defects increase the probability of crack initiation.

6.9 Materials: Possibilities, Constraints and Properties

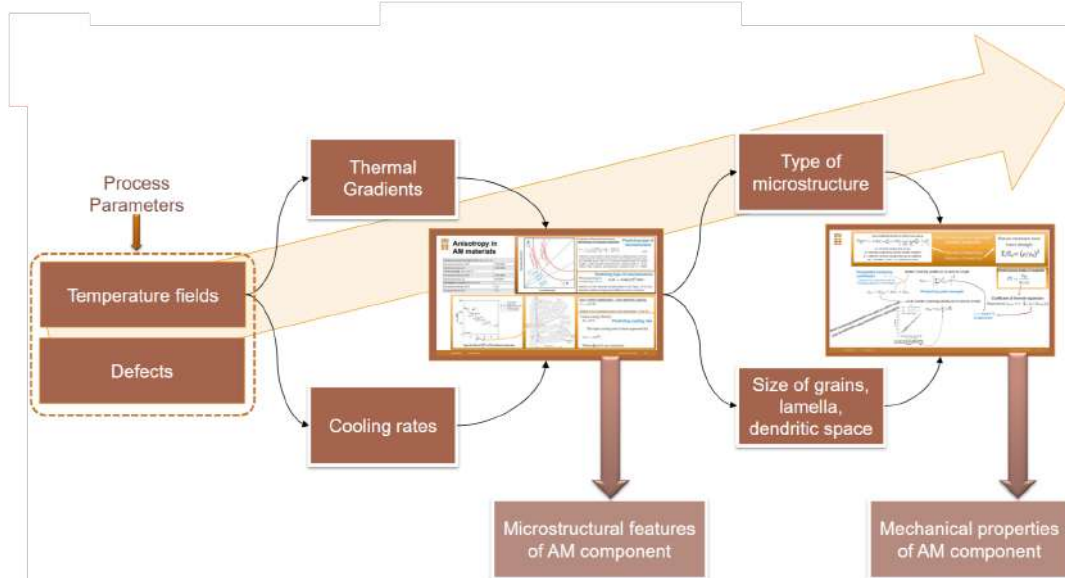


Figure 80: Properties-Microstructure-Properties establishing correlations

6.9.1 Understanding Metal Microstructure

Crystal structures Metals are composed of atoms arranged in periodic crystal lattices. The most common structures are:

- **Face-Centered Cubic (FCC)**: high packing density, high ductility (e.g. Al, Cu)
- **Body-Centered Cubic (BCC)**: lower packing density, higher strength, less ductile (e.g. Fe at room temperature)
- **Hexagonal Close-Packed (HCP)**: limited slip systems, lower ductility (e.g. Ti, Mg)

The crystal structure strongly influences mechanical properties such as strength and ductility.

Point defects Real crystals contain imperfections:

- **Vacancies**: missing atoms in the lattice
- **Interstitial defects**: atoms located between lattice sites
- **Substitutional defects**: foreign atoms replacing host atoms

These defects affect diffusion, strength and material behaviour.

Line defects (dislocations) Dislocations are linear defects responsible for plastic deformation:

- **Edge dislocation**: extra half-plane of atoms
- **Screw dislocation**: spiral arrangement of atoms

Plastic deformation occurs through dislocation motion under applied stress.

Elastic vs plastic deformation

- **Elastic deformation**: reversible deformation
- **Plastic deformation**: permanent deformation due to dislocation movement

Solidification and grain structure During solidification:

- Nucleation occurs at multiple sites
- Grains grow until they meet

This forms a **polycrystalline structure** with grains of different orientations.

Grain size influences:

- Strength (smaller grains $\rightarrow$ higher strength)
- Toughness
- Anisotropy

Nucleation and cooling effects Microstructure depends on:

- **Cooling rate:**
 - Fast cooling → fine grains
 - Slow cooling → coarse grains
- **Inoculants:**
 - Promote nucleation
 - Refine grain structure

In additive manufacturing, high cooling rates lead to fine microstructures and non-equilibrium phases.

6.9.2 Powder Bed Properties

Equivalent properties The effective properties of a powder bed depend on packing fraction:

$$k_{eq} = \emptyset^\beta k \quad (19)$$

Including gas effects (for dense powder beds > 0.9 packing fraction)

$$k_{eq} = \emptyset^\beta k + (1 - \emptyset)^\beta k_{gas} \quad (20)$$

Typical packing fractions

- Simple cubic: $\emptyset \approx 0.52$
- BCC: $\emptyset \approx 0.68$
- FCC: $\emptyset \approx 0.74$
- Random packing: $\emptyset \approx 0.55 - 0.64$

Key insight

- Powder beds have much lower thermal conductivity than bulk material
- Lower packing fraction → lower heat transfer
- Gas phase significantly reduces conductivity

Note More advanced models (e.g. Zehner–Schlünder–Damköhler) exist for less dense powder beds >0.7 packing fraction

$$\frac{k_{equivalent}}{k_f} = /1 - \sqrt{1 - \phi_0} \left(1 + \phi_0 \frac{k_r}{k_f} \right) + \sqrt{1 - \phi_0} \left(\frac{2}{1 - \frac{k_f}{k_s}} \left(\frac{1}{1 - \frac{k_f}{k_s}} \ln \left(\frac{k_s}{k_f} \right) - 1 \right) + \frac{k_r}{k_f} \right)$$

Where,

- k_f is the thermal conductivity of gas
- k_s is the thermal conductivity of bulk/powder metallic material
- k_r effective thermal conductivity du to radiation
- $\phi_0 = \text{porosity} = (1 - \emptyset) = (1 - \text{packing fraction})$

6.10 Characterization methods, powder, voids, microstructure and post-heat treatment

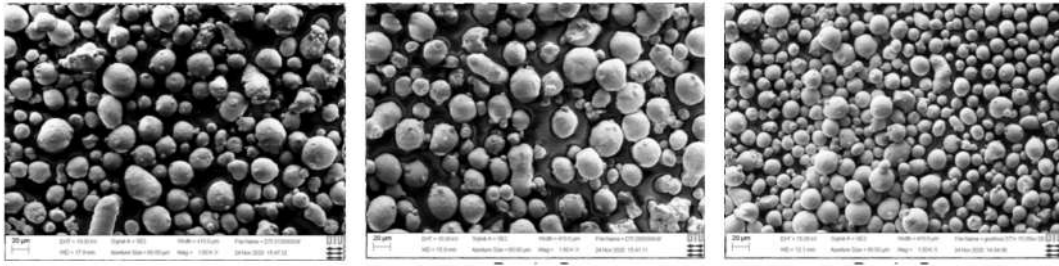


Figure 81: Comparison between nominally identical powders

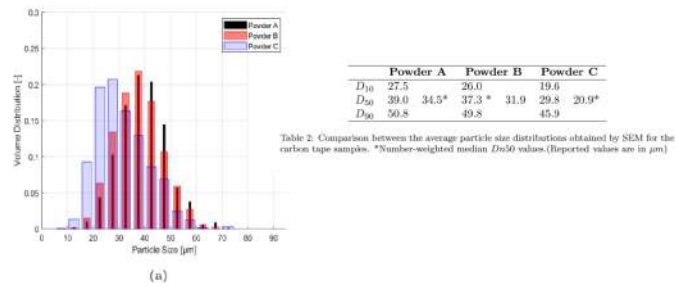


Figure 82: Comparison between nominally identical powders

6.10.1 Powder

Important parameters include,

- Size distribution
- Shape
- Surface roughness
- Interior pores

These parameters are characterized using,

- Sieve Analysis
- Laser diffraction
- SEM Quantitative and non-Quantitative
- X-Ray CT Scan

Laser Diffraction

Laser Diffraction Measurements



Figure 83: Laser Diffraction Measurements

SEM, two sample types used,

- Just pour the powder on tape sitting on a stub, and blow air over it
- Blend 5-10% powder into a resin and hot solidify

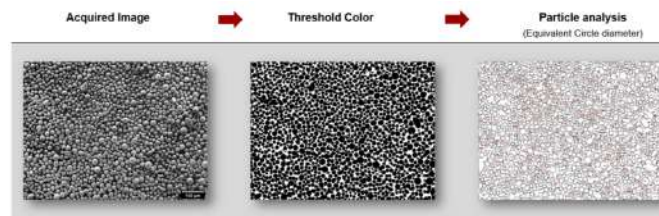


Figure 84: Analysis of SEM result (here powder on tape)

Some issues with this include, image quality, filtering, threshold and segmentation are key features for proper analysis.

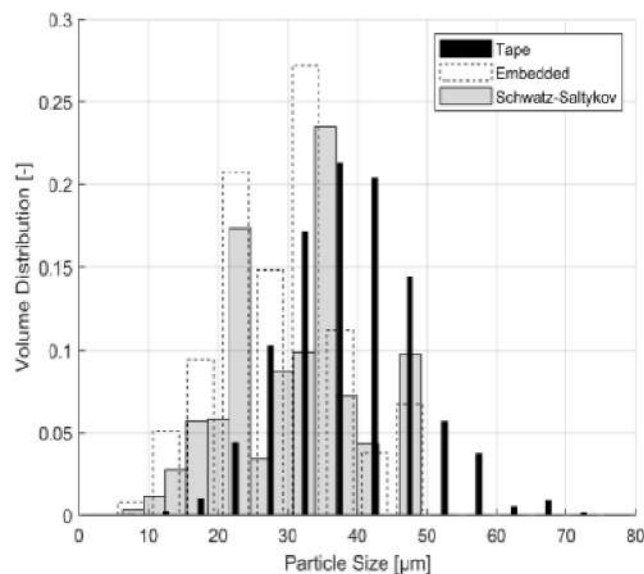


Figure 85: End result of particle distribution from different methods

X-Ray Absorption contrast tomography (XCT scanning), two sample types are used,

- Just pour the powder into a glass tube
- blend 5-10% powder into a resin and hot solidify

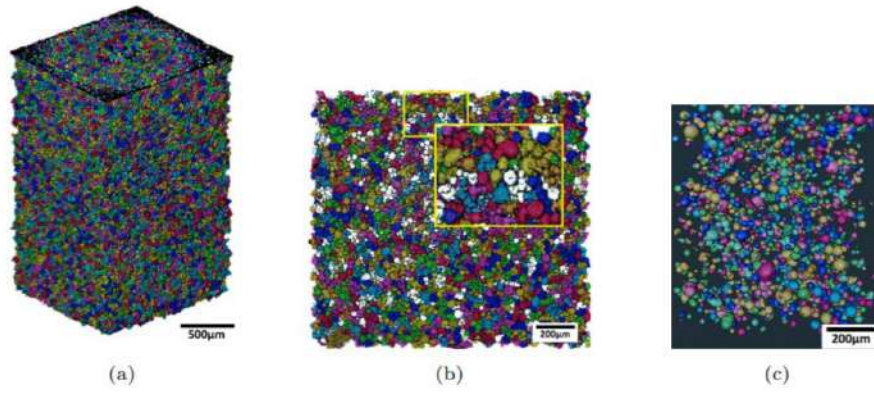


Figure 7: a) Reconstructed 3D volume of the powder A glass tube sample obtained from XCT b) Left: top-view of the 3D data, inaccurate segmentation of particles leading to overestimation of particle size. Right: side-view of embedded sample for powder A where the particles are well dispersed over the resin.

Figure 86: Analysis of XCT images from two types of samples

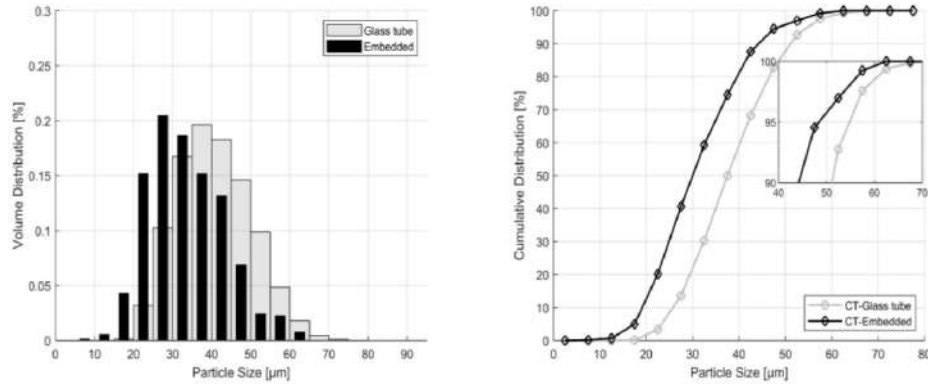


Figure 87: End result of particle distribution from different methods

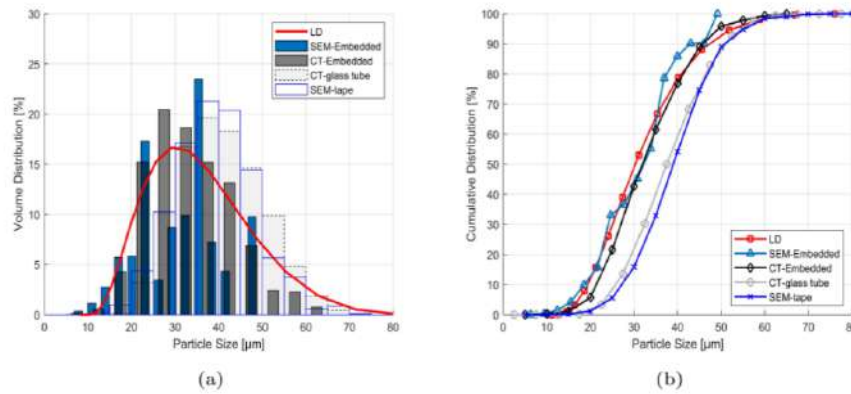


Figure 10: Particle size distribution for Powder sample A calculated with different characterization methods. Figure 9a plots the volume particle size distribution. Figure 9b displays the cumulative distribution obtained by the different methods discussed in this study. Schwartz-Saltykov 3D size distribution estimation from the embedded sample matches embedded and LD.

Figure 88: Comparison between methods from SEM and XCT

	Laser Diffraction (Dry)	SEM	X-Ray CT
Advantages			
	High reproducibility Fast analysis Simple sample preparation Compact instrument Standard procedures	Shape analysis Simple sample preparation Surface morphology Internal porosity (with preparation) Small amount of powder needed	Shape analysis High number of particles analyzed Non-destructive testing (NDT) Internal porosity analysis Density analysis Surface roughness evaluation 3D data generation
Limitations / Concerns			
	No shape analysis No internal porosity analysis	Time consuming Limited internal porosity detection Sensitive to threshold Small sample size	Time consuming Sample preparation required Sensitive to threshold

Table 6: Comparison of powder characterization methods.

Powder Particle analysis

- Several methods available, giving different results. Recommendation: stick to one method, but calibrate often and maybe against other methods
- Even nominally identical powders differ. Recommendation: check your powders on each delivery / each batch
- Even if the average particle size is similar, the size distribution and particle shapes may differ. Recommendation: check the parameters relevant for your printing

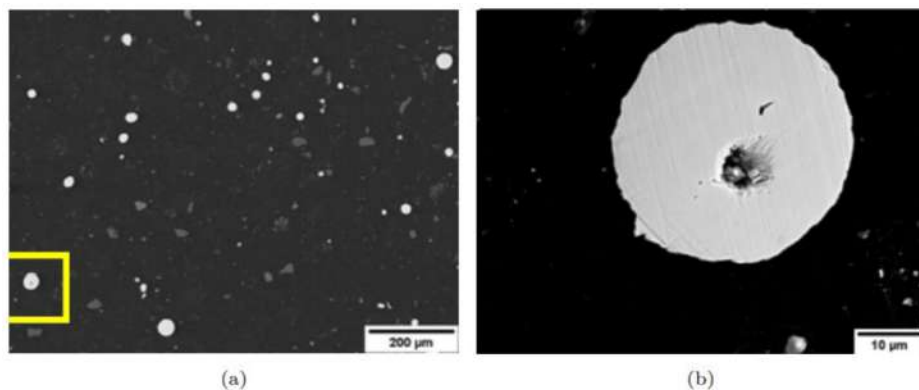


Figure 9: SEM secondary electron images showing powder A in the embedded sample. a) Micrograph at 100X showing an overview of the sectioned powder dispersed in the epoxy resin. b) At 1600X magnification, particle enclosed in yellow at a) shows internal porosity in one of the sliced particles.

Figure 89: SEM for voids

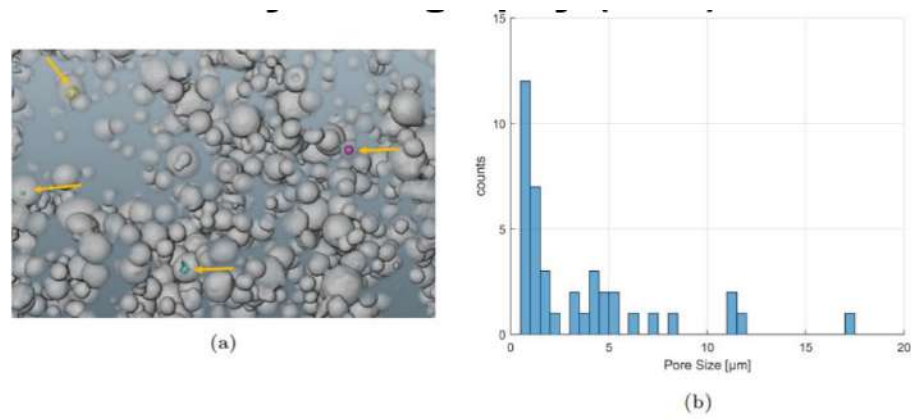


Figure 8: Porosity observed by XCT within the powder A particles in the Embedded sample
 a) 3D volume image of the powder with the reconstructed pores indicated by the yellow arrow and shown in colors; b) Size distribution (equivalent sphere diameter) of the reconstructed pores.

Figure 90: Absorption contrast X-ray tomography (XCT) for voids

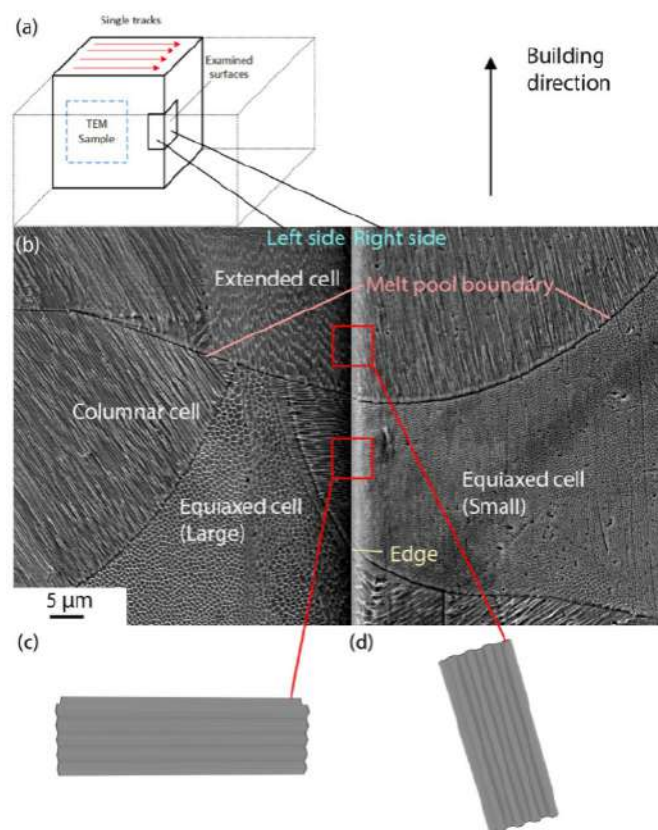


Table 1

Parameters used for LPBF fabrication of the specimens.

Specimen	Scanning strategy	Laser power /W	Laser speed /mm/s	Pitch /mm	Layer thickness /mm
Specimen A	X	250	1000	0.08	0.04
Specimen B	XY	350	1100	0.08	0.04
Specimen C	XY	200	1000	0.1	0.04

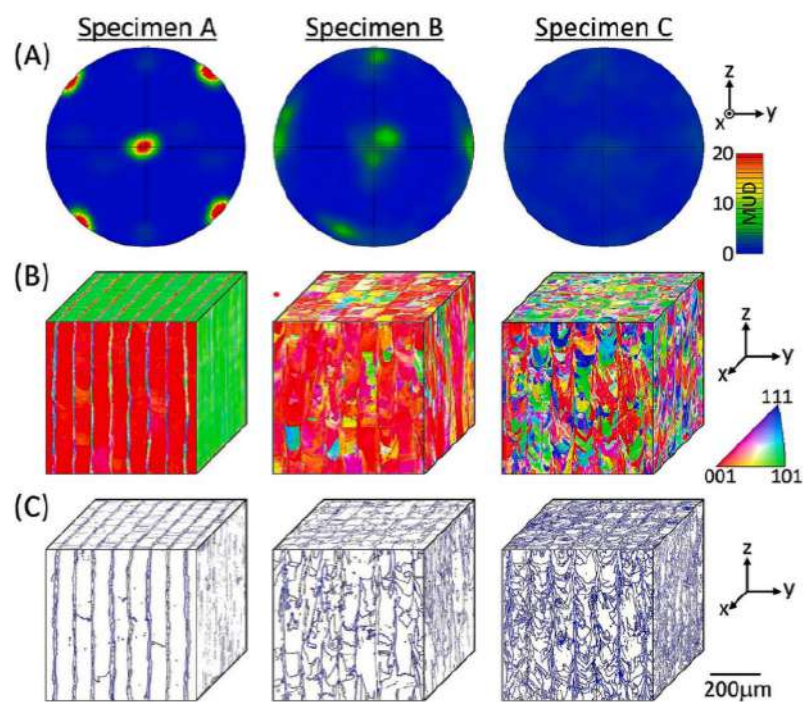
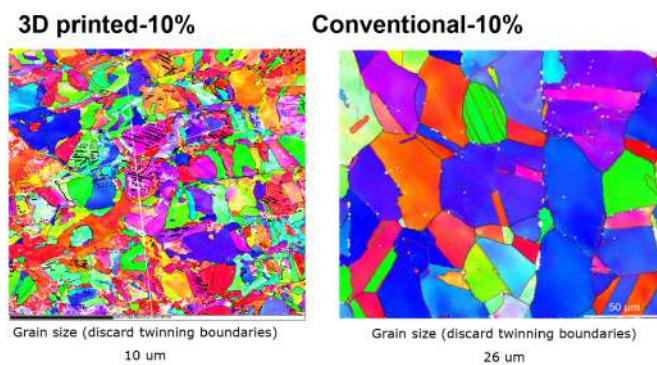


Fig. 1. Crystallographic texture and grain boundary distribution of the laser powder bed fusion (LPBF)-fabricated specimens. (A) {100} pole figures and (B) inverse pole figure maps for the yz-, xz-, and xy-planes, which were perpendicular to the x-, y-, and z-axes, respectively. (C) Maps for the grain boundaries with the misorientation angle of <15°. MUD: multiples of uniform density.



6.11 Solidification, microstructures and properties

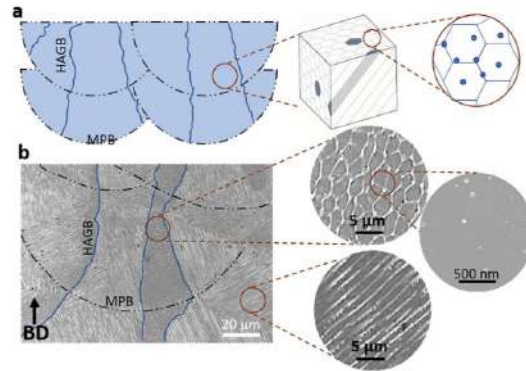


Fig. 4. Overview of the general microstructure showing a) schematic drawing of the melt pool boundaries (MPB), high angle grain boundaries (HAGB) that mainly show the columnar grain structure, cell structure in 3D (both hexagons and elongated sections), and nano-sized oxide particles, b) corresponding SEM micrograph of the general microstructural features in L-PBF 316 L.

Figure 91: 316L SS produced by PBF

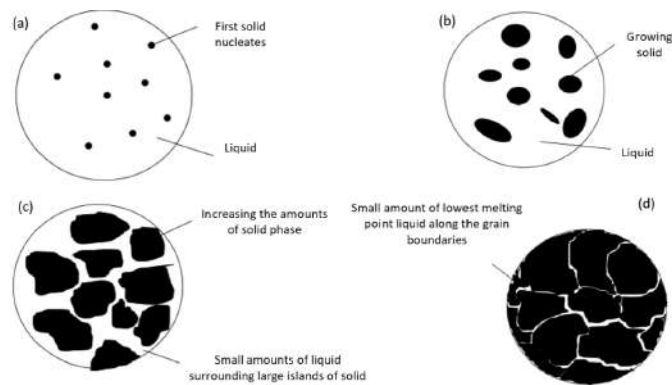


Figure 1. Schematic showing how an alloy solidifies. (a) The beginning of solidification (near liquidus temperature); (b) solid phase grows through the liquid phase; (c) the amount of solid increases by lowering temperature; (d) in the brittle temperature range, low amounts of liquid should remain along the grain boundaries of solid [32].

Source: 'Defect Prevention in Selective Laser Melting Components: Compositional and Process Effects' 2019

Figure 92: Solidification within a melt pool

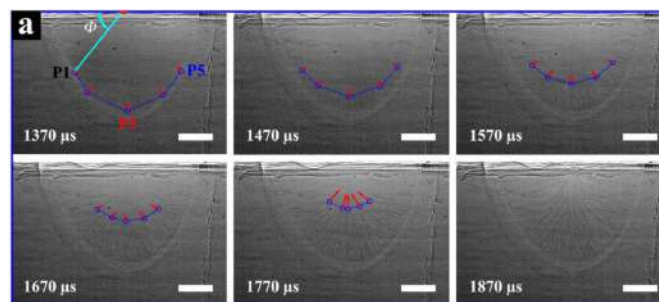


Figure 93: Solidification observed under real time X-ray imaging

6.11.1 Process – Signature – Product Relationship

Concept In metal additive manufacturing, part quality is determined through a chain:

Process Parameters → Process Signatures → Product Properties

- **Process parameters:** controllable inputs such as

- Laser power, scan speed, beam diameter
- Layer thickness
- Scanning pattern
- Inert gas flow
- **Process signatures:** physical phenomena occurring during the process
 - Melt pool temperature and thermal gradients
 - Melt pool geometry (width, depth, length)
 - Solidification conditions
- **Product properties:** resulting part characteristics
 - Geometric: dimensional accuracy
 - Mechanical: strength, hardness, fatigue resistance
 - Physical: residual stresses, porosity, surface roughness

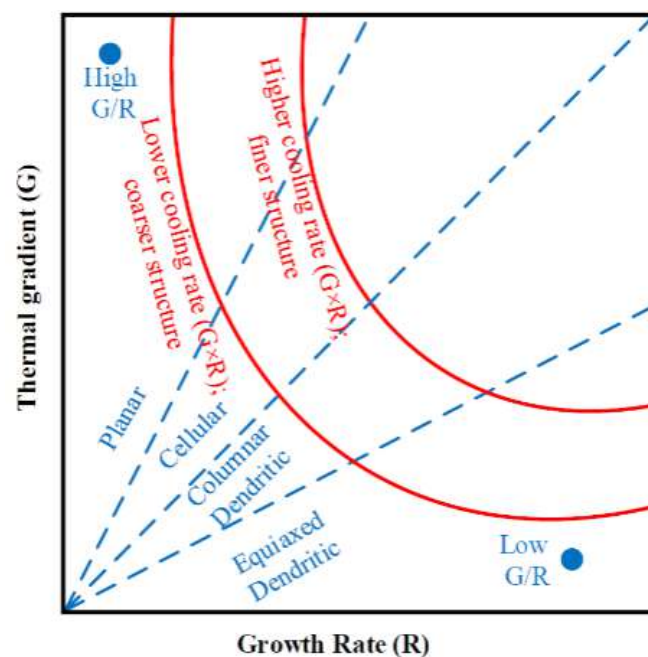
Key idea Process parameters do not directly determine part properties. Instead, they influence **process signatures**, which then determine the final product quality.

Melt pool and solidification relation The ratio between thermal gradient and solidification velocity is given by:

$$\frac{G}{R} = \frac{T_{mp}}{vL} \quad (21)$$

where:

- G = thermal gradient
- R = solidification (growth) velocity
- T_{mp} = melt pool temperature
- v = scan speed
- L = melt pool length (distance from heat source to rear of melt pool)



Physical interpretation

- High $G/R \rightarrow$ columnar grain growth
- Low $G/R \rightarrow$ equiaxed grain formation

Thus, process parameters indirectly control microstructure through their effect on melt pool behavior.

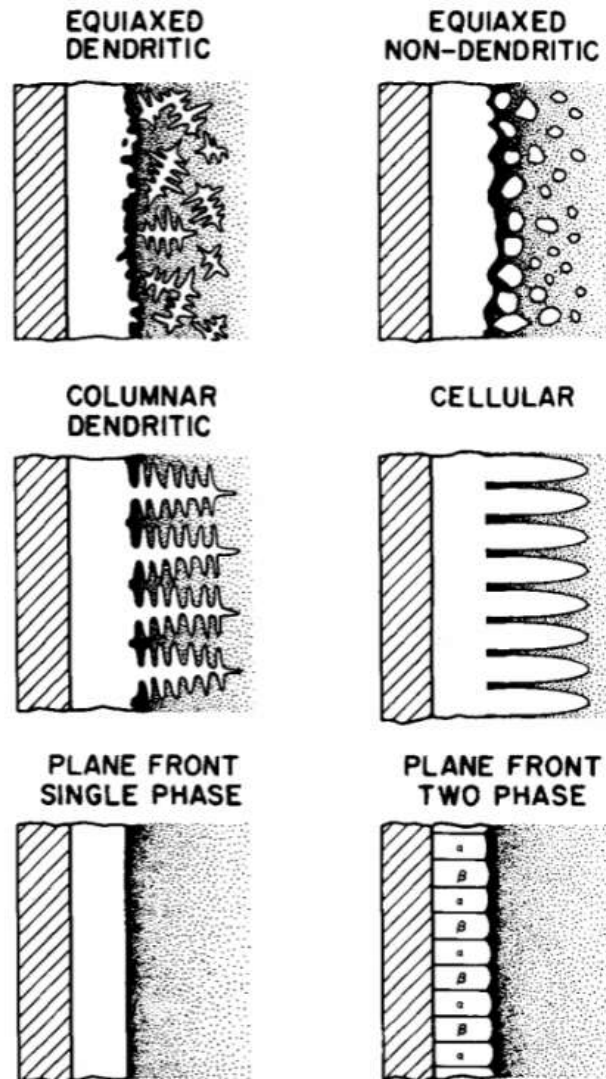
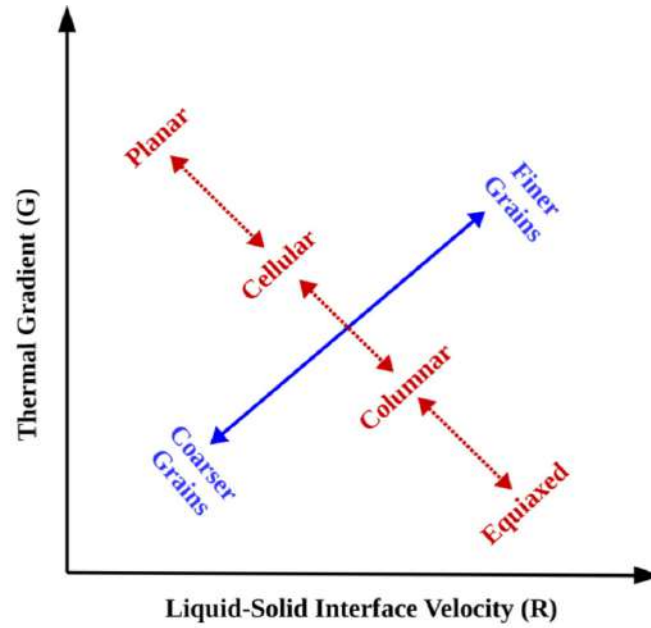


Figure 94: Types of solidification structures that can be obtained in directionally solidified binary alloys (of the non-faceting type)



Microstructure formation during solidification (predicting cooling rate)

Linear energy density,

$$Q' = P/v$$

The layer cooling rate is then expressed by,

$$G \cdot R = A/e^{aQ'}$$

Where a and A are constants.

Kurz-fisher relationship inter dendritic spacing,

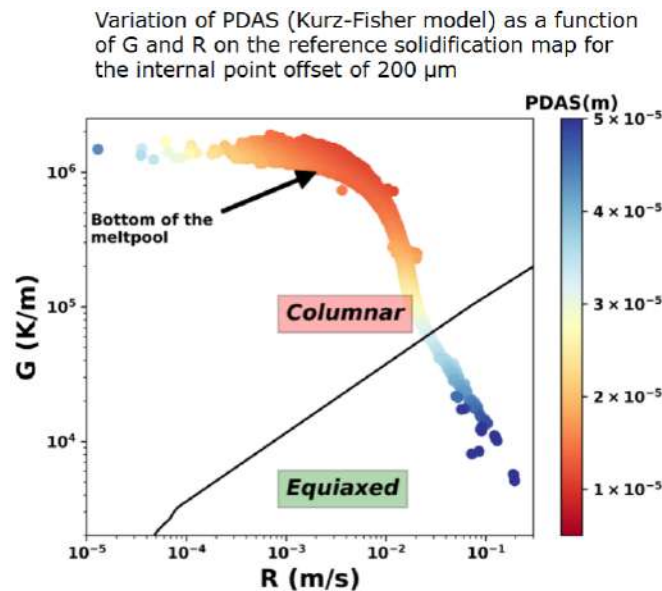
$$d = a(G \cdot R)^n$$

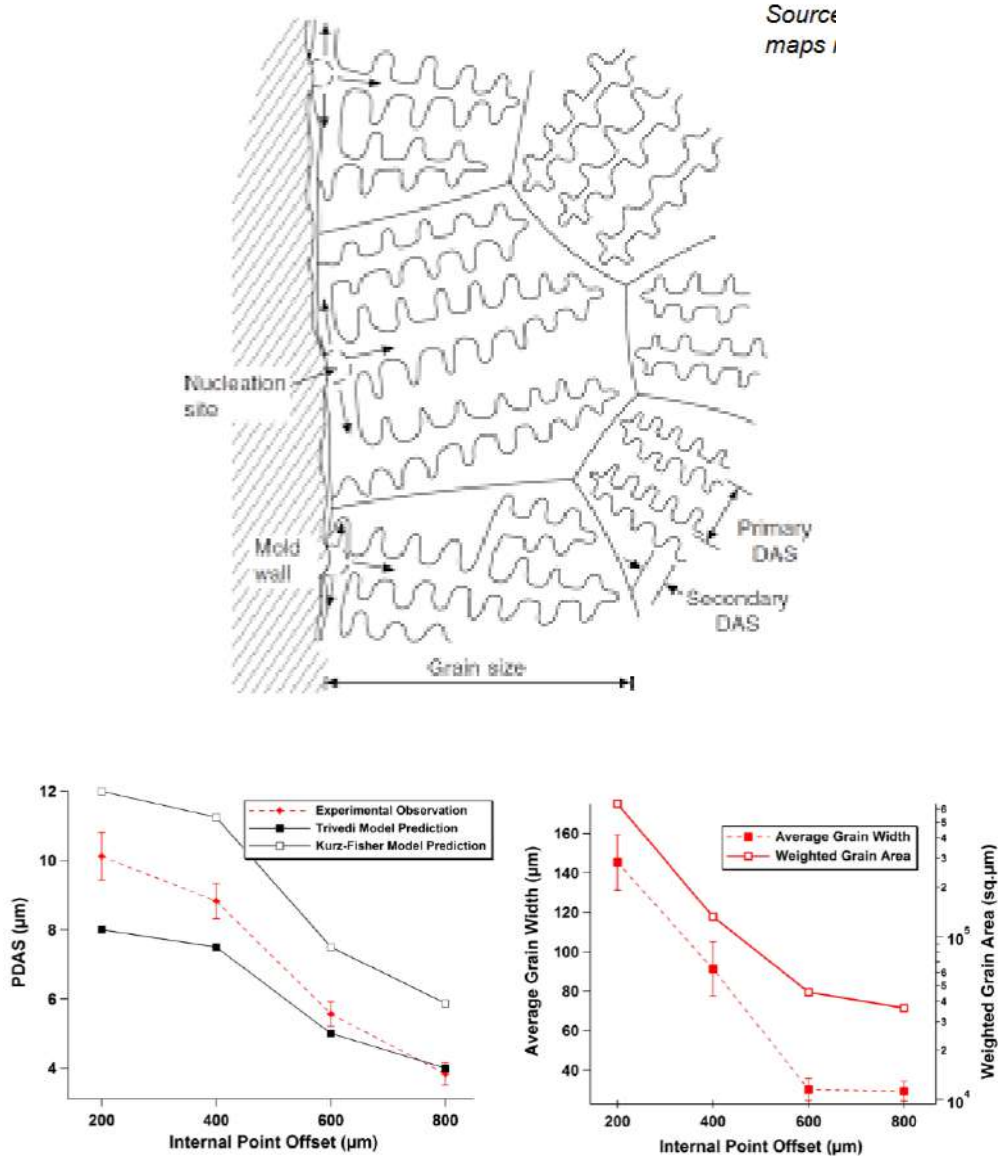
Where a is a constant and n can vary from $1/4$ to $1/2$.

Phenomenological morphology factor (Predicting type of Microstructure)

$$G/R = 2\pi k(\Delta T)^2 / BPv$$

Where ΔT is the solidification temperature range (STR) liquidus-solidus temperature difference, k is the thermal conductivity of the layer, and B is a constant





Columnar or Equiaxed microstructure (Predicting type of microstructure)

$$G = \frac{1}{1+n} \sqrt[3]{\frac{-4\pi N_0}{3 \ln(1-\phi)}} \Delta T_c \left(1 - \frac{\Delta T_n^{n+1}}{\Delta T^{n+1}}\right)$$

Where,

- N_0 is the nucleant volume density for equiaxed grains
- ΔT is the undercooling for columnar growth
- ΔT_n is the nucleation undercooling for equiaxed grains
- ΔT_c is the local undercooling
- n is a material constant

Based on several studies, a fully equiaxed microstructure is achieved with $\phi > 0.49$, while a fully columnar microstructure is achieved with $\phi < 0.0066$

6.11.2 Grain boundary hardening contribution to material strength

$$\Delta\sigma_{gb} = \sigma_o + K'/\sqrt{D}$$

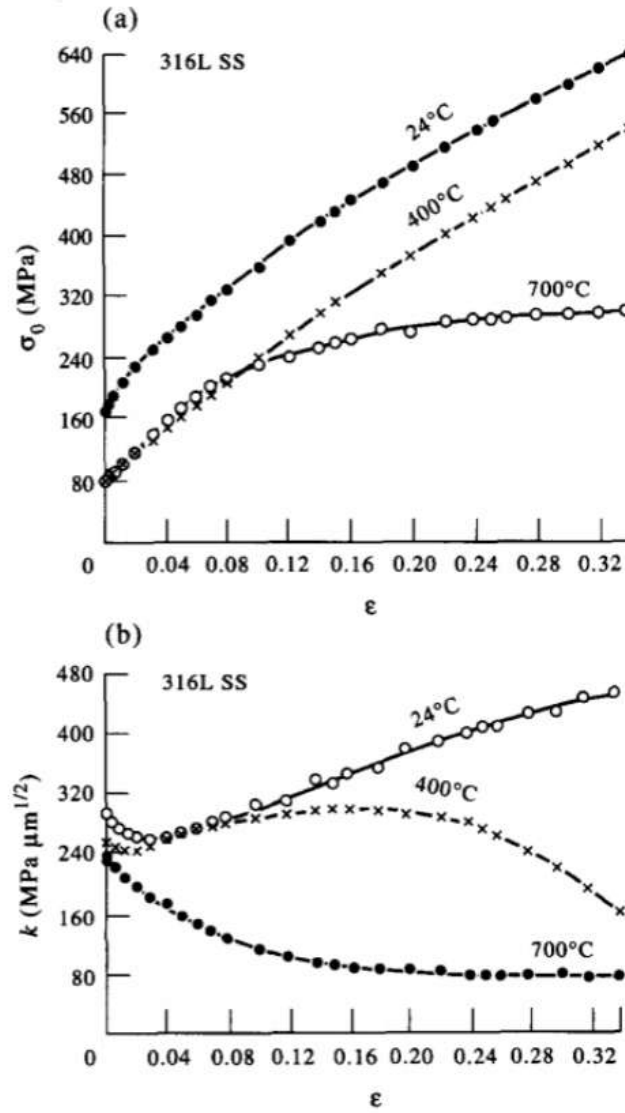


Figure 95: The dependence of H-P parameters σ_o (a) and k (b) on strain. Whereas σ_o increases with strain, k decreases or increases depending on strain level and text temperatures

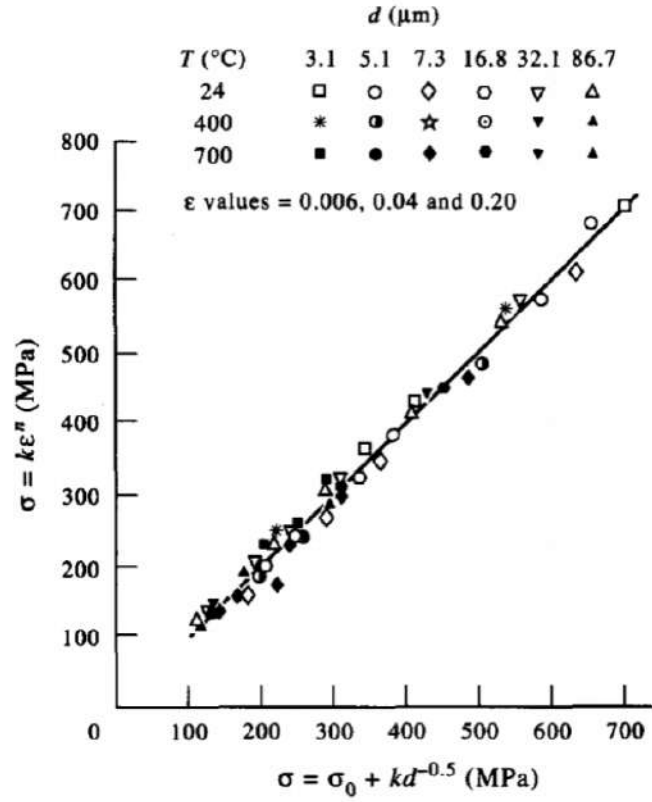


Figure 96: Flow stress using work hardening parameters vs flow stress using the H-P parameters. The flow stresses computed, using $\sigma = K\epsilon^n$ and $\sigma = \sigma_0 + kd^{-0.5}$ are comparable.

6.11.3 Performance Index (PI) of material

$$PI = \frac{\sigma_{ys}}{\alpha_{CTE}}$$

Where α is the coefficient of thermal expansion (also depended on alloy composition)

$$\text{Equivalent } \alpha_{CTE} \equiv \sum_i [c_i \cdot (\alpha_{CTE})_i], \quad c_i \text{ is the weight \% of elements}$$

$$\sigma_{ys} = \Delta\sigma_{gb} + \Delta\sigma_{ss} + \Delta\sigma_{pr}$$

Rule of mixtures

- Solution hardening contribution to material strength (c_i is the weight % of elements)

$$\Delta\sigma_{ss} = \left[\sum_i (k_{ss,i}^{3/2} c_i) \right]^{2/3}$$

- Grain boundary contribution to material strength,

$$\Delta\sigma_{gb} = \sigma_0 + K'/\sqrt{D}$$

- Precipitation hardening contribution (relevant for materials such as maraging steel and 17-4PH steel)

Recommendations

- Solution treating followed by aging at 500 °C for 6 hours has the highest tensile strength and the lowest elongation
- Aging at 480-500 °C has lower tensile strength and higher elongation
- Optimal heat treatments of printed samples differ from that of conventionally manufactured ones

6.11.4 Approximate thermal stress during powder bed fusion

$$\sigma_{thermal} = \left[\frac{E \cdot \alpha_{CTE}}{2(1 - \nu)} \right] \Delta T$$

Where,

- E is the Young's modulus
- α_{CTE} is the coefficient of thermal expansion
- ν is the Poisson's ratio
- ΔT is the temperature difference on the surface being heated

if $\sigma_{thermal} > \sigma_{ys}$, then the material yields during production $\Rightarrow$ residual stresses at the end.

6.11.5 Process-Materials-Properties-Performance links

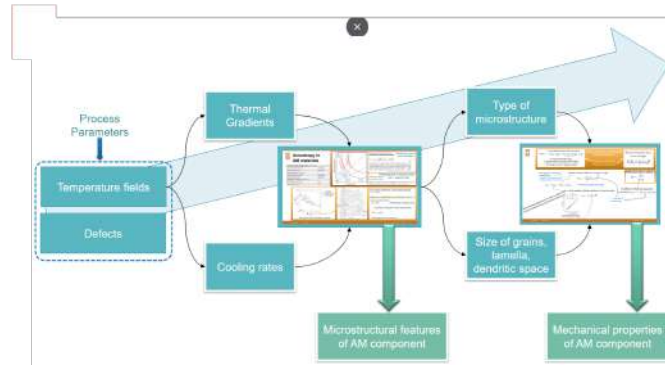


Figure 97: Process-Microstructure-Properties: Establishing Correlations

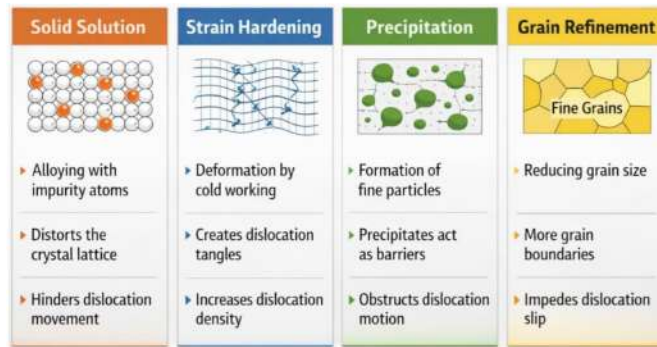


Figure 98: Strengthening in Metals: Solid Solution, Strain Hardening, Precipitation and Grain refinement

$$\sigma_{ys} = \Delta\sigma_{gb} + \Delta\sigma_{ss} + \Delta\sigma_{pr}$$

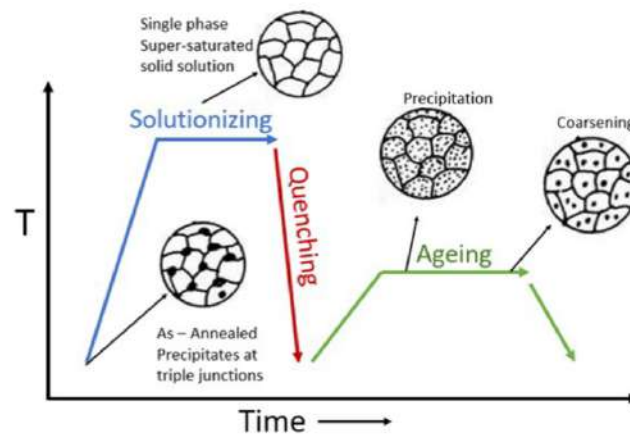


Figure 99: Aging

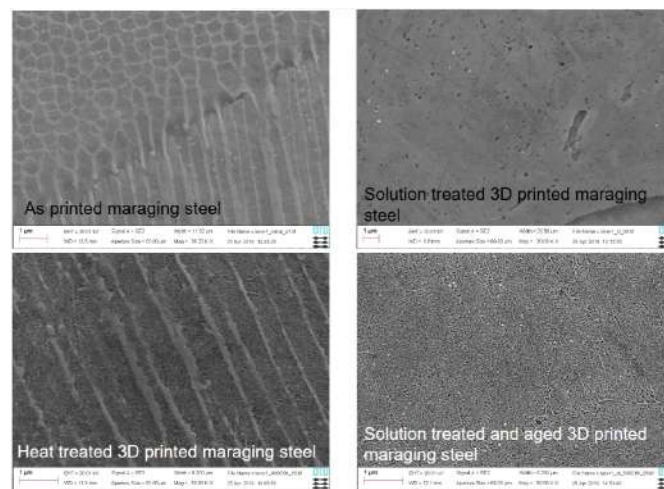


Figure 100: SEM images of maraging steel with multiple treatments

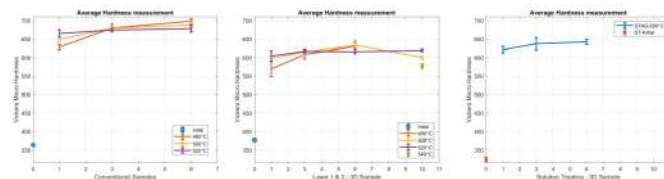


Figure 101: Hardness of sample after different post treatments

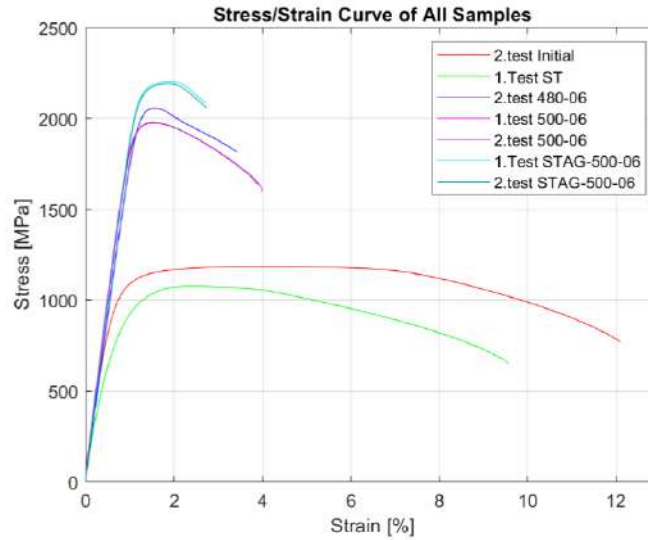


Figure 102: Effect on stress and strain on 3D printed maraging steel

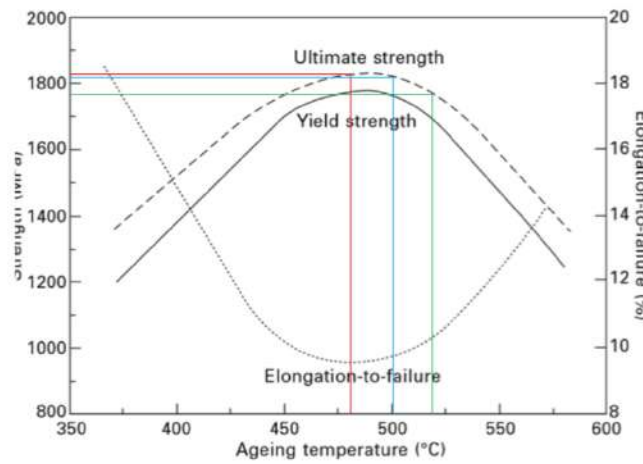


Figure 103: Effect on stress and strain on conventional maraging steel after various heat treatments

Recommendations

- Solution treating followed by aging at 500 °C for 6 hours: Has the highest tensile strength and the lowest elongation
- Aging at 480-500 °C for 6 hours: Has lower tensile strength and higher elongation
- Note: Optimal heat treatments of printed samples differ from that of conventionally manufactured ones

6.11.6 Approximate thermal stress during powder bed fusion

$$\sigma_{thermal} = \left[\frac{E \cdot \alpha_{CTE}}{2(1 - \nu)} \right] \Delta T$$

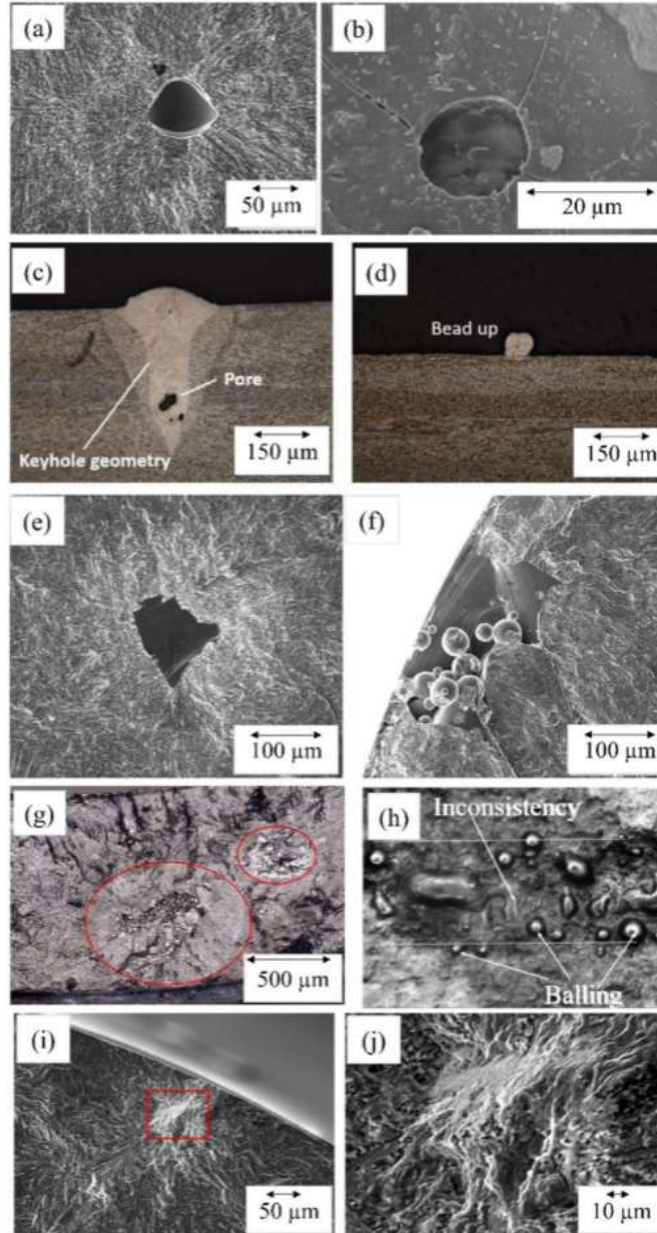
Where,

- E is the Young's Modulus
- α_{CTE} is the coefficient of thermal expansion

- ν is the Poisson's ratio
- ΔT is the temperature difference on the surface being heated

If $\sigma_{thermal} > \sigma_{ys}$, then the material yields during production $\Rightarrow$ residual stresses at end

6.11.7 Defects induced failure in PBF



- (a) Near circular pores caused by entrapped gas
- (b) initiation of fatigue cracks from circular gas pore
- (c) Keyholding pore in SLM Ti-6Al-4V
- (d) Bead up in SLM (Selective Laser Melting) Ti-6Al-4V
- (f) LOF defects caused by insufficient consolidation of the powder particles with unmelted particles
- (g) Fatigue crack initiation from LOF (Lack of Fusion) defects
- (h) inconsistency and balling in SLM Ti-6Al-4V sample

- (i) and (j) Fatigue crack initiation from α -phases

Figure 104: Surface defects

Correlations between fatigue life and square root of measured R_a at several stress levels from L-PBF Ti-6Al-4V: R is the ratio of minimum stress to maximum, (a) $R = -1$ loading and (b) $R = 0.1$ loading.

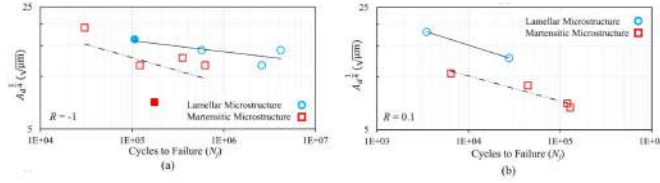


Figure 105: Porosities

Correlations between fatigue life and average critical defects size at stress amplitude of 300 MPa for PBF Ti-6Al-4V: R is the ratio of minimum stress to maximum, (a) $R = -1$ loading, the solid data points are from rotating bending fits, (b) $R = 0.1$ loading.

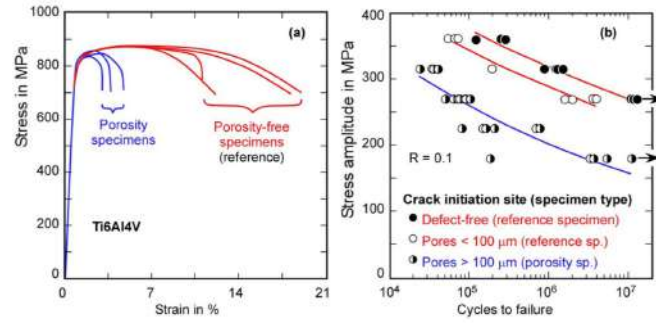


Figure 106: Porosities

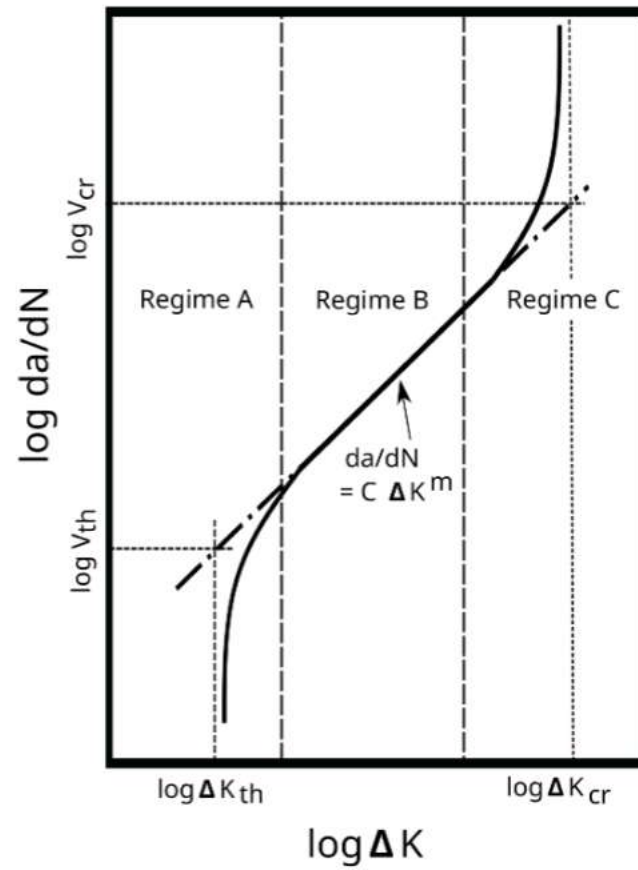


Figure 107: Impacts of Microstructure on behavior

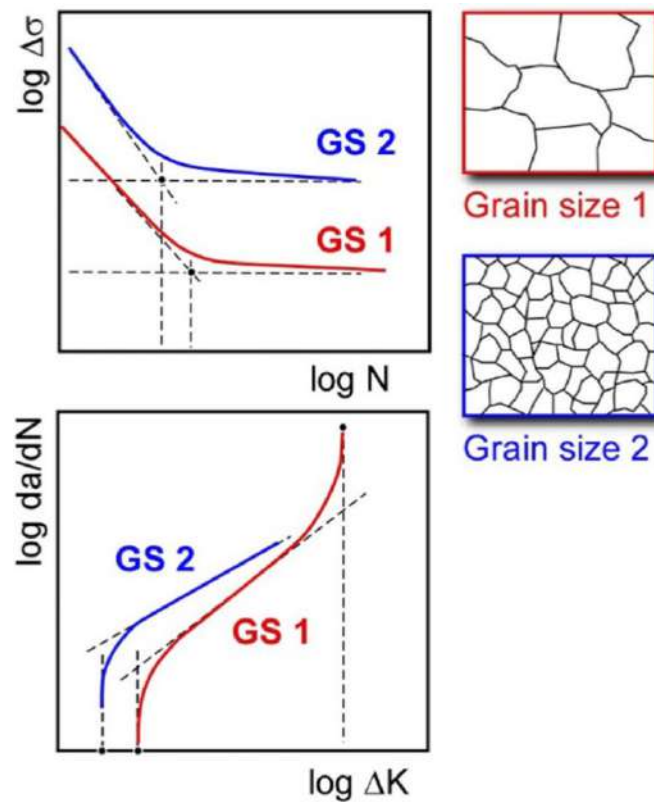


Figure 108: Examples of impact of grain size on behavior

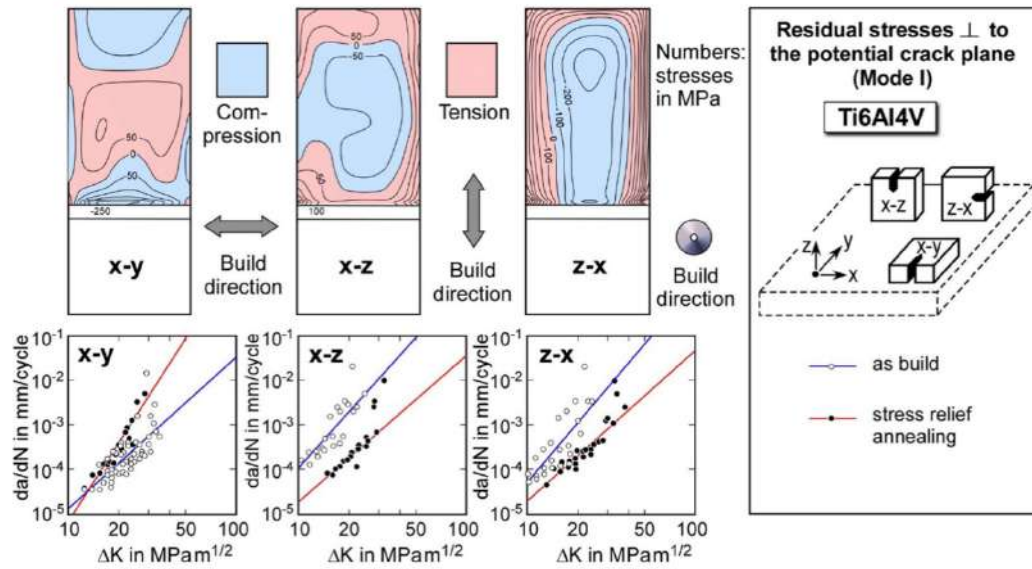
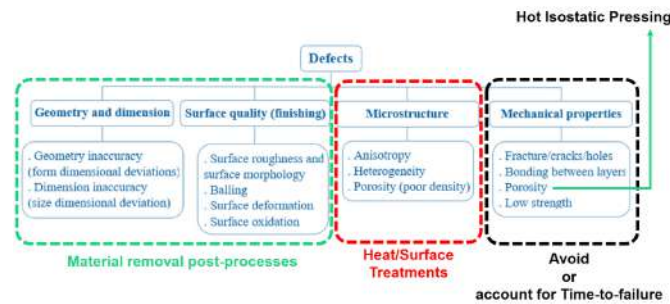


Figure 109: Impacts of Residual stress state on behavior

Residual stress distribution in L-PBF processed Ti-6Al-4V as-built at the crack planes in the C(T) specimens build in 3 different orientations and the effects of stress relieved heat treatment.

6.11.8 Post-processing of metal AM parts



Why we need finishing operations for AM parts

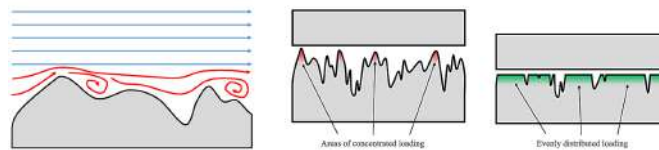


Figure 110: Surface peaks causing micro-turbulence in fluid flows and surface peak loading in bearing applications

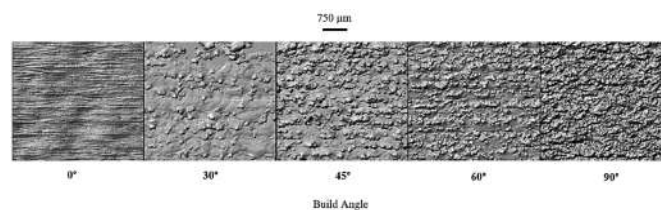


Figure 111: Example of the effect of build angle on surface texture in PBF parts

Challenges for finishing operations

- High initial (as-built) texture
- Surface texture variation across part after build
- Geometric complexity
- Part distortion
- Geometric variation from build to build on a specific machine.
- Machine-to-machine variation in build output

Common limitations of secondary finishing techniques form AM Features and geometries

1. Largely limited to rotational features, such as those with circular cross sections.
2. Only suitable for 2.5D features and details.
3. Minimum feature size limited by media size.
4. Limited to relatively large cylindrical bores, usually tens of millimetres in diameter or more.
5. Difficult to apply to freeform surfaces.
6. Difficult to generate required toolpaths on internal features.
7. Difficult to create a seal between an external part surface and equipment.
8. Process may round edges and remove fine details.
9. Shaped electrode required for complex surfaces, which may limit the complexity of internal features that can be accessed.
10. Difficult to process small features where grits can become stuck.
11. Limited effectiveness with shallow incidence angles.
12. Difficult to generate targeted magnetic field over small areas.
13. Difficult to manipulate magnetic field accurately within a component.

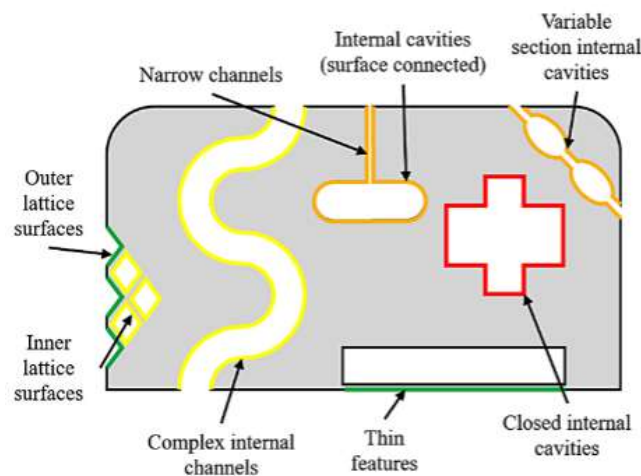


Figure 112: Typical geometrical features seen on AM components

Table 7: Conventional and non-conventional finishing methods

Conventional	Hybrid	Non-conventional
Mechanical		Mechanical
• Turning • Milling • Drilling • Broaching • Grinding • Honing • Lapping • Burnishing • Linishing	• ECM • Mag-AFM • Plasma electrolytic polishing	• Abrasive flow • Grit blasting • Media-based mass finishing
		Electrical
		• Electro-discharge machining
		Thermal
		• Laser re-melting • Laser ablation
		Chemical
		• Chemical etching

Table 8: Secondary finishing techniques and their strengths and limitations with respect to common AM feature types and geometries

Process	Strengths	Limitations	Suitable features				
			Simple external	Complex external	External details	Simple internal	Complex internal
Turning	High dimensional tolerances	Only rotational features	✓	①	X	✓	X
Milling	Complex external features	May require multiple operations	✓	✓	✓	✓	X
Drilling	Relatively cheap equipment	Primarily for through-holes	X	X	✓	✓	X
Broaching	Roughing to final finishing in a single pass	Expensive tools	✓	②	②	✓	✓
Grinding	Can be performed on a machine centre, manually or robotically	Tool access limited for small features	✓	✓	③	④	X
Honing	Good form retention	Expensive equipment	✓	⑤	X		X
Lapping	Achieves superfinishes	Requires a highly skilled operator	✓		X	⑥	X
Burnishing	Creates superior surface mechanical properties by material displacement	Requires relatively smooth surface to start	✓		X	✓	X
Linishing	Similar to grinding	Similar to grinding	✓	✓	③	④	X
Abrasive flow	Can access complex internal features	Expensive equipment and relatively long setup times	⑦	⑦	X	✓	✓
Chemical etching	Relatively fast process	Requires use of hazardous chemicals	✓	✓	⑧	✓	✓
Electro-discharge machining	Very high tolerances	Requires shaped electrodes for complex features	✓	✓		✓	⑨
Grit blasting	Low consumable cost	Low process control	✓	✓	⑩	⑪	X
Magnetic abrasive	–	–	✓	✓	⑫	✓	⑬

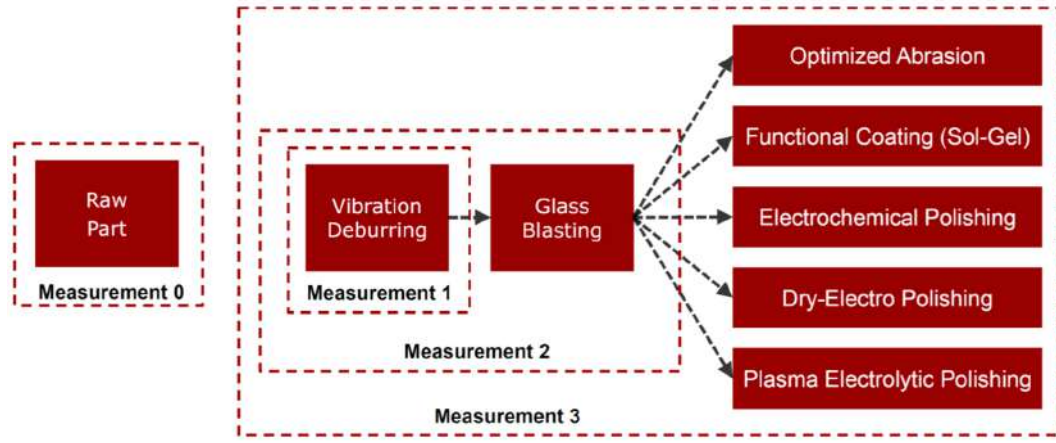


Figure 114: Specimens Post Processing and Measurements

Table 9: Secondary finishing techniques and their strengths and limitations with respect to common AM feature types and geometries

Process	Strengths	Limitations	Suitable features					
			Simple external	Complex external	External details	Simple internal	Complex internal	
Media-based mass finishing	Can finish multiple parts at once	Feature rounding, may require masking	✓	✓	③	③	X	
Ultrasonic polishing	No loss of detail through tool contact		✓	✓	✓	✓	X	
Laser re-melting	Fine detail and high process control; preserves overall geometry	May introduce thermal damage and oxidation, especially on thin features	✓	✓	✓	⑪	X	
Laser ablation	Fine detail and high process control	Relatively slow; thermal effects can be limited at the cost of processing speed	✓	✓	✓		X	
Plasma electrolytic polishing	Relatively short processing time	Requires special electrolyte solutions for each different metal	✓	✓	✓	✓	X	
Electrochemical machining	Achieves superfinites; suitable for hard-to-machine materials	Expensive equipment; requires use of hazardous chemicals	✓	✓	⑧	✓	✓	

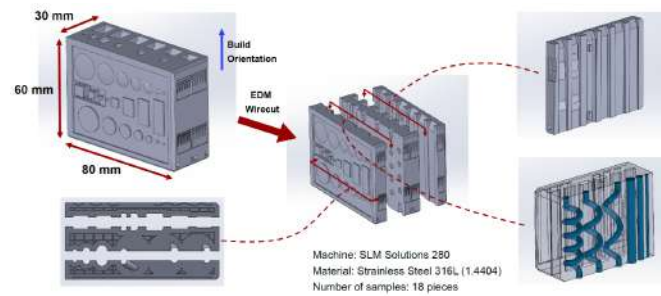


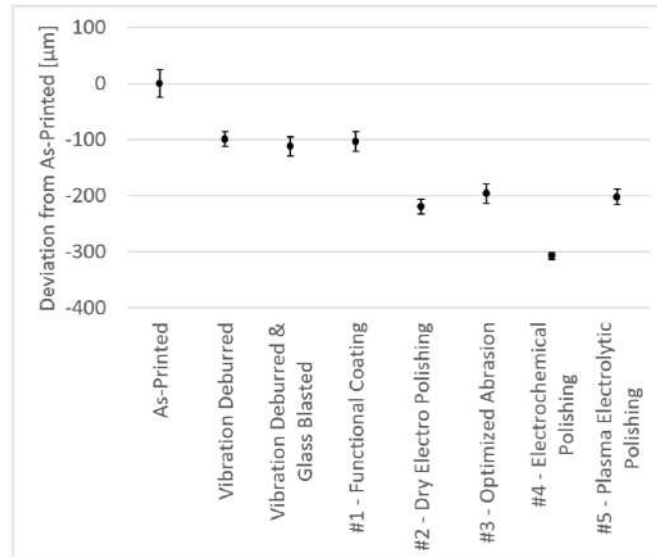
Figure 113: Benchmarking Speciment design and Manufacturing

Measurement Equipment and Strategy

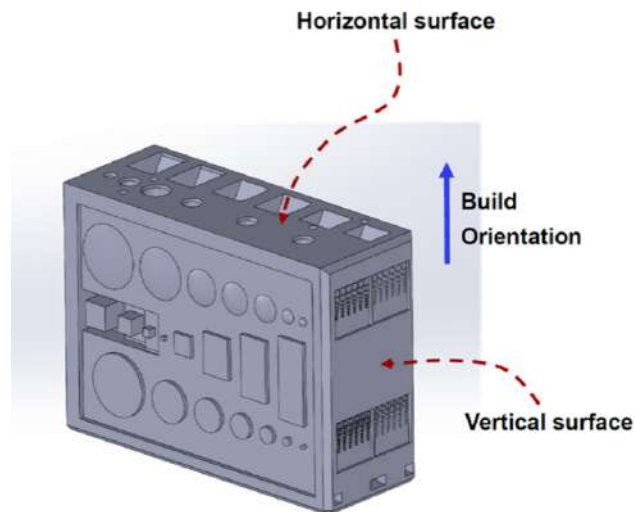
- Tactile CMM (Coordinate Measuring Machines), Zeiss PRISMO
- Scanning measurements, speed of $v = 2.5 \text{ mm/s}$
- Probe: Synthetic Ruby sphere, $\varnothing 3 \text{ mm}$ diameter
- Measurement area considering the full surface, with 2 mm of distance to edges to avoid influence of boundary effects

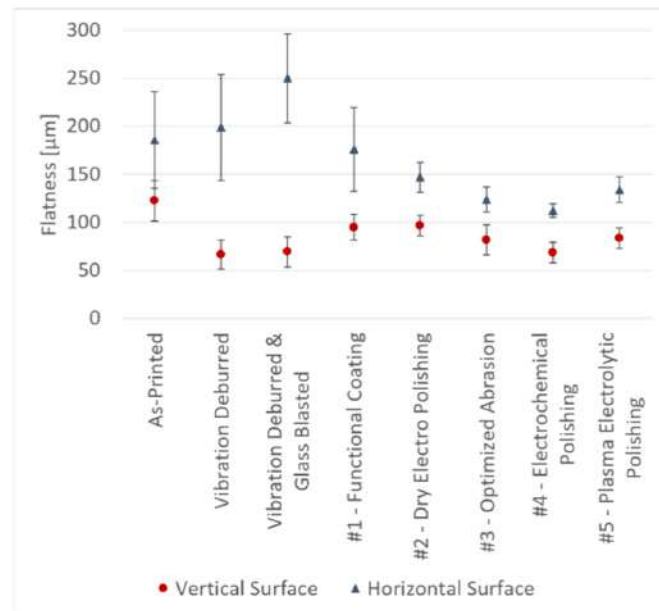
Dimensional Evaluation

- Nominal value of feature of As-Printed part highly dependent on several factors such as for example,
 - Material specific scaling to account for shrinkage
 - Beam compensation
- Comparison presented relative to As-printed specimen dimension
- Standard deviation shows as error bars (5 repeated measurements on 3 different specimens)



Surface flatness





Surface Characterization

- Food grade Sa -values
- 2 processes provide final roughness amplitude values below $1 \mu m$
- Deviations within $0.5 \mu m$

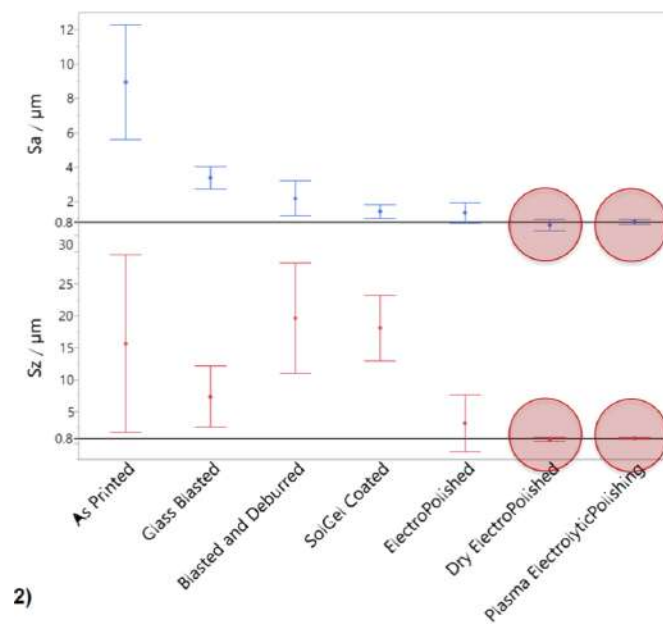


Figure 115: Surface Roughness analysis - Amplitude: Arithmetical mean height (Sa) and Maximum height of the surface (Sz)

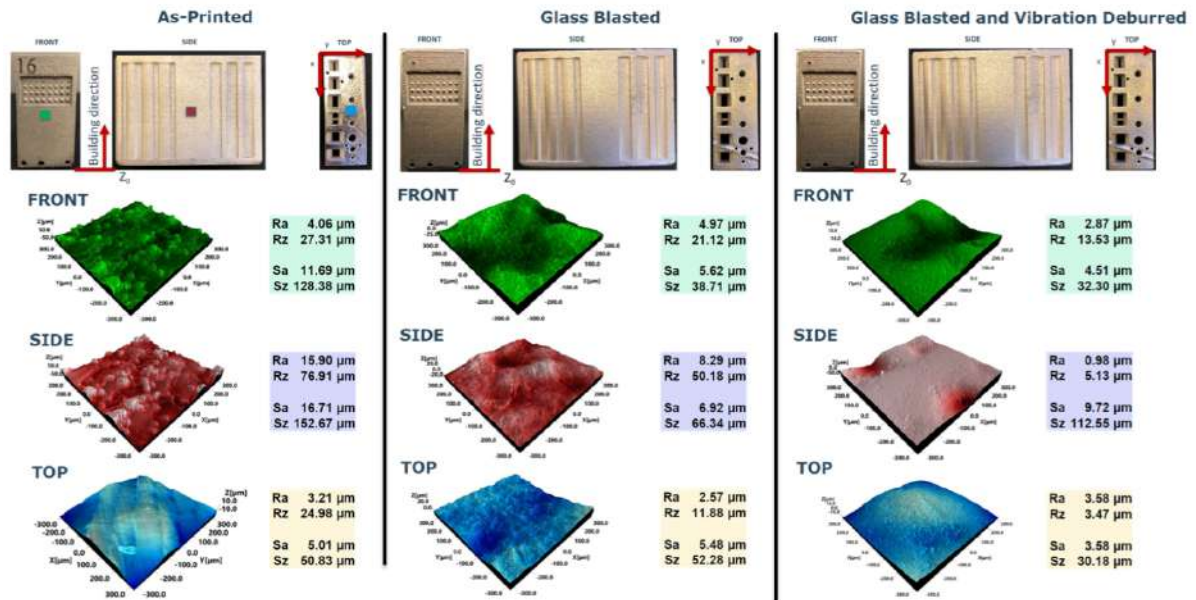


Figure 116: Results of the three initial stages: As-Printed, Glass blasted, Glass blasted and Vibration Deburred

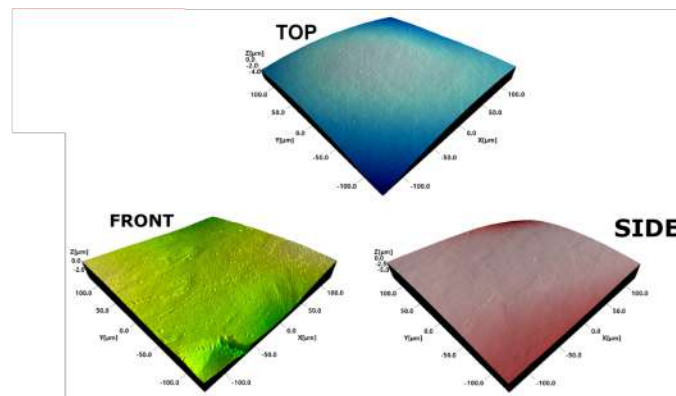


Figure 117: Plasma Electrolytic polishing (PEP) (50x magnification)

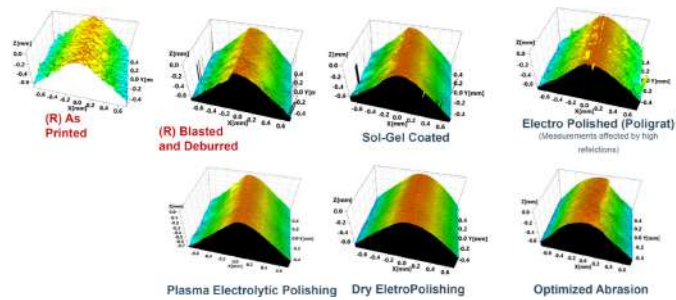


Figure 118: Corner angle analysis

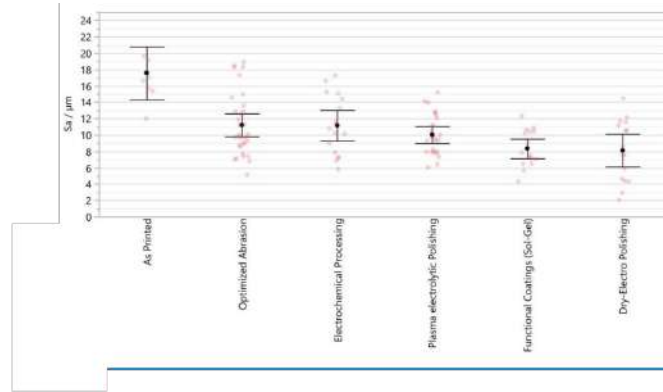
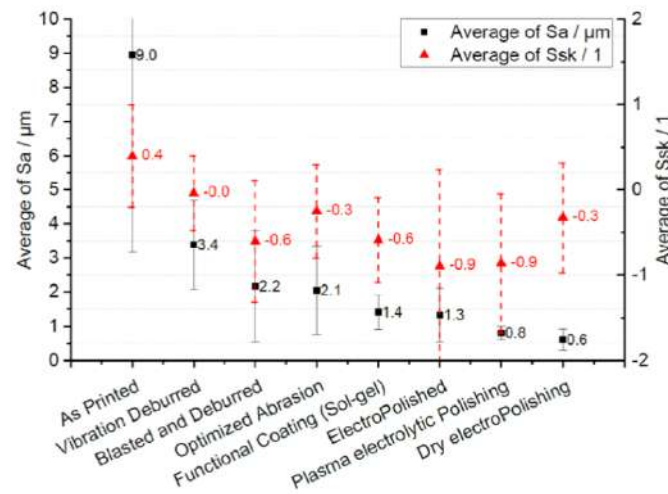


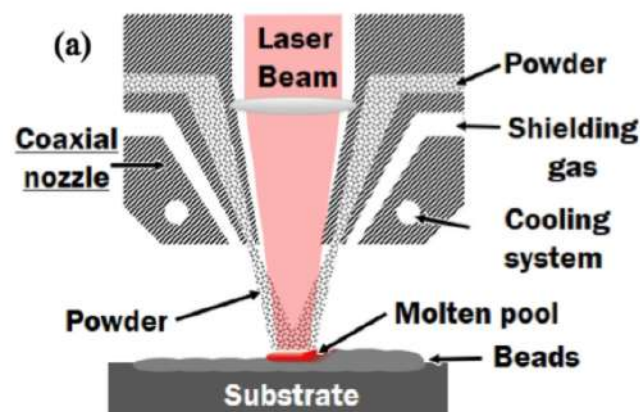
Figure 119: Characterization of Internal Channels



6.12 AI/ML for Additive Manufacturing

Research Hypotheses

- *Stability / Robustness / Replicability*: How does the process respond to parameter changes?
- *Monitoring / Process Quality*: How can defects be detected during manufacturing?
- *Property Prediction*: How do process conditions affect final part quality?
- *Variability / Uncertainty Management*: How can variability and uncertainty be reduced?



Modeling, observability, and scale gaps

- *High dimensional, non-stationary, nonlinear dynamics*: Process behavior emerges from tightly coupled thermal, fluid, material effects.
- *Latent variables, indirect observability*: Melt pool geometry, stability, microstructure are partially observed or observed offline
- *High fidelity simulation are expensive*: Monitoring and control require decisions at much faster time-scales than simulations can provide.
- *Limited generalization*: Process variability across machines and materials and variability across calibrated

6.12.1 Broad classification of Metal AM categories

Taxonomy of Metal AM processes: **Research Hypotheses**

- *Powder bed Fusion (PBF)*: Layer-wise melting of a powder bed using laser or electron beam. Material is pre-deposited as a powder and selectively melted layer by layer.
- *Directed Energy Deposition (DED)*: Continuous material deposition (powder or wire) into the melt pool during deposition.
- Other metal AM processes: binder jetting (printing + sintering), solid state processes
- *Key implications*: Trade-offs between spatial resolution, in-situ characterization, data availability, and control flexibility

Artificial intelligence and machine learning are useful in additive manufacturing because AM processes are difficult to describe only with simple analytical models. The process depends on many coupled physical phenomena, such as heat transfer, melt pool dynamics, material flow, solidification, residual stresses and microstructure evolution. These interactions are nonlinear, history-dependent and occur across different time and length scales.

6.12.2 PBF and DED from a data perspective

Powder Bed Fusion (PBF) and Directed Energy Deposition (DED) generate different types of data and therefore lead to different machine learning problems.

Table 10: Comparison between PBF and DED from an AI/ML perspective.

Powder Bed Fusion (PBF)	Directed Energy Deposition (DED)
Higher spatial resolution and smaller melt pools.	Lower geometric resolution and larger melt pools.
The material is already deposited as a powder bed and is selectively melted layer by layer.	Material is continuously deposited into the melt pool as powder or wire.
In-situ data is mainly optical or thermal imaging, often aggregated per scan or per layer.	In-situ monitoring can be continuous and synchronized during deposition.
Typical sensors include coaxial cameras, off-axis cameras, thermal cameras and infrared cameras.	Typical sensors include optical cameras, thermal sensors, laser power signals, powder/wire feed rate and machine position.
Limited direct access to melt pool depth, fluid flow, stability and microstructure during the process.	Richer in-situ access to melt pool behaviour, but with more continuous dynamic behaviour.
Data analysis usually focuses on defect detection, process stability and quality indicators.	Data analysis often focuses on dynamic modelling, state estimation, anomaly detection and closed-loop control.
Narrower process window for stable melting.	Larger process window, but stronger sensitivity to disturbances and parameter coupling.

The main difference is that PBF tends to produce indirect, image-based and defect-oriented data, while DED produces denser time-series data that is more suitable for monitoring and control.

6.12.3 Why AM is difficult to model

AM processes are difficult to model because the final part quality is not controlled by one parameter alone. Instead, the final result depends on the interaction between process parameters, process signatures and material response.

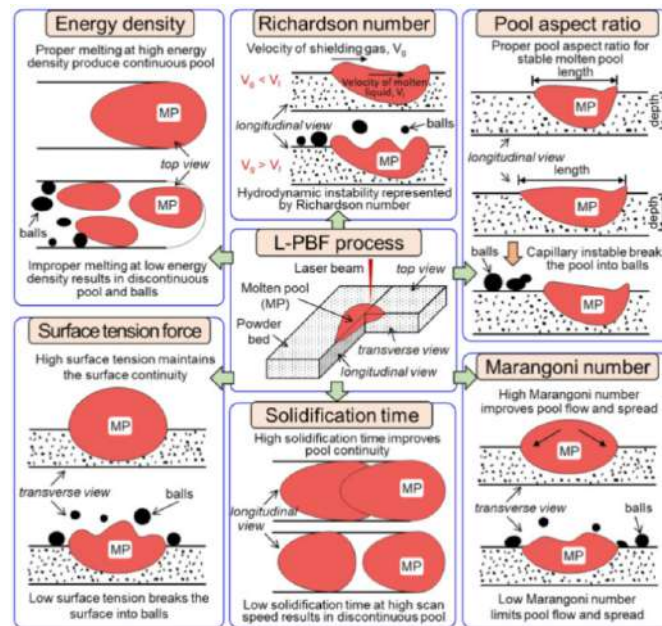


Figure 120: Physics-informed machine learning and mechanistic modeling of additive manufacturing to reduce defects

- **Multi-physics behaviour:** thermal, fluid, mechanical and metallurgical phenomena occur at the

same time.

- **Nonlinear dynamics:** small changes in laser power, scan speed, powder flow or temperature can produce large changes in the melt pool.
- **History dependence:** what happens in one layer depends on previous layers and previous thermal cycles.
- **Limited observability:** variables such as melt pool depth, solidification conditions and microstructure are often only indirectly measured.
- **Expensive simulations:** high-fidelity simulations are too slow for real-time monitoring and control.
- **Limited generalization:** models trained for one material, machine or parameter range may not work well for another.

Because of this, machine learning can be used as a fast surrogate model. Instead of solving the full physics every time, the model learns relationships between inputs and outputs from experimental, simulation or sensor data. These surrogate models can be used to predict temperature, melt pool instability, porosity, geometry errors or final material properties.

6.12.4 Types of AM data and suitable ML models

Different AM data types require different machine learning models.

Table 11: Typical AM data types and suitable machine learning models.

Data type	Example in AM	Typical models
Tabular data	Process parameter grids, such as laser power, scan speed and hatch spacing	Regression, MLPs, tree-based models
Time-series data	Pyrometer signals, laser power traces, photodiode signals, melt pool temperature	RNNs, LSTMs, ARIMA, temporal models
Images and thermal maps	Coaxial melt pool images, layer-wise optical images, IR thermal maps	CNNs, Vision Transformers
Sequential-spatial data	Videos, scan paths, layer sequences, bead build-up	CNN-RNN hybrids, Transformers
Graph-structured data	Connected process-structure-property relationships	Graph Neural Networks

6.12.5 Learning paradigms

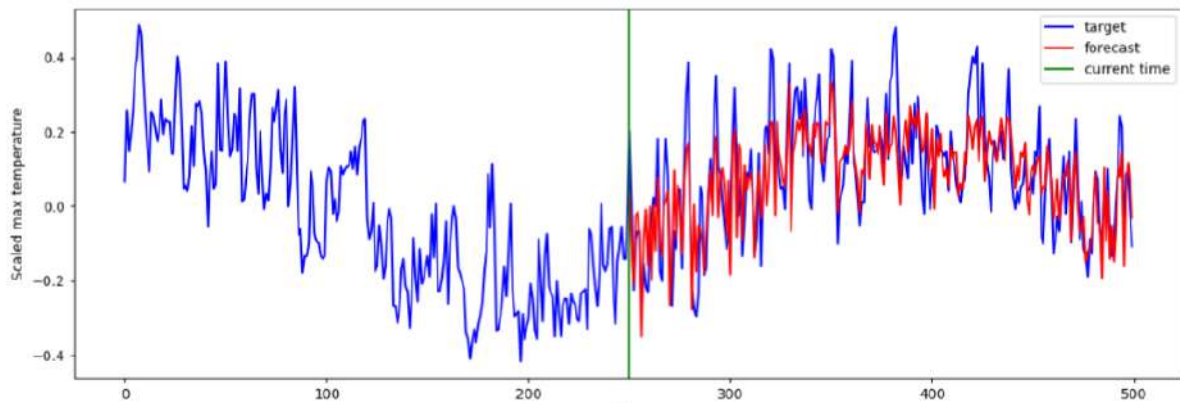
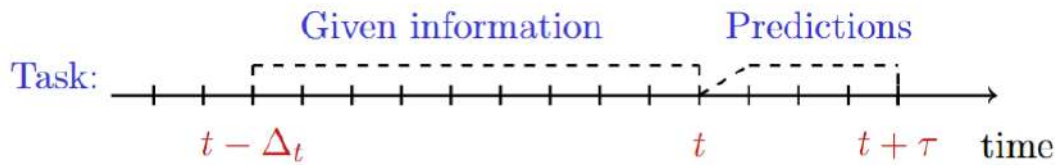
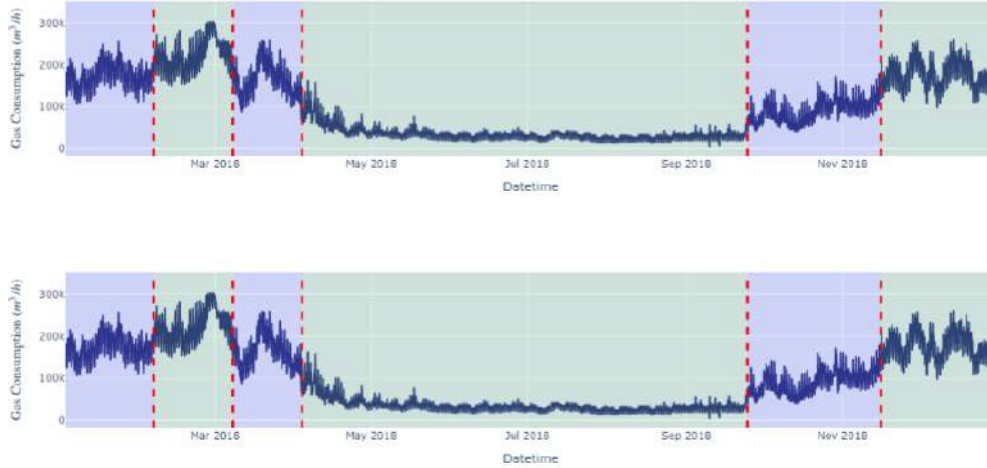
The choice of learning method depends on the problem definition and on the available data.

- **Supervised learning:** used when labelled data is available. Typical tasks include regression and classification, for example predicting density, roughness or whether a defect will occur.
- **Unsupervised learning:** used when labelled data is limited or unavailable. Typical tasks include clustering, anomaly detection and novelty detection.
- **Reinforcement learning:** used for sequential decision-making, where an agent learns how to act based on feedback. In AM, this can be relevant for adaptive process control.

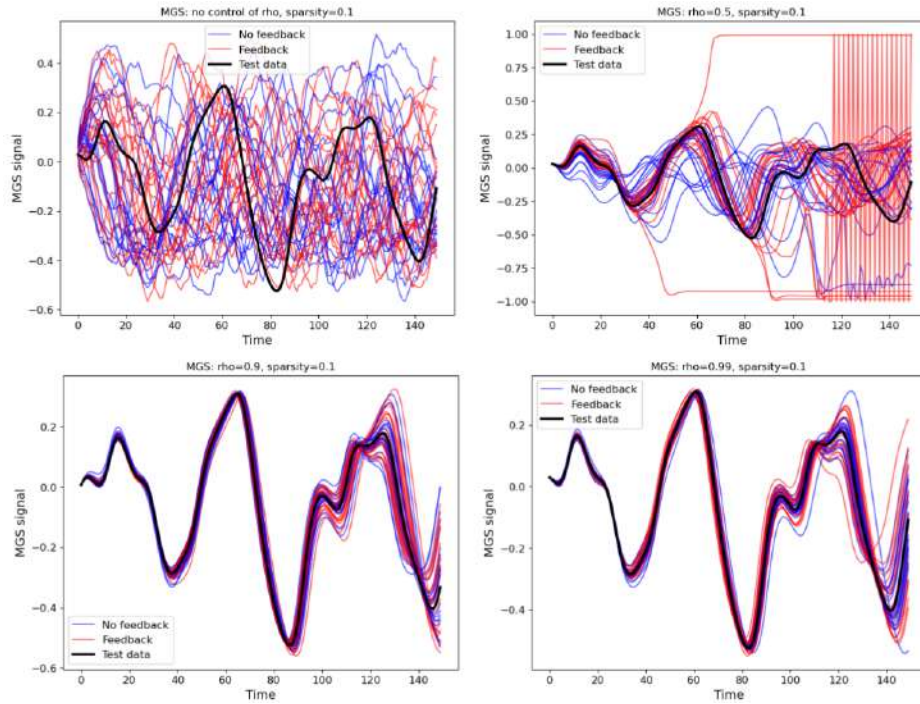
In supervised learning, the model is trained to minimize a chosen error or loss function. The error function can be based on different measures, such as Euclidean distance for regression or cross-entropy for classification.

6.12.6 Learning from time-series data

Time-series data is especially important for process monitoring. A model can be trained to predict the expected future process response from previous measurements.



- The model receives information from a previous time window.
- It predicts the future process state.
- The difference between the prediction and the actual measured signal is called the residual.
- Large residuals can indicate abnormal behaviour or process defects.



This is useful for anomaly detection because the model learns what a normal process signal should look like. If the real signal strongly deviates from the predicted normal signal, the process may be unstable.

6.12.7 Challenges of chaotic and non-stationary behaviour

A major challenge in AM monitoring is that the process can show chaotic and non-stationary behaviour.

- **Chaotic behaviour:** small differences in the initial condition can lead to very different future behaviour. This makes long-term prediction difficult.
- **Non-stationarity:** the data distribution changes over time. A model trained on older data may fail when the material, machine condition or process window changes.

To handle this, hybrid approaches can be used. These combine different tools, for example:

- statistical methods and machine learning,
- machine learning and physics-based models,
- feature selection using optimization methods,
- different models for different materials or parameter regions,
- symbolic rules for known process conditions.

6.12.8 Process Window Identification

Process Window Identification (PWI) is one important application of AI/ML in additive manufacturing. The goal is to identify which process parameters lead to stable production and which parameters lead to defects.

For PBF, relevant input features can include:

- laser power,

- scan speed,
- beam diameter,
- layer thickness,
- thermal diffusivity,
- alloy type.

For DED, relevant input features can include:

- laser power,
- scan speed,
- powder feed rate,
- single-scan-track length,
- single-scan-track height,
- alloy type.

The output is often a classification of the process condition, for example:

- non-defect,
- inconclusive,
- defect.

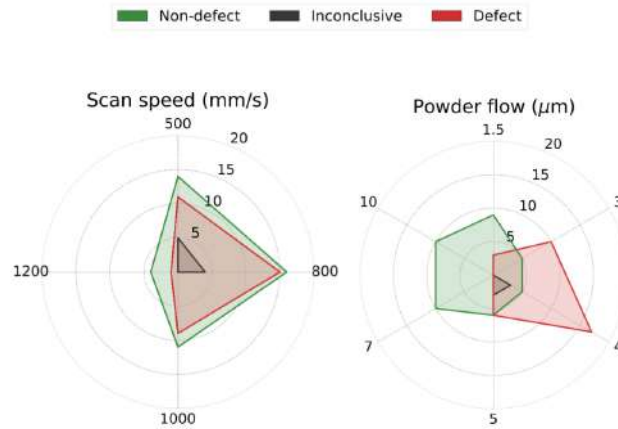


Figure 121: Different feature domains (even multi-scales, e.g., multi-frequency inputs)

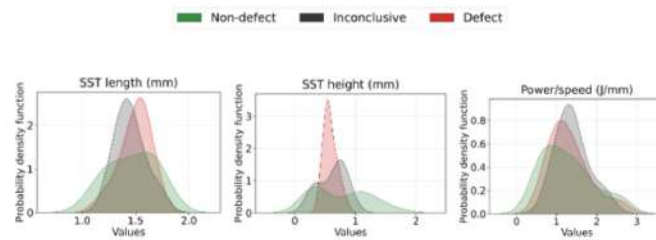


Figure 122: Different data distribution

Important challenges in PWI include limited data availability, noise in both input and output data, mixed labelled and unlabelled data, and poor generalization between materials.

6.12.9 Feature selection and redundancy

Feature selection is important because many process variables are correlated. For example, in DED, laser power, scan speed and track geometry can be strongly related. If correlated features are not considered carefully, the model may give misleading interpretations.

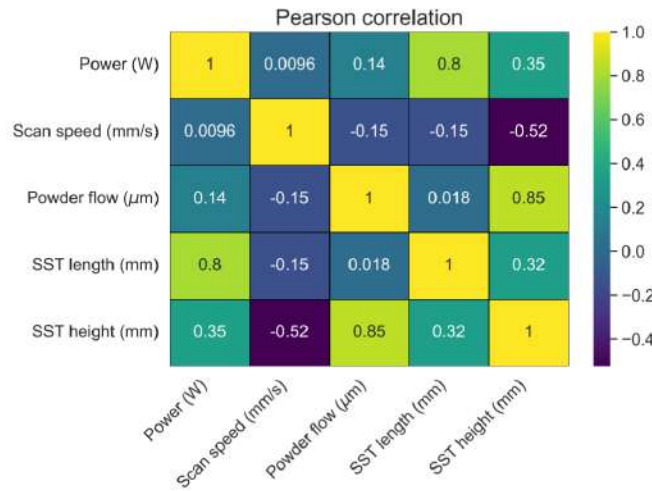


Figure 123: Structural relationships among features and redundancy

Feature selection helps to:

- remove redundant variables,
- reduce model complexity,
- improve generalization,
- make the model interpretation more reliable.

6.12.10 Decision boundaries and model sensitivity

Machine learning models often define a decision boundary between good and defective process regions. For example, Gaussian Mixture Models can be used to separate normal and anomalous process conditions.

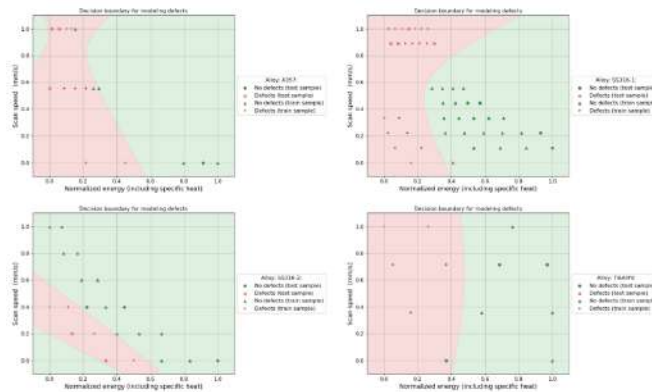


Figure 124: Example of prediction of anomalies using Gaussian Mixture Models

However, the predicted decision boundary can strongly depend on the chosen model architecture and hyperparameters. The same model type can give different results when changing:

- number of components,

- covariance assumptions,
- number of layers,
- number of neurons,
- training parameters,
- selected input features.

This is why hyperparameter optimization, neural architecture search, AutoML and meta-learning are important research topics in AI for AM.

6.12.11 Simpson's paradox in AM data analysis

A possible problem in AM data analysis is Simpson's paradox. This happens when a trend appears in aggregated data but disappears, or even reverses, when the data is divided into subgroups.

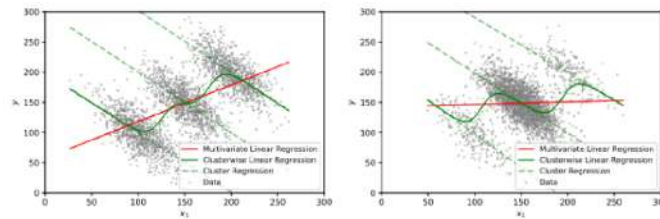


Figure 125: A trend that appears in several subgroups disappears when the subgroups are combined

- Red curve: regression in entire population
- Green dashed curve: regression in subgroups
- Green curve: unbiased regression
- Left curve: Balanced subgroups. Right curve: less balanced data

For example, a general regression model may show a weak relationship between a process parameter and a defect. However, when the same data is separated by alloy type, machine type or powder batch, stronger relationships may appear inside each subgroup.

Therefore, conclusions from AM data should be made carefully. It is often important to separate the data by:

- alloy type,
- machine type,
- powder batch,
- process category,
- build orientation,
- sensor setup.

Otherwise, the model may learn correlations that are not physically meaningful.

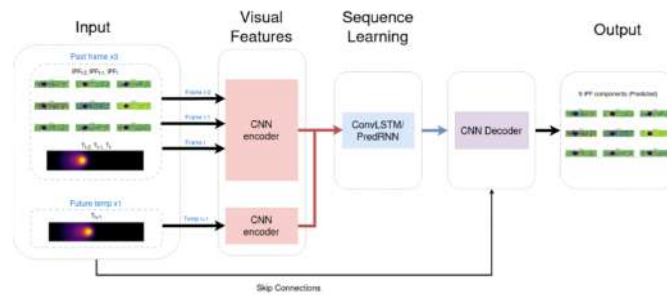
6.12.12 Predicting microstructure with AI

AI can also be used to predict microstructure evolution during metal AM. This is useful because microstructure depends on complex nonlinear interactions between laser heating, cooling rate, solidification and material composition.

Deep learning can be used as a surrogate model for expensive simulations or experiments. A trained model can quickly predict how grain morphology evolves under a given process condition. This allows faster exploration of parameter combinations.

A common approach is to combine:

- **CNNs** to extract spatial features from images or thermal maps,
- **RNNs or ConvLSTMs** to model the time evolution of the process,
- **decoders** to reconstruct or predict the future microstructure.



This type of model is useful when the input includes both spatial information and time-dependent information, such as a sequence of thermal fields during laser scanning.

6.12.13 Deep learning for X-ray tomography images

X-ray computed tomography is often used as a post-processing and non-destructive characterization technique for AM parts. It can reveal internal defects, pores and particle features after fabrication.

Deep learning can be used to analyze large XCT image datasets. Important tasks include:

- particle segmentation,
- pore detection,
- defect classification,
- particle tracking,
- quantification of internal porosity.

This is often a semi-supervised problem because large volumes of XCT images may be available, but only a small part of them are manually labelled. A typical deep learning workflow compares the input image, the ground truth mask and the predicted segmentation mask.

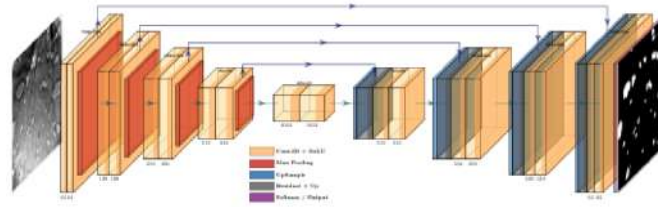


Figure 126: Example of experimental DL design

6.12.14 Main takeaway

AI and machine learning are useful in additive manufacturing because they can help connect process parameters, sensor data, defects, microstructure and final part quality. However, AM data is complex, noisy, nonlinear and often material-specific. Therefore, AI models should not be used blindly. They are most useful when combined with physical understanding, good experimental design, careful feature selection and validation across different materials and machines.

7 Precision Metal Additive Manufacturing

7.1 Introduction to Precision Metal Additive Manufacturing

Additive Manufacturing (AM), often referred to as 3D printing, is a manufacturing method where parts are produced directly from a digital 3D model by adding material, usually layer-by-layer. This is different from subtractive manufacturing, where material is removed, and formative manufacturing, where material is shaped using moulds or dies.

Although AM is often described as a new technology, its commercial development started in the 1980s, with the first commercial AM systems appearing in 1986. Initially, AM was mainly used for rapid prototyping. However, improvements in machine precision, repeatability and material range have allowed AM to move towards industrial production.

7.1.1 Why AM is important

AM is important because it allows:

- production of complex geometries,
- reduction of material waste,
- mass customisation,
- faster design changes,
- manufacturing without expensive tooling,
- integration of several parts into one component.

A key industrial example is the fuel injector nozzle used in modern aircraft engines. AM allowed the redesign of the internal channels and the consolidation of several parts into one complex metal component. This improved performance and reduced the need for assembly.

7.1.2 Limitations of AM

Despite its advantages, AM is still not fully equivalent to conventional manufacturing for many industrial applications. The main limitations are:

- limited dimensional accuracy compared with precision machining,
- high surface roughness in as-built parts,
- need for post-processing,
- process variability,
- difficulty in qualification and certification,
- limited understanding of defect formation,
- high cost for some metal AM applications.

AM must therefore be considered as part of a full manufacturing chain, including design, material supply, printing, post-processing, inspection and qualification.

7.1.3 Basic AM definitions

- **Additive Manufacturing (AM):** process of joining materials to make parts from 3D model data, usually layer upon layer.
- **AM system:** the complete machine and auxiliary equipment used for AM.
- **AM machine:** the main machine hardware, control software and accessories needed to complete a build cycle.
- **Rapid prototyping (RP):** use of AM to reduce the time needed for producing prototypes.
- **Rapid tooling (RT):** use of AM to produce tools or tooling components with reduced lead time.
- **3D printing:** commonly used as a general term for AM, but more specifically refers to material deposition using a print head, nozzle or similar printing technology.
- **STL file:** file format describing the surface geometry of a 3D object using triangular facets.
- **AMF file:** AM file format that can include geometry, colour, material, lattice information, texture and metadata.
- **Feedstock:** raw material supplied to the AM process, such as powder, wire, filament, sheet, liquid or suspension.

7.1.4 AM process categories

- **Binder Jetting (BJT):** a liquid binder is selectively deposited to join powder particles.
- **Directed Energy Deposition (DED):** focused thermal energy melts material as it is being deposited.
- **Material Extrusion (MEX):** material is selectively dispensed through a nozzle or orifice.
- **Material Jetting (MJT):** droplets of feedstock material are selectively deposited.
- **Powder Bed Fusion (PBF):** thermal energy selectively fuses regions of a powder bed.
- **Sheet Lamination (SHL):** sheets of material are bonded together to form a part.
- **Vat Photopolymerisation (VPP):** liquid photopolymer in a vat is selectively cured by light.

7.1.5 Other important terms

- **Slicing:** the digital process of dividing the 3D model into successive layers.
- **Build chamber:** enclosed region where the part is fabricated.
- **Build platform:** base surface on which the part is built.
- **Curing:** change in material properties due to a chemical reaction, for example light-induced polymerisation.
- **Hatch distance:** distance between the centres of two adjacent scan tracks.
- **Staircase effect:** stepped surface error caused by the layer-wise nature of AM.
- **Support structure:** auxiliary structure used to support the part during printing and later removed.
- **Lattice structure:** 3D arrangement of connected struts or cells.
- **Hybrid manufacturing:** combination of additive and subtractive processes.
- **Post-processing:** operations performed after printing to obtain the desired final properties.

7.1.6 Precision engineering in AM

Precision engineering is concerned with achieving high accuracy, repeatability and reliability. A useful way to describe precision is by comparing an error value with the full range of the machine:

$$\text{Precision ratio} \sim \frac{\text{Error}}{\text{System range}} \quad (22)$$

A system is often considered a precision engineering system when this ratio approaches:

$$10^{-6} \quad (23)$$

For example:

- A precision machine with repeatability of 100 nm over 100 mm gives approximately 10^{-6} .
- A metal AM machine with repeatability of 100 μm over 100 mm gives approximately 10^{-3} .

This shows that most current metal AM systems are still far from traditional precision engineering standards.

7.1.7 Sources of lack of precision in AM

Typical causes of lack of precision include:

- thermal effects,
- residual stresses,
- vibration,
- loose joints and machine compliance,
- limited actuator and sensor resolution,
- powder spreading variability,

- contamination,
- friction,
- surface roughness,
- post-processing variation.

The goal of precision metal AM is not only to correct errors after printing, but to understand the causes of errors and reduce them throughout the full process chain.

7.2 Development of Precision Additive Manufacturing Processes

This chapter focuses mainly on Laser Powder Bed Fusion (L-PBF), because it is one of the most important metal AM processes for producing precise metal components.

7.2.1 Why precision in metal AM matters

In metal AM, small errors may accumulate layer-by-layer. A small deviation in one layer can affect the next layers and lead to larger geometrical errors, poor surface quality or internal defects.

Precision process development aims to make metal AM:

- repeatable,
- predictable,
- stable,
- defect-minimising,
- suitable for industrial production.

Lack of precision can arise from:

- variation in material properties,
- strong interaction between process parameters,
- limited machine accuracy,
- lack of in-process control,
- thermal gradients,
- residual stresses,
- powder bed variability.

7.2.2 L-PBF process sequence

The L-PBF process follows these main steps:

1. A CAD model is created.
2. The CAD file is converted into STL format.
3. Process parameters are defined in pre-processing software.

4. The geometry is digitally sliced into layers.
5. A thin powder layer is spread over the build platform.
6. A laser selectively melts the powder according to the layer geometry.
7. The melt pool solidifies after cooling.
8. The build platform is lowered by one layer thickness.
9. A new powder layer is spread.
10. The process repeats until the full part is built.
11. The build platform is removed for heat treatment and finishing.
12. Excess powder is removed, sieved and reused when possible.

7.2.3 Setting priorities in AM process development

AM process development should be considered at two levels:

1. **Process level:** optimisation of the AM process itself, including process parameters, dimensional accuracy, surface texture, energy efficiency and material usage.
2. **System level:** optimisation of how AM fits into the full manufacturing chain, including lead time, logistics, cost, qualification and integration with other processes.

Therefore, precision AM process development must be holistic. Improving one parameter is not enough if the full production chain becomes more expensive or less reliable.

7.2.4 Significant L-PBF process parameters

The most important L-PBF process parameters can be grouped into four categories.

Laser-related parameters

- **Laser power P :** optical power delivered by the laser.
- **Spot size d :** diameter of the focused laser beam.
- **Focal offset distance:** distance between the laser focal plane and the powder bed surface.

Scan-related parameters

- **Scanning speed v :** speed of the laser beam over the powder bed.
- **Hatch spacing h :** distance between adjacent scan tracks.
- **Scan pattern:** path followed by the laser, for example stripes, rectangular cells or hexagonal cells.

Powder-related parameters

- **Particle size and distribution:** affects powder packing, flowability and laser absorption.
- **Powder bed density:** affects porosity, thermal conductivity and final surface quality.
- **Layer thickness t :** thickness of each deposited powder layer.

Build chamber-related parameters

- **Build chamber temperature:** affects residual stress, thermal gradients and mechanical properties.
- **Atmosphere:** inert gas atmosphere reduces oxidation and porosity.

7.2.5 Combined energy-density parameters

Since many process parameters interact, combined energy-density measures are commonly used.

$$E_V = \frac{P}{vht} \quad (24)$$

where E_V is the volumetric energy density.

$$E_I = \frac{4P}{v\pi d^2} \quad (25)$$

where E_I is the linear input energy density.

$$E_S = \frac{P}{vd} \quad (26)$$

where E_S is the surface energy density.

$$E_L = \frac{P}{v} \quad (27)$$

where E_L is the linear energy density.

The general energy-density regimes are:

- **Low energy density:** lack of fusion and unmelted powder.
- **Optimal energy density:** dense parts and stable melting.
- **High energy density:** keyhole formation, vaporisation and porosity.

7.2.6 AM performance indicators

Important AM performance indicators include:

- mechanical properties,
- dimensional accuracy,
- surface texture,
- part density,
- total build time,
- energy consumption,
- system-wide manufacturing KPIs.

Mechanical properties Commonly measured mechanical properties include:

- tensile strength,
- yield strength,
- hardness,
- fatigue strength,
- ductility.

Dimensional accuracy Dimensional accuracy describes how close the printed part is to the intended CAD geometry. It is strongly affected by:

- thermal gradients,
- residual stresses,
- part orientation,
- support structures,
- post-processing.

Residual stress formation During L-PBF, newly melted upper layers are hot and tend to expand. Colder underlying layers restrict this expansion. During cooling, tensile residual stresses may remain in the part. These can cause:

- warping,
- cracking,
- delamination,
- distortion after removal from the build plate.

Part orientation Part orientation affects surface quality and dimensional accuracy. Holes and cylindrical features are often printed most accurately when their axis is aligned with the build direction, because this reduces staircase effects.

Support structures Supports are used to:

- mechanically support overhangs,
- improve heat conduction to the build plate,
- reduce warping and deformation.

However, supports also:

- increase build time,
- require removal,
- can leave marks,
- may introduce burrs and surface defects.

Part density Internal defects such as pores, cracks, inclusions and lack-of-fusion regions reduce part density and degrade mechanical performance. A fully dense part is often considered to have density above 99%, although this definition is not universal.

System-wide KPIs In industrial AM, process quality must also be judged using manufacturing KPIs such as:

- on-time delivery,
- cycle time,
- changeover time,
- yield,
- throughput,
- overall equipment efficiency,
- manufacturing cost per unit.

Overall equipment efficiency is commonly written as:

$$OEE = \text{Availability} \times \text{Performance} \times \text{Quality} \quad (28)$$

7.2.7 Data-driven process improvement

Modern precision AM development is increasingly data-driven. A typical data-driven improvement cycle has five steps:

1. data collection,
2. data sharing,
3. data analysis,
4. optimisation,
5. feedback to the machine or operator.

Data may come from:

- in-situ sensors,
- cameras,
- thermal monitoring,
- machine logs,
- off-line inspection,
- mechanical testing.

Before modelling, AM data often needs to be:

- digitised,

- cleaned,
- normalised,
- filtered,
- checked for redundancy.

7.2.8 Design of Experiments

Design of Experiments (DoE) is used to study how process inputs affect output responses with a reduced number of tests.

A DoE includes:

- **factors:** input variables,
- **levels:** values assigned to the factors,
- **responses:** measured outputs.

For a full factorial design, the number of required experiments is:

$$n_r = n_l^{n_f} \quad (29)$$

where:

- n_r is the number of experimental runs,
- n_l is the number of levels,
- n_f is the number of factors.

Because this number increases very quickly, fractional factorial designs, central composite designs and Taguchi orthogonal arrays are often used.

7.2.9 Modelling and optimisation

Once data has been collected, process-performance models can be built. These models describe the relationship between input parameters and output responses.

Common modelling methods include:

- regression models,
- statistical analysis,
- artificial neural networks,
- surrogate models,
- machine learning models.

After a model is built, optimisation can be used to:

- maximise part density,
- minimise surface roughness,

- minimise dimensional deviation,
- reduce cost,
- reduce build time,
- improve mechanical properties.

7.2.10 Precision AM and Industry 4.0

Precision AM is strongly connected to Industry 4.0 because modern AM systems are increasingly:

- sensor-rich,
- connected,
- data-driven,
- automated,
- capable of real-time monitoring.

Real-time monitoring can help detect:

- lack of fusion,
- overheating,
- powder spreading defects,
- geometric distortion,
- unstable melt pools.

Artificial intelligence can support:

- defect recognition,
- prediction of part quality,
- automatic parameter adjustment,
- digital quality control.

7.3 Modelling Techniques to Enhance Precision in Metal AM

Modelling is used in metal AM to better understand, predict and control the process. Since direct observation of all process phenomena is difficult, simulations can help explain how process parameters influence melt pool behaviour, defects, residual stress, distortion and microstructure.

7.3.1 Why modelling is needed

Metal AM involves several coupled physical phenomena:

- heat transfer,
- laser-material interaction,
- melting and solidification,

- fluid flow in the melt pool,
- evaporation and recoil pressure,
- powder spreading,
- residual stress,
- phase transformation,
- microstructure evolution.

These phenomena occur over very different length and time scales. This makes full high-fidelity modelling very expensive.

7.3.2 Length and time scales

Metal AM can be modelled at different scales:

- **Macro-scale models:** part-level models used to predict thermal history, residual stress, distortion and warpage.
- **Meso-scale models:** powder-bed and melt-pool models used to study powder particles, pores, melt pool flow and local defects.
- **Micro-scale models:** metallurgical models used to study grain growth, phase formation and microstructure evolution.

7.3.3 Macro-scale modelling

Macro-scale models are used when the whole part or a large region of the part is being analysed. The goal is usually to predict:

- temperature history,
- residual stresses,
- part distortion,
- warpage,
- deformation after removal from the build plate.

Thermal history is important because it controls:

- cooling rate,
- residual stresses,
- microstructure,
- distortion.

Full thermo-mechanical models are accurate but computationally expensive. Therefore, simplified or reduced models are often used.

Types of macro-scale models

- **Full-scale models:** simulate detailed thermal and mechanical behaviour but are expensive.
- **Reduced-order models:** simplify the physics to reduce computational time.
- **Effective models:** use calibrated simplified assumptions to estimate distortion or residual stress.
- **Inherent strain models:** estimate final distortion using calibrated strain fields instead of simulating the full thermal cycle.

7.3.4 Meso-scale modelling

Meso-scale modelling focuses on powder particles, melt pool geometry and local defect formation.

Important topics include:

- powder bed formation,
- powder packing density,
- laser absorption,
- melt pool shape,
- keyhole formation,
- spatter,
- pore evolution,
- balling,
- surface roughness formation.

Powder bed formation The powder bed is not a continuous solid. It is made from many particles with different sizes, shapes and packing arrangements. This affects:

- local absorptivity,
- thermal conductivity,
- melt pool stability,
- porosity,
- surface quality.

Laser-material interaction When the laser interacts with the powder bed, energy is absorbed, reflected and scattered. The local energy input depends on:

- laser power,
- beam diameter,
- wavelength,
- powder morphology,
- powder absorptivity,
- surface angle,
- multiple reflections between particles.

Melt pool dynamics The melt pool is controlled by:

- surface tension,
- Marangoni flow,
- recoil pressure,
- evaporation,
- wetting behaviour,
- heat conduction into the substrate,
- interaction with surrounding powder.

Unstable melt pools can lead to:

- lack of fusion,
- keyhole porosity,
- spatter,
- balling,
- rough surfaces.

7.3.5 Microstructure simulations

Microstructure simulations are used to predict grain structure and phase formation. This is important because the microstructure strongly affects mechanical properties.

Important microstructural outputs include:

- grain size,
- grain morphology,
- columnar or equiaxed grains,
- crystallographic texture,
- phase fractions,
- local hardness,
- anisotropy.

Common modelling techniques include:

- cellular automata,
- phase-field models,
- Monte Carlo models,
- thermodynamic models,
- kinetic models,
- data-driven microstructure models.

7.3.6 Solidification in AM

The solidification structure depends strongly on:

- thermal gradient G ,
- solidification rate R ,
- cooling rate,
- melt pool geometry,
- alloy composition.

In general:

- high G/R promotes columnar grains,
- low G/R promotes equiaxed grains,
- high cooling rates produce fine microstructures.

7.3.7 Need for heat treatment

Due to rapid solidification and strong thermal gradients, as-built AM parts may contain:

- residual stresses,
- non-equilibrium phases,
- anisotropic microstructures,
- brittle regions,
- local hardness variations.

Heat treatment can be used to:

- relieve residual stresses,
- homogenise microstructure,
- modify phase fractions,
- improve ductility,
- improve fatigue performance.

7.3.8 Data-driven modelling for process windows

Data-driven models can be used to identify suitable process windows. A process window defines the parameter range where stable and high-quality printing is possible.

Input parameters may include:

- laser power,
- scan speed,
- hatch spacing,

- layer thickness,
- powder properties,
- build chamber temperature.

Outputs may include:

- density,
- roughness,
- porosity,
- distortion,
- tensile strength,
- defect probability.

7.3.9 Model verification, validation and uncertainty

For models to be useful in precision AM, they must be verified and validated.

- **Verification:** checking that the equations are solved correctly.
- **Validation:** checking that the model agrees with physical experiments.
- **Calibration:** adjusting uncertain parameters so the model matches measured results.
- **Uncertainty quantification:** estimating how uncertain inputs affect model outputs.

High-fidelity models are useful for understanding defects and developing new materials, but they are usually too slow for daily production control. Faster calibrated models are more useful for industrial process monitoring and feed-forward control.

7.4 Secondary Finishing Operations

Secondary finishing operations are post-build processes used to improve the final quality of AM parts. Metal AM parts often need finishing because as-built surfaces are usually rough, geometrically complex and affected by defects from the layer-wise process.

7.4.1 Why AM surfaces need finishing

AM surfaces may need finishing to:

- reduce surface roughness,
- improve fatigue strength,
- improve fluid flow,
- improve tribological performance,
- improve sealing and contact surfaces,
- improve dimensional tolerance,
- improve aesthetics,
- remove partially attached powder particles,
- remove support marks.

7.4.2 Effect of surface topography on function

Surface topography affects component performance. For example:

- high peaks can act as stress concentrators,
- rough surfaces can promote fatigue crack initiation,
- rough internal channels can increase flow resistance,
- surface peaks can disturb fluid flow and create micro-turbulence,
- rough bearing surfaces can cause local contact loading,
- poor surface finish can reduce sealing performance.

7.4.3 AM-specific finishing challenges

Finishing AM parts is difficult because of:

- high as-built roughness,
- surface texture variation across the part,
- partially melted particles,
- support witness marks,
- distortion,
- complex external geometry,
- internal channels,
- lattice structures,
- surfaces with no direct line-of-sight,
- small features that may be damaged by aggressive finishing.

7.4.4 Geometrical challenges

Typical geometries that are difficult to finish include:

- narrow channels,
- complex internal channels,
- internal cavities connected to the surface,
- variable cross-section channels,
- outer lattice surfaces,
- inner lattice surfaces,
- thin walls,
- closed internal cavities.

Closed internal cavities are especially difficult because finishing media cannot easily enter or leave the cavity.

7.4.5 Available finishing processes

Finishing processes can be grouped into conventional, non-conventional, hybrid and emerging AM-specific methods.

Table 12: Examples of secondary finishing operations for metal AM parts.

Category		Examples
Conventional methods	mechanical	Turning, milling, drilling, broaching, grinding, honing, lapping, burnishing, finishing
Media-based methods		Grit blasting, abrasive flow machining, media-based mass finishing
Electrical methods		Electrical discharge machining, electrochemical machining, electropolishing
Thermal methods		Laser re-melting, laser ablation
Chemical methods		Chemical etching, chemical polishing
Hybrid methods		Magnetic abrasive finishing, plasma electrolytic polishing, combined mass finishing and electropolishing

7.4.6 Strengths and limitations of selected finishing methods

- **Turning:** high dimensional accuracy, but mainly suitable for rotational features.
- **Milling:** suitable for complex external features, but tool access is required.
- **Drilling:** useful for through-holes, but limited for complex internal features.
- **Grinding:** improves surface finish, but tool access can be difficult for small features.
- **Honing:** good for internal cylindrical surfaces, but not suitable for complex freeform internal channels.
- **Lapping:** can produce very smooth surfaces, but requires skill and is limited in geometry.
- **Abrasive flow machining:** useful for complex internal channels, but setup can be expensive.
- **Chemical etching:** can access complex surfaces, but may use hazardous chemicals and reduce feature sharpness.
- **Electropolishing:** can improve surface smoothness, but requires suitable electrolyte and process control.
- **Laser re-melting:** offers local control and can preserve geometry, but may introduce thermal effects.
- **Laser ablation:** precise and non-contact, but relatively slow.
- **Mass finishing:** can finish many parts at once, but may round edges and damage fine details.

7.4.7 Finishing in precision applications

For precision AM parts, finishing must be selected carefully because improving surface texture may also change part geometry. A good finishing process should:

- reduce roughness,
- preserve functional dimensions,
- avoid excessive edge rounding,

- avoid damaging thin features,
- be repeatable,
- be compatible with inspection and tolerance requirements.

7.5 Non-Destructive Evaluation for Additive Manufacturing

Non-Destructive Evaluation (NDE) is used to inspect AM parts without damaging them. This is especially important for safety-critical components in aerospace, medical, defence and energy applications.

7.5.1 Purpose of NDE in AM

NDE is used to:

- detect internal and surface defects,
- support part qualification,
- verify structural integrity,
- reduce destructive testing,
- support process development,
- validate AM process stability.

7.5.2 Typical defects in AM

Typical defects include:

- gas porosity,
- lack-of-fusion porosity,
- keyhole pores,
- cracks,
- delamination,
- inclusions,
- unmelted powder,
- surface roughness defects,
- dimensional deviations,
- residual stress-related distortion.

Defect detectability depends mainly on:

- defect size,
- defect shape,
- defect location,
- defect orientation,

- material,
- surface condition,
- inspection method.

7.5.3 NDE challenges in AM

NDE is challenging for AM because:

- AM parts can have very complex geometries,
- internal channels and lattices may be difficult to inspect,
- rough surfaces affect many inspection methods,
- small defects may be critical for fatigue,
- defects may be irregular and lack simple shapes,
- standards are still developing,
- qualification routes can be expensive and slow.

7.5.4 Common NDE methods

Table 13: Common NDE methods for AM parts.

Method	Main use	Limitations
Visual inspection	Surface defects, obvious build failures	Only external and visible defects
Dye penetrant testing	Surface-breaking cracks and defects	Requires accessible surfaces and cleaning
Magnetic particle testing	Surface and near-surface defects in ferromagnetic materials	Only for magnetic materials
Ultrasonic testing	Internal defects and lack of bonding	Complex AM geometry and roughness can complicate signals
Radiography	Internal volumetric defects	Defect orientation and overlapping geometry can reduce detectability
X-ray CT	Internal and external geometry, porosity, inclusions, dimensional inspection	Can be slow and limited by part size, material density and artefacts
Eddy current testing	Surface and near-surface defects in conductive materials	Limited depth and sensitive to surface roughness
Thermography	Near-surface defects, heat-flow anomalies	Limited penetration depth and interpretation challenges
Acoustic/resonance methods	Global defect detection and part-to-part comparison	Usually less direct in locating specific defects

7.5.5 Post-process inspection

Post-process inspection is performed after the part is built. It can include:

- visual inspection,
- dimensional metrology,
- surface metrology,
- X-ray CT,
- ultrasonic testing,
- radiography,
- dye penetrant inspection,
- mechanical testing.

Post-process NDE is important for final part qualification, but it cannot correct a defect once the build is complete.

7.5.6 In-process inspection

In-process inspection is performed during the build. It can include:

- melt pool monitoring,
- thermal imaging,
- optical layer imaging,
- acoustic monitoring,
- recoater monitoring,
- laser power monitoring.

In-process NDE is useful because it may detect process deviations early and potentially allow corrective action before the part is finished.

7.5.7 NDE standardisation

NDE standardisation is important because AM parts need reliable inspection procedures. Standardisation helps define:

- defect terminology,
- inspection procedures,
- acceptance criteria,
- qualification methods,
- probability of detection,
- reporting requirements.

7.5.8 Probability of Detection

The reliability of an NDE method is commonly described using Probability of Detection (PoD). PoD describes the probability that a defect of a given size will be detected.

A general trend is:

- small defects have low detection probability,
- large defects have high detection probability,
- the transition depends on the inspection method and part condition.

PoD is important because a method that sometimes misses critical defects may not be acceptable for safety-critical AM components.

7.6 X-Ray Computed Tomography

X-ray Computed Tomography (CT) is a non-destructive and non-contact measurement method that can inspect both external and internal features of AM parts. It is especially useful for complex AM geometries with internal channels, lattices or inaccessible surfaces.

7.6.1 Why CT is useful for AM

CT can be used to analyse:

- internal defects,
- porosity,
- inclusions,
- lack-of-fusion regions,
- internal channels,
- lattice structures,
- dimensional accuracy,
- surface topography,
- powder feedstock.

Unlike tactile or optical measurement methods, CT can measure non-accessible internal features.

7.6.2 Industrial CT system components

A typical industrial CT system consists of:

- an X-ray source,
- an X-ray detector,
- a rotary stage or kinematic system,
- shielding,
- reconstruction software,
- analysis software.

X-ray source The X-ray source generates radiation that passes through the object. Important source parameters include:

- voltage,
- current,
- focal spot size,
- source stability,
- beam energy.

X-ray detector The detector records transmitted X-rays and converts them into projection images. Detector performance depends on:

- pixel size,
- dynamic range,
- noise,
- sensitivity,
- resolution.

Kinematic system The kinematic system rotates or moves the part so that projections can be acquired from many angles. Accuracy of the motion system affects the final CT reconstruction.

7.6.3 CT scanning process

The CT process typically follows these steps:

1. The part is placed between the X-ray source and detector.
2. X-ray projections are acquired from many angular positions.
3. The projections are reconstructed into a 3D voxel volume.
4. The material surface is determined by thresholding or surface determination methods.
5. The reconstructed volume is analysed for defects, dimensions or surface characteristics.

7.6.4 CT measurement errors and artefacts

CT measurements are affected by several error sources:

- beam hardening,
- scattered radiation,
- cone-beam artefacts,
- ring artefacts,
- noise,
- partial volume effects,

- detector errors,
- focal spot size,
- thermal drift,
- sample movement,
- wrong threshold selection,
- poor calibration.

Beam hardening Beam hardening occurs because low-energy X-rays are absorbed more strongly than high-energy X-rays. This can cause artificial density gradients and dimensional errors.

Partial volume effect The partial volume effect occurs when one voxel contains both material and air. This is especially important for small features and rough AM surfaces.

Ring artefacts Ring artefacts are circular artefacts caused by detector imperfections or calibration errors.

7.6.5 Traceability and uncertainty

For CT to be used as a metrology tool, measurement traceability and uncertainty must be considered.

Important topics include:

- system calibration,
- metrological performance verification,
- use of reference objects,
- uncertainty determination,
- repeatability tests,
- comparison with calibrated reference measurements.

7.6.6 Applications of CT in AM

Dimensional and geometrical verification CT can compare the printed part with the CAD model and identify:

- dimensional deviations,
- form errors,
- internal geometry errors,
- wall thickness variation,
- channel blockage,
- lattice deformation.

Internal defect analysis CT can detect and quantify:

- pore size,
- pore shape,
- pore volume fraction,
- pore location,
- inclusions,
- cracks if resolution and contrast are sufficient.

Surface topography characterisation CT can characterise both external and internal surface topography. This is useful for AM parts with internal channels that cannot be measured using optical or tactile methods.

Powder feedstock characterisation CT can also be used to inspect powder particles, including:

- particle size,
- particle morphology,
- internal porosity,
- satellite particles,
- particle defects.

Product development and process optimisation CT supports AM process development by linking process parameters to defects and geometry errors. It can therefore help improve:

- scan strategy,
- support design,
- orientation selection,
- heat treatment,
- finishing strategy,
- process window selection.

7.7 On-Machine Measurement, Monitoring and Control

On-machine measurement, monitoring and control are used to observe and improve the AM process during or immediately after printing. These methods are especially important in PBF processes, where many defects are formed during the build.

7.7.1 Basic terminology

- **On-machine sensing:** acquisition of data using sensors installed on the AM machine.
- **In-process measurement:** measurement performed during the build process itself.
- **On-machine measurement:** use of machine data to estimate product quality characteristics.
- **On-machine monitoring:** statistical monitoring of the process to detect abnormal conditions.
- **Process control:** closed-loop adjustment of process parameters to return the process to a target state.

7.7.2 Defect causes in PBF

Defects in PBF can originate from:

- feedstock material,
- equipment,
- design or job preparation,
- process settings.

Feedstock-induced defects Powder defects include:

- irregular particle morphology,
- entrapped gas porosity,
- satellite particles,
- broken particles,
- moisture contamination,
- oxide or nitride films,
- poor flowability.

Equipment-induced defects Equipment-related problems include:

- laser power instability,
- scanner errors,
- recoater defects,
- uneven powder spreading,
- gas flow instability,
- build platform errors.

Design and job preparation defects Defects can also come from:

- unsuitable support design,
- poor build orientation,
- excessive overhangs,
- poor scan strategy,
- insufficient compensation for shrinkage,
- wrong slicing settings.

Process setting defects Incorrect process parameters may cause:

- lack of fusion,
- keyhole porosity,
- overheating,
- balling,
- surface roughness,
- residual stress,
- cracking.

7.7.3 Types of defects monitored in PBF

Important defect types include:

- porosity,
- residual stresses,
- cracks,
- delamination,
- microstructural inhomogeneity,
- impurities,
- balling,
- dimensional and geometrical deviations,
- surface defects.

7.7.4 On-machine sensing methods

On-machine sensing can be based on different physical signals:

- optical emission,
- infrared radiation,
- thermal radiation,
- acoustic emission,
- photodiode signals,
- camera images,
- machine position signals,
- laser power signals,
- recoater force or motion data.

7.7.5 Spatially integrated and spatially resolved sensors

Spatially integrated sensors Spatially integrated sensors produce one signal or a small number of signals for a region. Examples include:

- photodiodes,
- pyrometers,
- acoustic sensors.

These sensors are fast but provide limited spatial information.

Spatially resolved sensors Spatially resolved sensors provide image-like data. Examples include:

- visible cameras,
- infrared cameras,
- thermal cameras,
- layer imaging systems.

These sensors provide more spatial information but produce larger data volumes.

7.7.6 Data gathering levels

AM monitoring data can be collected at different levels:

- point level,
- melt pool level,
- track level,
- layer level,

- part level,
- build level.

Each level gives different information. Melt pool data is useful for detecting local process instability, while layer-level imaging is useful for detecting recoating problems, spatter accumulation or geometric deviations.

7.7.7 Sensing architectures

Co-axial sensing In co-axial sensing, the sensor observes the process along the same optical path as the laser. This makes it possible to monitor the melt pool directly and follow the laser position.

Advantages:

- direct view of the laser interaction zone,
- good for melt pool monitoring,
- follows the scan path.

Limitations:

- small field of view,
- complex optical integration,
- limited information away from the melt pool.

Off-axis sensing In off-axis sensing, the sensor is placed at an angle relative to the process.

Advantages:

- larger field of view,
- easier integration,
- useful for layer and powder bed monitoring.

Limitations:

- perspective distortion,
- possible occlusion,
- less direct view of the melt pool.

7.7.8 On-machine measurement

On-machine measurement aims to estimate part properties during the build. Examples include:

- layer topography reconstruction,
- detection of powder bed irregularities,
- melt pool size estimation,
- estimation of surface defects,
- geometric deviation monitoring.

This can reduce reliance on post-process inspection, although it does not fully replace final qualification.

7.7.9 Statistical Process Monitoring

Statistical Process Monitoring (SPM), also called Statistical Process Control (SPC), is used to detect whether the process is behaving normally.

A monitoring system usually defines:

- a process signature,
- a reference or expected value,
- control limits,
- an alarm rule.

If the monitored signal exceeds the control limits, the process may be considered out of control.

Type I and Type II errors

- **Type I error:** false alarm. The system says there is a problem when the process is actually normal.
- **Type II error:** missed detection. The system fails to detect a real problem.

7.7.10 Control charts

Control charts are used to monitor process stability over time. They can be applied to:

- melt pool intensity,
- melt pool area,
- thermal emission,
- layer image features,
- recoater signals,
- acoustic emission,
- geometric measurements.

7.7.11 Process control

Process control uses monitoring data to adjust process parameters. Control actions may occur:

- point-to-point,
- track-to-track,
- layer-to-layer,
- build-to-build.

Possible control variables include:

- laser power,
- scan speed,

- hatch spacing,
- beam diameter,
- powder layer thickness,
- preheating temperature,
- scan strategy.

7.7.12 Main takeaway

On-machine measurement, monitoring and control are essential for moving metal AM from trial-and-error manufacturing towards stable, predictable and intelligent production. The long-term goal is to detect process errors early, estimate final part quality during the build and apply corrective actions before severe defects are formed.